Journal of Computational and Applied Mathematics 426 (2023) 115107

Contents lists available at ScienceDirect

Journal of Computational and Applied Mathematics

journal homepage: www.elsevier.com/locate/cam

Least squares parameter estimation and multi-innovation least squares methods for linear fitting problems from noisy data ☆

Feng Ding *

School of Electrical and Electronic Engineering, Hubei University of Technology, Wuhan 430068, PR China
College of Automation and Electronic Engineering, Qingdao University of Science and Technology, Qingdao 266061, PR China
School of Internet of Things Engineering, Jiangnan University, Wuxi 214122, PR China

ARTICLE INFO

Article history:
Received 24 October 2022
Received in revised form 5 February 2023

Keywords:
Least squares
Parameter estimation
System identification
Least squares review
Least squares survey
Multi-innovation identification

ABSTRACT

Least squares is an important method for solving linear fitting problems and quadratic optimization problems. This paper explores the properties of the least squares methods and the multi-innovation least squares methods. It demonstrates lemmas and theorems about the least squares and multi-innovation least squares parameter estimation algorithms after reviewing and surveying some important contributions in the area of system identification, such as the auxiliary model identification idea, the multi-innovation identification theory, the hierarchical identification principle, the coupling identification concept and the filtering identification idea. The results of the least squares and multi-innovation least squares algorithms for linear regressive systems with white noises can be extended to other systems with colored noises.

Contents

☆ This work was supported by the National Natural Science Foundation of China (No. 62273167) and the 111 Project (B23008).

* Correspondence to: College of Automation and Electronic Engineering, Qingdao University of Science and Technology, Qingdao 266061, PR China.
E-mail address: fding@jiangnan.edu.cn.

https://doi.org/10.1016/j.cam.2023.115107

Nomenclature

a.s.	almost surely
AR	Autoregressive
CAR	Controlled autoregressive
EE	Equation-error
FIR	Finite impulse response
GAE	Generalized attenuating excitation
GPE	Generalized persistent excitation
GWPE	Generalized WPE
LP-EE	Linear-parameter EE
LP-FIR	Linear-parameter FIR
LS	Least squares
LSE	LS estimate
m.s.	mean square
MILS	Multi-innovation LS
MIMO	Multi-input multi-output
RLS	Recursive LS
WPE	Weak persistent excitation

Mathematical notations

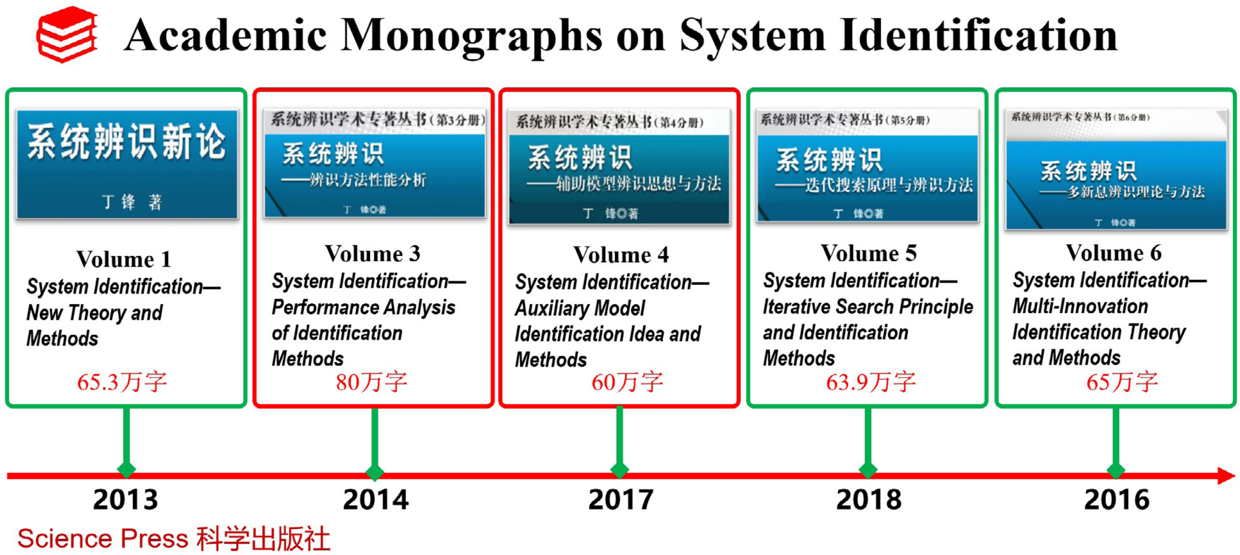
$\mathbf{0}$:	The zero vector or zero matrix of appropriate sizes.
$\mathbf{1}_{m\times n}$:	An $m\times n$ matrix whose entries are all 1.
$\mathbf{1}_n$:	An n-dimensional column vector whose entries are all 1.
$\boldsymbol{I}$ or $\boldsymbol{I}_n$:	The identity matrix of appropriate sizes or $n\times n$.
$\mathrm{tr}[\boldsymbol{X}]$:	The trace of the square matrix $\boldsymbol{X}$.
$\boldsymbol{X}^{\mathrm{T}}$:	The transpose of the vector/matrix $\boldsymbol{X}$.
$\\|\boldsymbol{X}\\|$:	$\\|\boldsymbol{X}\\|^2 := \mathrm{tr}[\boldsymbol{X}\boldsymbol{X}^{\mathrm{T}}] = \mathrm{tr}[\boldsymbol{X}^{\mathrm{T}}\boldsymbol{X}]$.
$\lambda_{\max}[\boldsymbol{X}]$	The greatest eigenvalue of the symmetric real matrix $\boldsymbol{X}\in\mathbb{R}^{n\times n}$.
$\lambda_{\min}[\boldsymbol{X}]$	The smallest eigenvalue of the symmetric real matrix $\boldsymbol{X}\in\mathbb{R}^{n\times n}$.
$A =: X$:	X is defined by A.
$X := A$:	X is defined by A.
t:	The time variable $t = 1, 2, 3, \ldots$
$\hat{\boldsymbol{\theta}}$:	The estimate of the parameter vector/matrix $\boldsymbol{\theta}$.
$\hat{\boldsymbol{\theta}}(t)$:	The estimate of the parameter vector/matrix $\boldsymbol{\theta}$ at time t.
z^{-1}:	The unit backward shift operator, e.g., $z^{-1}y(t) = y(t-1)$.
p_0:	A large positive constant, e.g., $p_0 = 10^6$.

1. Introduction

Mathematical models are the foundation of natural science and social science. All scientific research processes are building mathematical models describing the laws of motion and evolution of things. Control science is a special discipline, and its general research methods support the research and development of other disciplines. System identification is the theory and methods of investigating and building the mathematical models of various systems [1–6] such as linear systems [7–10], bilinear systems [11–15] and nonlinear systems [16–20]. The gradient search and the Newton search are two basic methods for solving nonlinear optimization problems. For quadratic optimization problems, the Newton methods degenerate to the least squares methods. For linear-parameter fitting problems, the least squares is undoubtedly a best method. These search schemes have been used in system identification for exploring new identification methods of linear stochastic systems and nonlinear stochastic systems with colored noises [21–24].

In general, the iterative search schemes are suitable for solving static optimization problems and are used for offline identification, which update the parameter estimates by means of a batch of observation data [25–28]. The recursive search schemes are suitable for solving dynamical optimization problems and are used for online identification, which update the parameter estimates in real time while collecting data [29–33]. System identification includes two large categories of identification methods, namely iterative identification methods and recursive identification methods. The reference book *System Identification—Iterative Search Principle and Identification Methods* is an early iterative identification monograph in the world [4]. Recent published monographs about system identification also include *System Identification—New Theory and Methods* [1], *System Identification—Performances Analysis for Identification Methods* [2], *System Identification—Auxiliary Model Identification Idea and Methods* [3], *System Identification—Multi-Innovation Identification Theory and Methods* [5], which involve the auxiliary model identification idea, multi-innovation identification theory, hierarchical identification principle, coupling identification concept, and filtering identification idea proposed first by the author of this paper.

Some identification ideas and identification principles can combine the least squares to derive new parameter estimation methods, such as the auxiliary model identification idea [1,3,34–36], the multi-innovation identification theory [1,5,37–40], the hierarchical identification principle [1,6,41–49], the coupling identification concept [1,50–53], and the filtering identification idea [5,36,54]. The multi-innovation identification theory is beneficial for deriving more accurate estimation algorithms by expanding the innovation from a scalar to a vector and/or from a vector to a matrix, which can be used for linear systems [55–60] and nonlinear systems [61–67]. The hierarchical identification principle is the decomposition-based identification, whose key is to decompose a system into several subsystems which can be identified more easily by using the gradient and the least squares methods. The hierarchical identification is suitable for large-scale systems for reducing computational complexity. For example, the typical applications of the hierarchical identification principle are the separable projection algorithms and the separable least squares algorithms [68–72], and the separable gradient algorithms and the separable Newton algorithms [73–77]. The hierarchical identification principle can be used to investigate the hierarchical gradient-based algorithms and the hierarchical least squares-based iterative algorithms for solving general matrix equations and coupled matrix equations [35,78–84].



Considering the widespread use of the least squares methods with high estimation accuracy and rapid convergence, this paper focuses on the least squares and multi-innovation least squares parameter estimation algorithms and their convergence properties for linear regressive models.

2. Preliminary linear regressive models

Let $t = 1, 2, 3, \ldots$. Suppose that $y(t)$, $x_1(t)$, $x_2(t)$, $\ldots$, $x_n(t)$ are $n + 1$ independent observations for each t, and they satisfy a linear regression,

$$y(t) = \theta_1 x_1(t) + \theta_2 x_2(t) + \cdots + \theta_n x_n(t) + v(t), \tag{2.1}$$

with unknown coefficients (i.e., model parameters) θ_1, θ_2, $\ldots$, θ_n, where $\{v(t)\}$ is an independent white noise sequence with mean zero and variance σ^2, and the integer n is the number of the parameters.

Define the parameter vector $\boldsymbol{\theta}$ and the information vector $\boldsymbol{\varphi}(t)$ as

$$\boldsymbol{\theta} := [\theta_1, \theta_2, \ldots, \theta_n]^{\mathrm{T}} \in \mathbb{R}^n,$$

$$\boldsymbol{\varphi}(t) := [x_1(t), x_2(t), \ldots, x_n(t)]^{\mathrm{T}} \in \mathbb{R}^n.$$

Then Eq. (2.1) can be written as a linear regressive model,

$$y(t) = \boldsymbol{\varphi}^{\mathrm{T}}(t)\boldsymbol{\theta} + v(t). \tag{2.2}$$

Eq. (2.2) is called an identification model or an identification representation in system identification. Its feature is that the observation $y(t)$ is a linear function of the parameter vector $\boldsymbol{\theta}$ and thus this is a linear model or linear-parameter model and a linear fitting problem. The fitting from data is to determine the parameters of the given model and its generalization is system identification. System identification is the theory and methods of establishing the mathematical models of systems.

Eq. (2.2) is a linear-parameter model and can describe some nonlinear systems with having linear parameters. For example, the nonlinear system [1],

$$y(t) = a_1 y^2(t-1) + a_2 y(t-1)u(t) + \mathrm{e}^{b_1}u(t) + b_2 u(t-1)u(t-2) + v(t),$$

can be written as the form of (2.2) through letting

$$\begin{aligned}
\boldsymbol{\theta} &:= [a_1, a_2, \mathrm{e}^{b_1}, b_2]^{\mathrm{T}},\\
\boldsymbol{\varphi}(t) &:= [y^2(t-1), y(t-1)u(t), u(t), u(t-1)u(t-2)]^{\mathrm{T}}.
\end{aligned}$$

Of course, the following nonlinear system is a linear-parameter system:

$$\begin{aligned}
y(t) &= \theta_1 f_1(x_1(t), x_2(t), \ldots, x_n(t)) + \theta_2 f_2(x_1(t), x_2(t), \ldots, x_n(t)) + \cdots\\
&\quad + \theta_n f_n(x_1(t), x_2(t), \ldots, x_n(t)) + v(t)\\
&= [\theta_1, \theta_2, \ldots, \theta_n]\begin{bmatrix} f_1(x_1(t), x_2(t), \ldots, x_n(t))\\ f_2(x_1(t), x_2(t), \ldots, x_n(t))\\ \vdots\\ f_n(x_1(t), x_2(t), \ldots, x_n(t))\end{bmatrix} + v(t)\\
&= \boldsymbol{\theta}^{\mathrm{T}}\boldsymbol{\varphi}(t) + v(t)\\
&= \boldsymbol{\varphi}^{\mathrm{T}}(t)\boldsymbol{\theta} + v(t),
\end{aligned} \tag{2.3}$$

with

$$\boldsymbol{\varphi}(t) = \begin{bmatrix} f_1(x_1(t), x_2(t), \ldots, x_n(t))\\ f_2(x_1(t), x_2(t), \ldots, x_n(t))\\ \vdots\\ f_n(x_1(t), x_2(t), \ldots, x_n(t))\end{bmatrix} \in \mathbb{R}^n, \quad \boldsymbol{\theta} = \begin{bmatrix}\theta_1\\ \theta_2\\ \vdots\\ \theta_n\end{bmatrix} \in \mathbb{R}^n,$$

$f_i(x_1(t), x_2(t), \ldots, x_n(t))$ is the nonlinear function of $x_1(t), x_2(t), \ldots, x_n(t)$.

The objective of identification is to determine the model parameter vector $\boldsymbol{\theta}$ from given observation data $\{y(t), \boldsymbol{\varphi}(t)$: $t = 1, 2, \ldots, L\}$, where L is the data length. The following drives the least squares estimation algorithm for identifying the parameter vector of the linear regressive systems.

3. Least squares (LS) methods

This section derives the LS estimates and the LS algorithm for linear regressive models, and investigates the properties and distributions of the LS estimates.

3.1. LS estimates

Define the stacked output vector $\boldsymbol{Y}_t$, the stacked information matrix $\boldsymbol{H}_t$ and the stacked noise vector $\boldsymbol{V}_t$ as

$$\boldsymbol{Y}_t := \begin{bmatrix} y(1)\\ y(2)\\ \vdots\\ y(t)\end{bmatrix} \in \mathbb{R}^t, \quad \boldsymbol{H}_t := \begin{bmatrix} \boldsymbol{\varphi}^{\mathrm{T}}(1)\\ \boldsymbol{\varphi}^{\mathrm{T}}(2)\\ \vdots\\ \boldsymbol{\varphi}^{\mathrm{T}}(t)\end{bmatrix} \in \mathbb{R}^{t\times n}, \quad \boldsymbol{V}_t := \begin{bmatrix} v(1)\\ v(2)\\ \vdots\\ v(t)\end{bmatrix} \in \mathbb{R}^t,$$

whose dimensions increase with the data length t increasing.

From (2.2) or (2.3), we have

$$\boldsymbol{Y}_t = \boldsymbol{H}_t\boldsymbol{\theta} + \boldsymbol{V}_t. \tag{3.1}$$

According to the least squares principle, define a quadratic criterion function of the squared sum of the errors by using the observation data $\{y(t), \boldsymbol{\varphi}(t)\}$ as

$$\begin{aligned} J_1(\boldsymbol{\theta}) &:= \sum_{j=1}^{t} v^2(j) = \sum_{j=1}^{t}[y(j) - \boldsymbol{\varphi}^{\mathrm{T}}(j)\boldsymbol{\theta}]^2 \\ &= \boldsymbol{V}_t^{\mathrm{T}}\boldsymbol{V}_t = (\boldsymbol{Y}_t - \boldsymbol{H}_t\boldsymbol{\theta})^{\mathrm{T}}(\boldsymbol{Y}_t - \boldsymbol{H}_t\boldsymbol{\theta}) \\ &= \|\boldsymbol{Y}_t - \boldsymbol{H}_t\boldsymbol{\theta}\|^2. \end{aligned}$$

Assume that $J_1(\boldsymbol{\theta})$ achieves a minimum when $\boldsymbol{\theta} = \hat{\boldsymbol{\theta}}(t)$. Letting the partial derivative of $J_1(\boldsymbol{\theta})$ with respect to $\boldsymbol{\theta}$ be zero gives

$$\left.\frac{\partial J_1(\boldsymbol{\theta})}{\partial \boldsymbol{\theta}}\right|_{\theta=\hat{\theta}(t)} = -2\boldsymbol{H}_t^{\mathrm{T}}(\boldsymbol{Y}_t - \boldsymbol{H}_t\boldsymbol{\theta})\Big|_{\theta=\hat{\theta}(t)} = \mathbf{0}.$$

This means

$$(\boldsymbol{H}_t^{\mathrm{T}}\boldsymbol{H}_t)\hat{\boldsymbol{\theta}}(t) = \boldsymbol{H}_t^{\mathrm{T}}\boldsymbol{Y}_t. \tag{3.2}$$

This gives the normal equation. Under the persistent excitation condition, i.e., $(\boldsymbol{H}_t^{\mathrm{T}}\boldsymbol{H}_t)$ is a positive definite matrix for large t, we can compute the least squares estimate (LSE or LS estimate) of the parameter vector $\boldsymbol{\theta}$ as

$$\hat{\boldsymbol{\theta}}(t) = (\boldsymbol{H}_t^{\mathrm{T}}\boldsymbol{H}_t)^{-1}\boldsymbol{H}_t^{\mathrm{T}}\boldsymbol{Y}_t. \tag{3.3}$$

Using the definitions of $\boldsymbol{H}_t$ and $\boldsymbol{Y}_t$, it follows that the LS estimate can be expressed as

$$\hat{\boldsymbol{\theta}}(t) = \left[\sum_{j=1}^{t}\boldsymbol{\varphi}(j)\boldsymbol{\varphi}^{\mathrm{T}}(j)\right]^{-1}\left[\sum_{j=1}^{t}\boldsymbol{\varphi}(j)y(j)\right]. \tag{3.4}$$

This is the least squares (LS) algorithm, which is also called the one-shot identification algorithm or direct identification algorithm.

When collecting a batch of data, we can use the LS algorithm in (3.3) or (3.4) for computing the LS estimate $\hat{\boldsymbol{\theta}}(t)$ of $\boldsymbol{\theta}$. This is an offline identification algorithm. Its shortcoming is that we have to compute the inverse matrix for each t, which leads to a heavy computational burden, especially when the dimension of $\boldsymbol{\theta}$ is very large. Thus the LS algorithm is not suitable for online identification. The following discusses the recursive computation relations of the LS algorithm in (3.3), i.e., the recursive least squares algorithm. It can be used for online identification.

In general, the observation data include stochastic errors, which are called noises, the LS estimate $\hat{\boldsymbol{\theta}}(t)$ depends on the data $\{y(t), \boldsymbol{\varphi}(t)\}$ and the data length t. Thus it is desired that the LS estimate $\hat{\boldsymbol{\theta}}(t)$ converges to its true value $\boldsymbol{\theta}$ with the increasing of the data length t. Once one obtains the LS estimate $\boldsymbol{\theta} = \hat{\boldsymbol{\theta}}(t)$ from available data $\{y(t), \boldsymbol{\varphi}(t)\}$, the minimum value of the criterion function can be obtained from

$$\begin{aligned} J_1(\hat{\boldsymbol{\theta}}(t)) &= [\boldsymbol{Y}_t - \boldsymbol{H}_t\hat{\boldsymbol{\theta}}(t)]^{\mathrm{T}}[\boldsymbol{Y}_t - \boldsymbol{H}_t\hat{\boldsymbol{\theta}}(t)] \\ &= [\boldsymbol{Y}_t - \boldsymbol{H}_t(\boldsymbol{H}_t^{\mathrm{T}}\boldsymbol{H}_t)^{-1}\boldsymbol{H}_t^{\mathrm{T}}\boldsymbol{Y}_t]^{\mathrm{T}}[\boldsymbol{Y}_t - \boldsymbol{H}_t(\boldsymbol{H}_t^{\mathrm{T}}\boldsymbol{H}_t)^{-1}\boldsymbol{H}_t^{\mathrm{T}}\boldsymbol{Y}_t] \\ &= \boldsymbol{Y}_t^{\mathrm{T}}[\boldsymbol{I}_t - \boldsymbol{H}_t(\boldsymbol{H}_t^{\mathrm{T}}\boldsymbol{H}_t)^{-1}\boldsymbol{H}_t^{\mathrm{T}}]^{\mathrm{T}}[\boldsymbol{I}_t - \boldsymbol{H}_t(\boldsymbol{H}_t^{\mathrm{T}}\boldsymbol{H}_t)^{-1}\boldsymbol{H}_t^{\mathrm{T}}]\boldsymbol{Y}_t \\ &= \boldsymbol{Y}_t^{\mathrm{T}}[\boldsymbol{I}_t - \boldsymbol{H}_t(\boldsymbol{H}_t^{\mathrm{T}}\boldsymbol{H}_t)^{-1}\boldsymbol{H}_t^{\mathrm{T}}]^2\boldsymbol{Y}_t \\ &= \boldsymbol{Y}_t^{\mathrm{T}}[\boldsymbol{I}_t - \boldsymbol{H}_t(\boldsymbol{H}_t^{\mathrm{T}}\boldsymbol{H}_t)^{-1}\boldsymbol{H}_t^{\mathrm{T}}]\boldsymbol{Y}_t. \end{aligned} \tag{3.5}$$

The symbol $\boldsymbol{I}_t$ represents an identity matrix of size $t \times t$.

3.2. Properties of LS estimates

The concepts of an "independent sequence" and "white noise" play an important role in the modeling of stochastic processes [85].

A discrete-time stochastic process $\{X(t), t \in \mathbb{N}\}$ is said to be an independent sequence if for any set $(t_1, t_2, \ldots, t_k) \in \mathbb{N}$, the corresponding random variables $X(t_1)$, $X(t_2)$, ..., $X(t_k)$ are independent; i.e., the joint distribution function F can be factored as

$$F(X(t_1), X(t_2), \ldots, X(t_k)) = F_1(X(t_1))F_2(X(t_2))\ldots F_k(X(t_k)),$$

$F_i(X(t_i))$ is the marginal distribution function of $X(t_i)$, $i = 1, 2, \ldots, k$. This means

$$\mathrm{E}[X(t_1), X(t_2), \ldots, X(t_k)] = \mathrm{E}[X(t_1)]\mathrm{E}[X(t_2)]\ldots\mathrm{E}[X(t_k)].$$

Further, if $F_1(X(t_1))$, $F_2(X(t_2))$, ..., $F_k(X(t_k))$ are identical functions, then the sequence is said to be an independent and identically distributed (i.i.d.) sequence.

A discrete-time stochastic process $\{X(t), t \in \mathbb{N}\}$ is said to be white noise if the covariance matrix can be expressed in the form $\mathrm{cov}[X(t), X(s)] = \mathrm{E}\{(X(t) - \mathrm{E}[X(t)])(X(s) - \mathrm{E}[X(s)])^{\mathrm{T}}\} = \delta_{ts}\Sigma_t$, where Σ_t is nonnegative definite.

The independent sequence is defined through the distribution functions or probability density functions. The white noise sequence is defined through the covariance function/matrix or the correlation functions.

About the statistical properties of the least squares estimate $\hat{\boldsymbol{\theta}}(t)$ in (3.3), we have the following lemmas and theorems [1–3,5,85].

Lemma 3.1. *Suppose that b_i and c_i, $i = 1, 2, \ldots, n$, satisfy the relation:*

$$b_i c_j = \delta_{ij} = \begin{cases} 1, & i = j, \\ 0, & i \neq j, \end{cases}$$

where δ_{ij} is the Kronecker delta function. Then the following equality holds:

$$\left[\sum_{i=1}^{n} a_i b_i\right]\left[\sum_{i=1}^{n} c_i d_i\right] = \sum_{i=1}^{n} a_i d_i.$$

Theorem 3.1 (*Unbiasedness Theorem*)**.** *For the linear regressive system in (2.2) or (3.1), suppose that $\{v(t)\}$ is a white noise sequence with zero mean, which implies that $v(t)$ is uncorrelated with $v(s)$, $s \neq t$, and satisfies*

$$\text{(C1)}\ \ \mathrm{E}[v(t)] = 0,\ \ \mathrm{E}[v(t)v(s)] = 0,\ s \neq t,\ \ \mathrm{E}[v^2(t)] = \sigma^2 \text{ for any } t,$$

the information vector $\boldsymbol{\varphi}(t) \in \mathbb{R}^n$ is uncorrelated with $v(t)$, i.e.,

$$\text{(C2)}\ \ \mathrm{E}[\boldsymbol{\varphi}(t)v(t)] = \mathbf{0},\ \text{for any } t,$$

or more strictly, the information vector $\boldsymbol{\varphi}(t) \in \mathbb{R}^n$ is statistically independent of $v(t)$, which means

$$\text{(C3)}\ \ \mathrm{E}[\boldsymbol{\varphi}(t)v(s)] = \mathrm{E}[\boldsymbol{\varphi}(t)]\mathrm{E}[v(s)] = \mathbf{0},\ \text{for any } t,\ s.$$

Then $\hat{\boldsymbol{\theta}}(t)$ is the unbiased estimate of the parameter vector $\boldsymbol{\theta}$, i.e., $\mathrm{E}[\hat{\boldsymbol{\theta}}(t)] = \boldsymbol{\theta}$.

Proof. Assumptions (C1) and (C2) imply that $\boldsymbol{V}_t \in \mathbb{R}^t$ is a white noise vector with mean zero and covariance matrix $\mathrm{cov}[\boldsymbol{V}_t] = \mathrm{E}[\boldsymbol{V}_t\boldsymbol{V}_s^{\mathrm{T}}] = \sigma^2\delta_{st}\boldsymbol{I}_t$, and $\boldsymbol{H}_t$ are uncorrelated with $\boldsymbol{V}_t$, i.e., $\mathrm{E}[\boldsymbol{H}_t^{\mathrm{T}}\boldsymbol{V}_t] = \mathrm{E}[\sum_{j=1}^{t}\boldsymbol{\varphi}(j)v(j)] = \mathbf{0}$. Substituting (3.1) into (3.3) gives

$$\begin{aligned}\hat{\boldsymbol{\theta}}(t) &= (\boldsymbol{H}_t^{\mathrm{T}}\boldsymbol{H}_t)^{-1}\boldsymbol{H}_t^{\mathrm{T}}(\boldsymbol{H}_t\boldsymbol{\theta} + \boldsymbol{V}_t) \\ &= \boldsymbol{\theta} + (\boldsymbol{H}_t^{\mathrm{T}}\boldsymbol{H}_t)^{-1}\boldsymbol{H}_t^{\mathrm{T}}\boldsymbol{V}_t. \end{aligned} \tag{3.6}$$

Taking the expectation to both sides gives

$$\begin{aligned}\mathrm{E}[\hat{\boldsymbol{\theta}}(t)] &= \mathrm{E}[(\boldsymbol{H}_t^{\mathrm{T}}\boldsymbol{H}_t)^{-1}\boldsymbol{H}_t^{\mathrm{T}}\boldsymbol{Y}_t] \\ &= \mathrm{E}[(\boldsymbol{H}_t^{\mathrm{T}}\boldsymbol{H}_t)^{-1}\boldsymbol{H}_t^{\mathrm{T}}(\boldsymbol{H}_t\boldsymbol{\theta} + \boldsymbol{V}_t)] \\ &= \mathrm{E}[\boldsymbol{\theta} + (\boldsymbol{H}_t^{\mathrm{T}}\boldsymbol{H}_t)^{-1}\boldsymbol{H}_t^{\mathrm{T}}\boldsymbol{V}_t] \\ &= \mathrm{E}[\boldsymbol{\theta}] + \mathrm{E}[(\boldsymbol{H}_t^{\mathrm{T}}\boldsymbol{H}_t)^{-1}\boldsymbol{H}_t^{\mathrm{T}}\boldsymbol{V}_t] = \boldsymbol{\theta}. \quad \square \end{aligned}$$

The unbiasedness requires $\mathrm{E}[(\boldsymbol{H}_t^{\mathrm{T}}\boldsymbol{H}_t)^{-1}\boldsymbol{H}_t^{\mathrm{T}}\boldsymbol{V}_t] = \mathbf{0}$. That is to say that $(\boldsymbol{H}_t^{\mathrm{T}}\boldsymbol{H}_t)^{-1}\boldsymbol{H}_t^{\mathrm{T}}$ and $\boldsymbol{V}_t$ are uncorrelated or $\boldsymbol{H}_t$ is deterministic (not random). This condition is very weak and does not require that $(\boldsymbol{H}_t^{\mathrm{T}}\boldsymbol{H}_t)^{-1}\boldsymbol{H}_t^{\mathrm{T}}$ and $\boldsymbol{V}_t$ are independent.

Theorem 3.2 (*Covariance Theorem*)**.** *For the linear regressive system in (3.1), suppose that $\boldsymbol{V}_t$ is a random vector with mean zero and covariance matrix $\mathrm{cov}[\boldsymbol{V}_t] := \mathrm{E}[\boldsymbol{V}_t\boldsymbol{V}_t^{\mathrm{T}}] = \boldsymbol{R}_v$ and is statistically independent of $\boldsymbol{H}_t$. Then the covariance matrix $\bar{\boldsymbol{P}}(t)$ of the parameter estimation error vector $\tilde{\boldsymbol{\theta}}(t) := \hat{\boldsymbol{\theta}}(t) - \boldsymbol{\theta}$ satisfies*

$$\bar{\boldsymbol{P}}(t) := \mathrm{cov}[\tilde{\boldsymbol{\theta}}(t)] = \mathrm{E}[(\boldsymbol{H}_t^{\mathrm{T}}\boldsymbol{H}_t)^{-1}\boldsymbol{H}_t^{\mathrm{T}}\boldsymbol{R}_v\boldsymbol{H}_t(\boldsymbol{H}_t^{\mathrm{T}}\boldsymbol{H}_t)^{-1}].$$

Proof. From Theorem 3.1, we have

$$\tilde{\boldsymbol{\theta}}(t) = \hat{\boldsymbol{\theta}}(t) - \boldsymbol{\theta} = (\boldsymbol{H}_t^{\mathrm{T}}\boldsymbol{H}_t)^{-1}\boldsymbol{H}_t^{\mathrm{T}}\boldsymbol{V}_t.$$

Hence, $\mathrm{E}[\tilde{\boldsymbol{\theta}}(t)] = \mathbf{0}$ and

$$\begin{aligned}\bar{\boldsymbol{P}}(t) &:= \mathrm{cov}[\tilde{\boldsymbol{\theta}}(t)] = \mathrm{E}[\tilde{\boldsymbol{\theta}}(t)\tilde{\boldsymbol{\theta}}^{\mathrm{T}}(t)] \\ &= \mathrm{E}\{[\hat{\boldsymbol{\theta}}(t) - \boldsymbol{\theta}][\hat{\boldsymbol{\theta}}(t) - \boldsymbol{\theta}]^{\mathrm{T}}\} \\ &= \mathrm{E}[(\boldsymbol{H}_t^{\mathrm{T}}\boldsymbol{H}_t)^{-1}\boldsymbol{H}_t^{\mathrm{T}}\boldsymbol{V}_t\boldsymbol{V}_t^{\mathrm{T}}\boldsymbol{H}_t(\boldsymbol{H}_t^{\mathrm{T}}\boldsymbol{H}_t)^{-1}] \\ &= \mathrm{E}\{(\boldsymbol{H}_t^{\mathrm{T}}\boldsymbol{H}_t)^{-1}\boldsymbol{H}_t^{\mathrm{T}}\mathrm{E}[\boldsymbol{V}_t\boldsymbol{V}_t^{\mathrm{T}}]\boldsymbol{H}_t(\boldsymbol{H}_t^{\mathrm{T}}\boldsymbol{H}_t)^{-1}\} \\ &= \mathrm{E}[(\boldsymbol{H}_t^{\mathrm{T}}\boldsymbol{H}_t)^{-1}\boldsymbol{H}_t^{\mathrm{T}}\boldsymbol{R}_v\boldsymbol{H}_t(\boldsymbol{H}_t^{\mathrm{T}}\boldsymbol{H}_t)^{-1}]. \quad \square \end{aligned}$$

Furthermore, if (C1) and (C2) hold, then we have $\mathrm{cov}[\boldsymbol{V}_t]=\boldsymbol{R}_v=\sigma^2\boldsymbol{I}_t$, and

$$\begin{aligned}\mathrm{cov}[\tilde{\boldsymbol{\theta}}(t)] &= \mathrm{E}[(\boldsymbol{H}_t^{\mathrm{T}}\boldsymbol{H}_t)^{-1}\boldsymbol{H}_t^{\mathrm{T}}\sigma^2\boldsymbol{H}_t(\boldsymbol{H}_t^{\mathrm{T}}\boldsymbol{H}_t)^{-1}]\\ &= \sigma^2\mathrm{E}[(\boldsymbol{H}_t^{\mathrm{T}}\boldsymbol{H}_t)^{-1}].\end{aligned} \tag{3.7}$$

Taking the trace to both sides gives the mean square parameter estimation error

$$\begin{aligned}\mathrm{E}[\|\hat{\boldsymbol{\theta}}(t)-\boldsymbol{\theta}\|^2] &= \mathrm{tr}\{\mathrm{cov}[\tilde{\boldsymbol{\theta}}(t)]\}\\ &= \sigma^2\mathrm{tr}\{\mathrm{E}[(\boldsymbol{H}_t^{\mathrm{T}}\boldsymbol{H}_t)^{-1}]\}.\end{aligned}$$

This shows that if the smallest eigenvalue of the data product moment matrix $(\boldsymbol{H}_t^{\mathrm{T}}\boldsymbol{H}_t)$ goes to infinity as the data length t increases, then the mean square parameter estimation error approaches zero, i.e., $\mathrm{E}[\|\hat{\boldsymbol{\theta}}(t)-\boldsymbol{\theta}\|^2]\to 0$, or $\hat{\boldsymbol{\theta}}(t)\to\boldsymbol{\theta}$, m.s.

Theorem 3.3 (*Mean Square Convergence Theorem and Consistent Convergence Theorem*)**.** *For the linear regressive system in (2.2) or (3.1), suppose that (C1) and (C2) hold, and there exist two positive numbers $0<\alpha\leqslant\beta$ such that the following weak persistent excitation (WPE) condition holds,*

$$\text{(A1)}\quad \alpha\boldsymbol{I}_n\leqslant\frac{1}{t}(\boldsymbol{H}_t^{\mathrm{T}}\boldsymbol{H}_t)\leqslant\beta\boldsymbol{I}_n,\ \text{a.s.},\ \textit{for large } t,$$

and the data are stationary and ergodic:

$$\text{(A2)}\qquad \lim_{t\to\infty}\frac{1}{t}(\boldsymbol{H}_t^{\mathrm{T}}\boldsymbol{H}_t)=\lim_{t\to\infty}\frac{1}{t}\sum_{j=1}^{t}\boldsymbol{\varphi}(j)\boldsymbol{\varphi}^{T}(j)=\boldsymbol{R}>0,\ \text{a.s.},$$

$$\text{(C4)}\qquad \lim_{t\to\infty}\frac{1}{t}(\boldsymbol{H}_t^{\mathrm{T}}\boldsymbol{V}_t)=\lim_{t\to\infty}\frac{1}{t}\sum_{j=1}^{t}\boldsymbol{\varphi}(j)v(j)=\boldsymbol{0},\ \text{a.s.},$$

where $\boldsymbol{R}$ is a positive-definite matrix. Then the LS estimate $\hat{\boldsymbol{\theta}}(t)$ in (3.3) mean squarely and consistently converges to its true value $\boldsymbol{\theta}$, i.e., $\hat{\boldsymbol{\theta}}(t)\to\boldsymbol{\theta}$, m.s. (a.s.), or $\lim_{t\to\infty}\hat{\boldsymbol{\theta}}(t)=\boldsymbol{\theta}$, m.s. (a.s.)

Proof. Using the WPE conditions (A1) and (A2), we have

$$\begin{aligned}0=\lim_{t\to\infty}\frac{\sigma^2}{t}\frac{\boldsymbol{I}_n}{\beta} &\leqslant \lim_{t\to\infty}\mathrm{cov}[\tilde{\boldsymbol{\theta}}(t)]\leqslant\lim_{t\to\infty}\frac{\sigma^2}{t}\frac{\boldsymbol{I}_n}{\alpha}=\boldsymbol{0},\\ 0\leqslant\lim_{t\to\infty}\mathrm{cov}[\tilde{\boldsymbol{\theta}}(t)] &= \lim_{t\to\infty}\sigma^2\mathrm{E}[(\boldsymbol{H}_t^{\mathrm{T}}\boldsymbol{H}_t)^{-1}]\\ &= \lim_{t\to\infty}\frac{\sigma^2}{t}\mathrm{E}\left[\left(\frac{1}{t}\boldsymbol{H}_t^{\mathrm{T}}\boldsymbol{H}_t\right)^{-1}\right]\\ &= \lim_{t\to\infty}\frac{\sigma^2}{t}\boldsymbol{R}^{-1}=\boldsymbol{0}.\end{aligned}$$

This means $\lim_{t\to\infty}\mathrm{E}[\|\tilde{\boldsymbol{\theta}}(t)\|^2]=\lim_{t\to\infty}\mathrm{tr}\{\mathrm{cov}[\|\,\tilde{\boldsymbol{\theta}}(t)]\}=0$, or $\tilde{\boldsymbol{\theta}}(t)\to\boldsymbol{0}$, m.s. Moreover, the lower-bound α and upper-bound β of the WPE conditions can determine the lower-bound and upper-bound of the parameter estimation error, i.e., $n\sigma^2/(\beta t)\leqslant\mathrm{E}[\|\tilde{\boldsymbol{\theta}}(t)\|^2]\leqslant n\sigma^2/(\alpha t)$ for large t.

Pre-multiplying (3.1) by $\boldsymbol{H}_t^{\mathrm{T}}$ gives

$$\boldsymbol{H}_t^{\mathrm{T}}\boldsymbol{Y}_t=\boldsymbol{H}_t^{\mathrm{T}}\boldsymbol{H}_t\boldsymbol{\theta}+\boldsymbol{H}_t^{\mathrm{T}}\boldsymbol{V}_t.$$

Using (3.2), we have

$$\frac{1}{t}(\boldsymbol{H}_t^{\mathrm{T}}\boldsymbol{H}_t)[\hat{\boldsymbol{\theta}}(t)-\boldsymbol{\theta}]=\frac{1}{t}\boldsymbol{H}_t^{\mathrm{T}}\boldsymbol{V}_t.$$

Taking the limits to both sides and using (A2) and (C4) give

$$\boldsymbol{R}[\hat{\boldsymbol{\theta}}(t)-\boldsymbol{\theta}]=\boldsymbol{0},\ \text{a.s.}$$

This means $\lim_{t\to\infty}\hat{\boldsymbol{\theta}}(t)=\boldsymbol{\theta}$, a.s., or $\tilde{\boldsymbol{\theta}}(t)\to\boldsymbol{0}$, a.s. □

From Theorem 3.3, it can be seen that under the WPE condition (A1) or (A2), the parameter estimation error $\tilde{\boldsymbol{\theta}}(t)$ converges to zero. In fact, the convergence of the least squares estimate can be achieved under the WPE conditions, i.e., the smallest eigenvalue of the data product moment matrix $(\boldsymbol{H}_t^{\mathrm{T}}\boldsymbol{H}_t)$ goes to infinity as the data length t increases. For example, under the attenuating excitation condition $(\boldsymbol{H}_t^{\mathrm{T}}\boldsymbol{H}_t)\geqslant C\boldsymbol{I}_n\ln\ln t$, a.s., ($C$ is a constant), we have

$$\lim_{t\to\infty}\mathrm{E}[\|\hat{\boldsymbol{\theta}}(t)-\boldsymbol{\theta}\|^2]=\lim_{t\to\infty}\sigma^2\mathrm{tr}\{\mathrm{E}[(\boldsymbol{H}_t^{\mathrm{T}}\boldsymbol{H}_t)^{-1}]\}$$

$$\begin{aligned}&\leqslant \lim_{t\to\infty}\sigma^2\mathrm{tr}\{\mathrm{E}[(C\boldsymbol{I}_n\ln\ln t)^{-1}]\}\\&=\lim_{t\to\infty}\frac{n\sigma^2}{C\ln\ln t}=0.\end{aligned}$$

Theorem 3.4 (*Noise Variance Estimation Theorem*)**.** *For the linear regressive system in (2.2) or (3.1), suppose that (C1) to (C3) hold. Then the variance σ^2 of the noise $\{v(t)\}$ can be estimated through*

$$\hat{\sigma}^2(t)=\frac{J_1(\hat{\boldsymbol{\theta}}(t))}{t-n},\quad \textit{for large } t, \tag{3.8}$$

where $n := \dim\boldsymbol{\theta}$ is the dimension of the parameter vector $\boldsymbol{\theta}$, $J_1(\hat{\boldsymbol{\theta}}(t))$ is the criterion function value corresponding LS estimate $\hat{\boldsymbol{\theta}}(t)$ in (3.3) and is computed by (3.5).

Proof. Define the output residual $\varepsilon(t)$ and the residual vector $\boldsymbol{\varepsilon}_t$ as

$$\begin{aligned}&\varepsilon(j) := y(j)-\boldsymbol{\varphi}^{\mathrm{T}}(j)\hat{\boldsymbol{\theta}}(t),\ j=1,2,\ldots,t,\\&\boldsymbol{\varepsilon}_t := \begin{bmatrix}\varepsilon(1)\\ \varepsilon(2)\\ \vdots\\ \varepsilon(t)\end{bmatrix}=\boldsymbol{Y}_t-\boldsymbol{H}_t\hat{\boldsymbol{\theta}}(t).\end{aligned}$$

Using (3.1) and (3.3), we have

$$\begin{aligned}\boldsymbol{\varepsilon}_t&=\boldsymbol{Y}_t-\boldsymbol{H}_t(\boldsymbol{H}_t^{\mathrm{T}}\boldsymbol{H}_t)^{-1}\boldsymbol{H}_t^{\mathrm{T}}\boldsymbol{Y}_t\\&=[\boldsymbol{I}_t-\boldsymbol{H}_t(\boldsymbol{H}_t^{\mathrm{T}}\boldsymbol{H}_t)^{-1}\boldsymbol{H}_t^{\mathrm{T}}]\boldsymbol{Y}_t\\&=[\boldsymbol{I}_t-\boldsymbol{H}_t(\boldsymbol{H}_t^{\mathrm{T}}\boldsymbol{H}_t)^{-1}\boldsymbol{H}_t^{\mathrm{T}}][\boldsymbol{H}_t\boldsymbol{\theta}+\boldsymbol{V}_t]\\&=[\boldsymbol{I}_t-\boldsymbol{H}_t(\boldsymbol{H}_t^{\mathrm{T}}\boldsymbol{H}_t)^{-1}\boldsymbol{H}_t^{\mathrm{T}}]\boldsymbol{V}_t=:\boldsymbol{Q}\boldsymbol{V}_t,\end{aligned}$$

where $\boldsymbol{Q} := \boldsymbol{I}_t-\boldsymbol{H}_t(\boldsymbol{H}_t^{\mathrm{T}}\boldsymbol{H}_t)^{-1}\boldsymbol{H}_t^{\mathrm{T}}$. Since $\boldsymbol{Q}^2=\boldsymbol{Q}$, $\boldsymbol{Q}^{\mathrm{T}}=\boldsymbol{Q}$, that is $\boldsymbol{Q}$ is an idempotent matrix, we have

$$\mathrm{E}[\boldsymbol{\varepsilon}_t^{\mathrm{T}}\boldsymbol{\varepsilon}_t]=\mathrm{E}[\boldsymbol{V}_t^{\mathrm{T}}\boldsymbol{Q}\boldsymbol{Q}\boldsymbol{V}_t]=\mathrm{E}[\boldsymbol{V}_t^{\mathrm{T}}\boldsymbol{Q}\boldsymbol{V}_t].$$

Using the properties of the trace: $\mathrm{tr}[\boldsymbol{AB}]=\mathrm{tr}[\boldsymbol{BA}]$, $\mathrm{E}\{\mathrm{tr}[\boldsymbol{A}]\}=\mathrm{tr}\{\mathrm{E}[\boldsymbol{A}]\}$, $\mathrm{tr}[\boldsymbol{A}^{\mathrm{T}}]=\mathrm{tr}[\boldsymbol{A}]$, and Assumptions (C1) to (C3), we have

$$\begin{aligned}\mathrm{E}\{J_1(\hat{\boldsymbol{\theta}}(t))\}&=\mathrm{E}[\boldsymbol{\varepsilon}_t^{\mathrm{T}}\boldsymbol{\varepsilon}_t]=\mathrm{E}\{\mathrm{tr}[\boldsymbol{Q}\boldsymbol{V}_t\boldsymbol{V}_t^{\mathrm{T}}]\}\\&=\mathrm{tr}\{\mathrm{E}[\boldsymbol{Q}\boldsymbol{V}_t^{\mathrm{T}}\boldsymbol{V}_t]\}=\sigma^2\mathrm{tr}[\mathrm{E}(\boldsymbol{Q})]\\&=\sigma^2\mathrm{E}\{\mathrm{tr}[\boldsymbol{I}_t-\boldsymbol{H}_t(\boldsymbol{H}_t^{\mathrm{T}}\boldsymbol{H}_t)^{-1}\boldsymbol{H}_t^{\mathrm{T}}]\}\\&=\sigma^2\mathrm{E}\{\mathrm{tr}[\boldsymbol{I}_t]-\mathrm{tr}[\boldsymbol{H}_t(\boldsymbol{H}_t^{\mathrm{T}}\boldsymbol{H}_t)^{-1}\boldsymbol{H}_t^{\mathrm{T}}]\}\\&=\sigma^2(t-\dim\boldsymbol{\theta}).\end{aligned}$$

Thus we have

$$\sigma^2=\frac{\mathrm{E}[\boldsymbol{\varepsilon}_t^{\mathrm{T}}\boldsymbol{\varepsilon}_t]}{t-\dim\boldsymbol{\theta}}=\frac{\mathrm{E}[J_1(\hat{\boldsymbol{\theta}}(t))]}{t-\dim\boldsymbol{\theta}}.$$

This indicates that the estimate of the noise variance σ^2 can be computed through (3.8). □

3.3. Distributions of LS estimates

Lemma 3.2. *If $\boldsymbol{v}$ is a random vector with mean $\boldsymbol{\mu}$ and covariance matrix Σ, then $\boldsymbol{y}=\boldsymbol{Ax}+\boldsymbol{v}$ is a random vector with mean $\boldsymbol{Ax}+\boldsymbol{\mu}$ and covariance matrix Σ.*

Lemma 3.3. *If $\boldsymbol{x}$ is a random vector with mean $\mathrm{E}[\boldsymbol{x}]=\boldsymbol{\mu}$ and covariance matrix $\mathrm{cov}[\boldsymbol{x}]=\mathrm{E}[(\boldsymbol{x}-\boldsymbol{\mu})(\boldsymbol{x}-\boldsymbol{\mu})^{T}]=\Sigma$, then $\boldsymbol{y}=\boldsymbol{Ax}+\boldsymbol{b}$ is a random vector with mean $\mathrm{E}[\boldsymbol{y}]=\boldsymbol{A\mu}+\boldsymbol{b}$ and covariance matrix $\mathrm{cov}[\boldsymbol{y}]=\boldsymbol{A}\Sigma\boldsymbol{A}^{T}$.*

Lemma 3.4. *If $\boldsymbol{x}$ is a random vector with the normal distribution $\boldsymbol{x}\sim\mathrm{N}(\boldsymbol{\mu},\Sigma)$, then $\boldsymbol{y}=\boldsymbol{Ax}+\boldsymbol{b}$ is a normal random vector with the normal distribution $\boldsymbol{y}\sim\mathrm{N}(\boldsymbol{A\mu}+\boldsymbol{b},\boldsymbol{A}\Sigma\boldsymbol{A}^{T})$.*

Lemma 3.5. *If $\boldsymbol{x}_i$, $i=1,2,\ldots,m$, are independent random vectors with the normal distribution $\boldsymbol{x}_i\sim\mathrm{N}(\boldsymbol{\mu}_i,\Sigma_i)$, then $\boldsymbol{y}=c_1\boldsymbol{x}_1+c_2\boldsymbol{x}_2+\cdots+c_m\boldsymbol{x}_m$ is a normal random vector with the normal distribution*

$$\boldsymbol{y}\sim\mathrm{N}(c_1\boldsymbol{\mu}_1+c_2\boldsymbol{\mu}_2+\cdots+c_m\boldsymbol{\mu}_m,c_1^2\Sigma_1+c_2^2\Sigma_2+\cdots+c_m^2\Sigma_m).$$

Lemma 3.6. *If $\boldsymbol{x}_i$, $i = 1, 2, \ldots, m$, are independent random vectors with the normal distribution $\boldsymbol{x}_i \sim \mathrm{N}(\boldsymbol{\mu}_i, \Sigma_i)$, then $\boldsymbol{y} = c_1\boldsymbol{A}_1\boldsymbol{x}_1 + c_2\boldsymbol{A}_2\boldsymbol{x}_2 + \cdots + c_m\boldsymbol{A}_m\boldsymbol{x}_m$ is a normal random vector with the normal distribution*

$$\boldsymbol{y} \sim \mathrm{N}(c_1\boldsymbol{\mu}_1 + c_2\boldsymbol{\mu}_2 + \cdots + c_m\boldsymbol{\mu}_m, c_1^2\boldsymbol{A}_1\Sigma_1\boldsymbol{A}_1^T + c_2^2\boldsymbol{A}_2\Sigma_2\boldsymbol{A}_2^T + \cdots + c_m^2\boldsymbol{A}_m\Sigma_m\boldsymbol{A}_m^T).$$

This lemma shows that a linear combination of the normal random variables is still a normal random variable.

Lemma 3.7. *If the random variable X follows the uniform distribution over the interval $[a, b]$, i.e., $X \sim \mathrm{U}(a, b)$, then X has mean value $\mu = \frac{a+b}{2}$ and variance $\sigma^2 = \frac{(b-a)^2}{12}$. $Y = kX + d$ is a uniformly distributed random variable with $\mathrm{E}[Y] = k\mathrm{E}[X] + d$ and $\mathrm{var}[Y] = k^2\mathrm{var}[X]$.*

Proof. The probability density function of X is given by

$$p(x) = \begin{cases} \frac{1}{b-a}, & a \leqslant x \leqslant b, \\ 0, & x < a, \ x > b. \end{cases}$$

The mean and variance of X are given by

$$\begin{aligned}
\mu &= \mathrm{E}[X] = \int_a^b xp(x)\mathrm{d}x = \int_a^b \frac{x}{b-a}\mathrm{d}x = \frac{1}{2}\frac{x^2}{b-a}\bigg|_a^b = \frac{a+b}{2}, \\
\sigma^2 &= \mathrm{E}[(X-\mu)^2] = \int_a^b \left(x - \frac{a+b}{2}\right)^2 p(x)\mathrm{d}x = \int_a^b \left(x - \frac{a+b}{2}\right)^2 \frac{1}{b-a}\mathrm{d}x \\
&= \frac{1}{3(b-a)}\left(x - \frac{a+b}{2}\right)^3\bigg|_a^b = \frac{1}{24(b-a)}[(b-a)^3 - (a-b)^3] = \frac{(b-a)^2}{12}.
\end{aligned}$$

$$\begin{aligned}
\mathrm{E}[X^2] &= \int_a^b x^2 p(x)\mathrm{d}x = \int_a^b \frac{x^2}{b-a}\mathrm{d}x \\
&= \frac{1}{3(b-a)}x^3\bigg|_a^b = \frac{1}{3(b-a)}[b^3 - a^3] = \frac{1}{3}[a^2 + ab + b^2], \\
\sigma^2 &= \mathrm{var}[X] = \mathrm{E}[(X-\mu)^2] = \mathrm{E}[X^2] - (\mathrm{E}[X])^2 \\
&= \frac{1}{3}[a^2 + ab + b^2] - \left(\frac{b+a}{2}\right)^2 = \frac{(b-a)^2}{12}. \quad \square
\end{aligned}$$

When $a = 0$ and $b = 1$, X is a standard uniform distribution $X \sim \mathrm{U}(0, 1)$ with mean $\frac{1}{2}$ and standard deviation $\frac{1}{2\sqrt{3}}$. Furthermore, $Y = cX$ has mean $\mathrm{E}[Y] = \frac{c(a+b)}{2}$ and variance $\mathrm{var}[Y] = \frac{c^2(b-a)^2}{12}$. Therefore, if $X \sim \mathrm{U}(0, 1)$, then $Y = \sqrt{12}(X - \frac{1}{2})$ has mean $\mathrm{E}[Y] = 0$ and variance $\mathrm{var}[Y] = 1$. That is, $X \sim \mathrm{U}(-\sqrt{3}, \sqrt{3})$ is a uniformly distributed random variable with zero mean and unit variance.

The distribution of the least squares estimate $\hat{\boldsymbol{\theta}}(t)$ depends on the distribution of the disturbance noise $v(t)$.

Theorem 3.5. *Suppose that the conditions of Theorem 3.1 hold, $\boldsymbol{V}_t$ is an uncorrelated white noise vector with the normal distribution $\boldsymbol{V}_t \sim \mathrm{N}(\boldsymbol{0}, \sigma^2\boldsymbol{I}_t)$, the information matrix $\boldsymbol{H}_t$ is statistically independent of $\boldsymbol{V}_t$. Then $\hat{\boldsymbol{\theta}}(t)$ follows the normal distribution*

$$\hat{\boldsymbol{\theta}}(t) \sim \mathrm{N}(\boldsymbol{\theta}, \sigma^2\mathrm{E}[(\boldsymbol{H}_t^T\boldsymbol{H}_t)^{-1}]).$$

Proof. Because the information matrix $\boldsymbol{H}_t$ is statistically independent of $\boldsymbol{V}_t$, applying Lemma 3.3 to (3.6) and using (3.7), we directly obtain the conclusion of Theorem 3.5. $\square$

Theorem 3.5 indicates that only when the information matrix $\boldsymbol{H}_t$ is statistically independent of $\boldsymbol{V}_t$, $\hat{\boldsymbol{\theta}}(t)$ follows the normal distribution. Otherwise, this conclusion does not hold. See the following example.

Example 3.1. For the following first-order autoregressive (AR) system,

$$y(t) = ay(t-1) + v(t), \ y(0) = 0,$$

suppose that the independent white noise $v(t)$ follows the normal distribution $v(t) \sim \mathrm{N}(0, 1)$ for $t = 1, 2, 3, \ldots$. Then $y(1) = v(1)$ follows the normal distribution. According to Lemma 3.5, $y(2) = ay(1) + v(2)$ also follows the normal distribution $y(2) \sim \mathrm{N}(0, a^2 + 1)$. Thus we have $y(3) \sim \mathrm{N}(0, a^4 + a^2 + 1)$ and $y(t)$ follows the normal distribution

$$y(t) \sim \mathrm{N}(0, 1 + a^2 + a^4 + \cdots + a^{2t-2}).$$

Also, we can obtain this result from

$$y(t) = \frac{1}{1 - az^{-1}}v(t)$$

$$\begin{aligned}
&= (1 + az^{-1} + a^2z^{-2} + a^3z^{-3} + \cdots + a^{t-1}z^{-t+1})v(t)\\
&= v(t) + av(t-1) + a^2v(t-2) + a^3v(t-3) + \cdots + a^{t-1}v(1),\\
\mathrm{var}[y(t)] &= 1 + a^2 + a^4 + \cdots + a^{2t-2}.
\end{aligned}$$

In this first-order AR model, we have $y(t) = \varphi(t)a + v(t)$, $\varphi(t) = y(t-1)$, and

$$\boldsymbol{Y}_t := \begin{bmatrix} y(1)\\ y(2)\\ \vdots\\ y(t)\end{bmatrix} \in \mathbb{R}^t, \quad \boldsymbol{H}_t := \begin{bmatrix} y(0)\\ y(1)\\ \vdots\\ y(t-1)\end{bmatrix} \in \mathbb{R}^t, \quad \boldsymbol{V}_t := \begin{bmatrix} v(1)\\ v(2)\\ \vdots\\ v(t)\end{bmatrix} \in \mathbb{R}^t.$$

So the LS estimate $\hat{a}(t) = (\boldsymbol{H}_t^{\mathrm{T}}\boldsymbol{H}_t)^{-1}\boldsymbol{H}_t^{\mathrm{T}}\boldsymbol{Y}_t$ of the parameter a in the first-order AR model is given by

$$\begin{aligned}
\hat{a}(2) &= \frac{y(2)y(1)}{y^2(1)} = \frac{y(2)}{y(1)},\\
\hat{a}(3) &= \frac{y(3)y(2) + y(2)y(1)}{y^2(2) + y^2(1)},\\
\hat{a}(4) &= \frac{y(4)y(3) + y(3)y(2) + y(2)y(1)}{y^2(3) + y^2(2) + y^2(1)},\\
&\ \ \vdots\\
\hat{a}(t) &= \frac{y(t)y(t-1) + y(t-1)y(t-2) + \cdots + y(2)y(1)}{y^2(t-1) + y^2(t-2) + y^2(t-3) + \cdots + y^2(1)}.
\end{aligned}$$

It is obvious that $\hat{a}(t)$ does not follow the normal distribution because $\varphi(t) = y(t-1)$ and $v(s)$ are not statistically independent. That is, Condition (C3) does not hold for the AR model, for example, $\mathrm{E}[\varphi(3)v(2)] = \mathrm{E}[y(2)v(2)] = \mathrm{E}[(ay(1) + v(2))v(2)] = \mathrm{E}[(av(1) + v(2))v(2)] = 1 \neq 0$. Therefore, the LS parameter estimates of the n-th-order AR system $y(t) = a_1y(t-1) + a_2y(t-2) + \cdots + a_ny(t-n) + v(t)$ do not follow the normal distribution.

Example 3.2. For the following first-order controlled autoregressive (CAR) system,

$$y(t) = ay(t-1) + bu(t-1) + v(t), \ \ y(0) = 0,$$

the input $u(t)$ is a random signal with zero mean and is independent of the white noise $v(t)$ with the normal distribution $v(t) \sim \mathrm{N}(0, 1)$, the input $u(t)$ is an uncorrelated random sequence with mean zero and variance σ^2 independent of $v(t)$. According to Lemma 3.3, $y(t)$ follows the normal distribution

$$y(t) \sim \mathrm{N}(0, (1 + a^2 + a^4 + \cdots + a^{2t-2})(1 + b^2\sigma^2)).$$

Example 3.3. For the following finite impulse response (FIR) system,

$$y(t) = b_1u(t-1) + b_2u(t-1) + \cdots + b_nu(t-n) + v(t),$$

the input $u(t)$ is independent of the white noise $v(t)$ with the normal distribution $v(t) \sim \mathrm{N}(0, 1)$ for $t = 1, 2, 3, \ldots$. It is easy to conclude that the LS estimates of the parameters b_1, b_2, …, b_n in the FIR model follow the normal distribution.

Example 3.4. For the following CAR model,

$$y(t) = a_1y(t-1) + a_2y(t-1) + \cdots + a_ny(t-n) + b_1u(t-1) + b_2u(t-1) + \cdots + b_nu(t-n) + v(t),$$

although the input $u(t)$ is independent of the white noise $v(t)$ with the normal distribution $v(t) \sim \mathrm{N}(0, 1)$ for $t = 1, 2, 3, \ldots$, the LS estimates of the parameters a_1, a_2, …, a_n, b_1, b_2, …, b_n in the CAR model do not follow the normal distribution.

Example 3.5. For the following fraction system [4],

$$y(t) = \frac{b_1\psi_1(t) + b_2\psi_2(t)}{1 + a_1\phi_1(t) + a_2\phi_2(t)} + v(t),$$

if $\psi_1(t)$, $\psi_2(t)$, $\phi_1(t)$ and $\phi_2(t)$ composed of the input $u(t-i)$ are all statistically independent of $v(t)$, and $v(t) \sim \mathrm{N}(0, 1)$, then the estimates of the parameters b_1 and b_2 follow the normal distribution but any estimates of the parameters a_1 and a_2 cannot follow the normal distribution. If at least one of $\psi_1(t)$, $\psi_2(t)$, $\phi_1(t)$ and $\phi_2(t)$ are correlated with the output $y(t-i)$, then the estimates of the parameters a_1, a_2, b_1 and b_2 cannot follow the normal distribution.

Example 3.6. Suppose that $y(t)$ is the output of the system, $u(t)$ is the input of the system, and $v(t)$ follows the normal distribution $v(t) \sim \mathrm{N}(0, \sigma^2)$. For the following equation-error systems:

$$
\begin{aligned}
&\text{(i)} \quad A(z)y(t) = B(z)u(t) + v(t),\\
&\text{(ii)} \quad A(z)y(t) = B(z)u(t) + D(z)v(t),\\
&\text{(iii)} \quad A(z)y(t) = B(z)u(t) + \frac{1}{C(z)}v(t),\\
&\text{(iv)} \quad A(z)y(t) = B(z)u(t) + \frac{D(z)}{C(z)}v(t)
\end{aligned}
$$

and output-error systems:

$$
\begin{aligned}
&\text{(v)} \quad y(t) = \frac{B(z)}{A(z)}u(t) + v(t),\\
&\text{(vi)} \quad y(t) = \frac{B(z)}{A(z)}u(t) + D(z)v(t),\\
&\text{(vii)} \quad y(t) = \frac{B(z)}{A(z)}u(t) + \frac{1}{C(z)}v(t),\\
&\text{(viii)} \quad y(t) = \frac{B(z)}{A(z)}u(t) + \frac{D(z)}{C(z)}v(t),\\
&\text{(ix)} \quad A(z)y(t) = \frac{B(z)}{F(z)}u(t) + v(t),\\
&\text{(x)} \quad A(z)y(t) = \frac{B(z)}{F(z)}u(t) + D(z)v(t),\\
&\text{(xi)} \quad A(z)y(t) = \frac{B(z)}{F(z)}u(t) + \frac{1}{C(z)}v(t),\\
&\text{(xii)} \quad A(z)y(t) = \frac{B(z)}{F(z)}u(t) + \frac{D(z)}{C(z)}v(t),\\
&\qquad A(z) := 1 + a_1z^{-1} + a_2z^{-2} + \cdots + a_{n_a}z^{-n_a} \in \mathbb{R},\\
&\qquad B(z) := b_1z^{-1} + b_2z^{-2} + \cdots + b_{n_b}z^{-n_b} \in \mathbb{R},\\
&\qquad C(z) := 1 + c_1z^{-1} + c_2z^{-2} + \cdots + c_{n_c}z^{-n_c} \in \mathbb{R},\\
&\qquad D(z) := 1 + d_1z^{-1} + d_2z^{-2} + \cdots + d_{n_d}z^{-n_d} \in \mathbb{R},\\
&\qquad F(z) := 1 + f_1z^{-1} + f_2z^{-2} + \cdots + f_{n_f}z^{-n_f} \in \mathbb{R},
\end{aligned}
$$

the estimates of the parameters a_i, b_i, c_i, d_i, f_i cannot follow the normal distributions.

Example 3.7. Let $x_{j:t} := x(j:t) = (x(j), x(j+1), \ldots, x(t))$, and suppose that $y(t)$ is the output of the system, $u(t)$ is the input of the system, $v(t)$ is a normally distributed white noise with zero mean and variance σ^2, $\phi_i(u_{j:t}, y_{j:t})$ and $\psi_i(u_{j:t}, y_{j:t})$ are the scalar functions of $u_{j:t}$ and $y_{j:t}$, $\boldsymbol{\theta} := [\theta_1, \theta_2, \ldots, \theta_n]^{\mathrm{T}}$ and $\boldsymbol{\rho} := [\rho_1, \rho_2, \ldots, \rho_n]^{\mathrm{T}}$ are the parameter vectors to be identified. For the following linear-parameter fraction systems [4]:

$$
\begin{aligned}
&\text{(i)} \quad y(t) = \frac{\boldsymbol{\phi}^{\mathrm{T}}(u_{t-n:t}, y_{t-n:t-1})\boldsymbol{\theta}}{1 + \boldsymbol{\psi}^{\mathrm{T}}(u_{t-n:t}, y_{t-n:t-1})\boldsymbol{\rho}} + v(t)\\
&\qquad\quad = \frac{\sum_{i=1}^{n} \phi_i(u_{t-n:t}, y_{t-n:t-1})\theta_i}{1 + \sum_{i=1}^{n} \psi_i(u_{t-n:t}, y_{t-n:t-1})\rho_i} + v(t),\\
&\text{(ii)} \quad y(t) = \frac{\boldsymbol{\phi}^{\mathrm{T}}(u_{t-n:t}, y_{t-n:t-1})\boldsymbol{\theta}}{\psi_0(u_{t-n:t}, y_{t-n:t-1}) + \boldsymbol{\psi}^{\mathrm{T}}(u_{t-n:t}, y_{t-n:t-1})\boldsymbol{\rho}} + v(t),\\
&\text{(iii)} \quad A(z)y(t) = \frac{\boldsymbol{\phi}^{\mathrm{T}}(u_{t-n:t}, y_{t-n:t-1})\boldsymbol{\theta}}{1 + \boldsymbol{\psi}^{\mathrm{T}}(u_{t-n:t}, y_{t-n:t-1})\boldsymbol{\rho}} + v(t),\\
&\text{(iv)} \quad A(z)y(t) = \frac{\boldsymbol{\phi}^{\mathrm{T}}(u_{t-n:t}, y_{t-n:t-1})\boldsymbol{\theta}}{\psi_0(u_{t-n:t}, y_{t-n:t-1}) + \boldsymbol{\psi}^{\mathrm{T}}(u_{t-n:t}, y_{t-n:t-1})\boldsymbol{\rho}} + v(t),\\
&\text{(v)} \quad A(z)y(t) = \frac{\boldsymbol{\phi}^{\mathrm{T}}(u_{t-n:t}, y_{t-n:t-1})\boldsymbol{\theta}}{1 + \boldsymbol{\psi}^{\mathrm{T}}(u_{t-n:t}, y_{t-n:t-1})\boldsymbol{\rho}} + \frac{D(z)}{C(z)}v(t),
\end{aligned}
$$

$$\text{(vi)}\quad A(z)y(t)=\frac{\boldsymbol{\phi}^{\mathrm{T}}(u_{t-n:t},y_{t-n:t-1})\boldsymbol{\theta}}{\psi_0(u_{t-n:t},y_{t-n:t-1})+\boldsymbol{\psi}^{\mathrm{T}}(u_{t-n:t},y_{t-n:t-1})\boldsymbol{\rho}}+\frac{D(z)}{C(z)}v(t),$$

their parameter estimates cannot follow the normal distributions.

3.4. LS algorithm with recursive covariance

Lemma 3.8 (*Matrix Inversion Lemma*). *For $\boldsymbol{A}\in\mathbb{R}^{n\times n}$, $\boldsymbol{B}\in\mathbb{R}^{n\times r}$, $\boldsymbol{C}\in\mathbb{R}^{r\times n}$, if the matrices $\boldsymbol{A}$ and $(\boldsymbol{I}_r+\boldsymbol{C}\boldsymbol{A}^{-1}\boldsymbol{B})$ are invertible, then the following relation holds,*

$$(\boldsymbol{A}+\boldsymbol{B}\boldsymbol{C})^{-1}=\boldsymbol{A}^{-1}-\boldsymbol{A}^{-1}\boldsymbol{B}(\boldsymbol{I}_r+\boldsymbol{C}\boldsymbol{A}^{-1}\boldsymbol{B})^{-1}\boldsymbol{C}\boldsymbol{A}^{-1}.$$

Proof. In order to prove $\boldsymbol{X}^{-1}=\boldsymbol{Y}$, it can be transformed to prove $\boldsymbol{X}\boldsymbol{Y}=\boldsymbol{I}_n$ and $\boldsymbol{Y}\boldsymbol{X}=\boldsymbol{I}_n$. Based on this idea, we have

$$\begin{aligned}
&(\boldsymbol{A}+\boldsymbol{B}\boldsymbol{C})[\boldsymbol{A}^{-1}-\boldsymbol{A}^{-1}\boldsymbol{B}(\boldsymbol{I}_r+\boldsymbol{C}\boldsymbol{A}^{-1}\boldsymbol{B})^{-1}\boldsymbol{C}\boldsymbol{A}^{-1}]\\
&\quad=\boldsymbol{I}_n-\boldsymbol{B}(\boldsymbol{I}_r+\boldsymbol{C}\boldsymbol{A}^{-1}\boldsymbol{B})^{-1}\boldsymbol{C}\boldsymbol{A}^{-1}+\boldsymbol{B}\boldsymbol{C}\boldsymbol{A}^{-1}\\
&\qquad-\boldsymbol{B}\boldsymbol{C}\boldsymbol{A}^{-1}\boldsymbol{B}(\boldsymbol{I}_r+\boldsymbol{C}\boldsymbol{A}^{-1}\boldsymbol{B})^{-1}\boldsymbol{C}\boldsymbol{A}^{-1}\\
&\quad=\boldsymbol{I}_n+\boldsymbol{B}\boldsymbol{C}\boldsymbol{A}^{-1}-\boldsymbol{B}(\boldsymbol{I}_r+\boldsymbol{C}\boldsymbol{A}^{-1}\boldsymbol{B})^{-1}\boldsymbol{C}\boldsymbol{A}^{-1}\\
&\qquad-\boldsymbol{B}\boldsymbol{C}\boldsymbol{A}^{-1}\boldsymbol{B}(\boldsymbol{I}_r+\boldsymbol{C}\boldsymbol{A}^{-1}\boldsymbol{B})^{-1}\boldsymbol{C}\boldsymbol{A}^{-1}\\
&\quad=\boldsymbol{I}_n+\boldsymbol{B}\boldsymbol{C}\boldsymbol{A}^{-1}-\boldsymbol{B}(\boldsymbol{I}_r+\boldsymbol{C}\boldsymbol{A}^{-1}\boldsymbol{B})(\boldsymbol{I}_r+\boldsymbol{C}\boldsymbol{A}^{-1}\boldsymbol{B})^{-1}\boldsymbol{C}\boldsymbol{A}^{-1}\\
&\quad=\boldsymbol{I}_n+\boldsymbol{B}\boldsymbol{C}\boldsymbol{A}^{-1}-\boldsymbol{B}\boldsymbol{I}_r\boldsymbol{C}\boldsymbol{A}^{-1}=\boldsymbol{I}_n.
\end{aligned}$$

Similarly, we can prove that

$$[\boldsymbol{A}^{-1}-\boldsymbol{A}^{-1}\boldsymbol{B}(\boldsymbol{I}_r+\boldsymbol{C}\boldsymbol{A}^{-1}\boldsymbol{B})^{-1}\boldsymbol{C}\boldsymbol{A}^{-1}](\boldsymbol{A}+\boldsymbol{B}\boldsymbol{C})=\boldsymbol{I}_n.\qquad\square\quad\square$$

Define the vector $\boldsymbol{\xi}(t)$ and the covariance matrix $\boldsymbol{P}(t)$ as

$$\boldsymbol{\xi}(t):=\boldsymbol{H}_t^{\mathrm{T}}\boldsymbol{Y}_t=\sum_{j=1}^{t}\boldsymbol{\varphi}(j)y(j)=\boldsymbol{\xi}(t-1)+\boldsymbol{\varphi}(t)y(t),\tag{3.9}$$

$$\boldsymbol{P}^{-1}(t):=\boldsymbol{H}_t^{\mathrm{T}}\boldsymbol{H}_t=\sum_{j=1}^{t}\boldsymbol{\varphi}(j)\boldsymbol{\varphi}^{\mathrm{T}}(j)=\boldsymbol{P}^{-1}(t-1)+\boldsymbol{\varphi}(t)\boldsymbol{\varphi}^{\mathrm{T}}(t).\tag{3.10}$$

Eq. (3.3) can be written as

$$\hat{\boldsymbol{\theta}}(t)=\boldsymbol{P}(t)\boldsymbol{\xi}(t).\tag{3.11}$$

Applying Lemma 3.8 to (3.10) gives

$$\boldsymbol{P}(t)=\boldsymbol{P}(t-1)-\frac{\boldsymbol{P}(t-1)\boldsymbol{\varphi}(t)\boldsymbol{\varphi}^{\mathrm{T}}(t)\boldsymbol{P}(t-1)}{1+\boldsymbol{\varphi}^{\mathrm{T}}(t)\boldsymbol{P}(t-1)\boldsymbol{\varphi}(t)}.\tag{3.12}$$

Combining (3.11), (3.9) and (3.12) gives the least squares (LS) algorithm [1,5]:

$$\hat{\boldsymbol{\theta}}(t)=\boldsymbol{P}(t)\boldsymbol{\xi}(t),\tag{3.13}$$

$$\boldsymbol{\xi}(t)=\boldsymbol{\xi}(t-1)+\boldsymbol{\varphi}(t)y(t),\quad \boldsymbol{\xi}(0)=\boldsymbol{0},\tag{3.14}$$

$$\boldsymbol{P}(t)=\boldsymbol{P}(t-1)-\frac{\boldsymbol{P}(t-1)\boldsymbol{\varphi}(t)\boldsymbol{\varphi}^{\mathrm{T}}(t)\boldsymbol{P}(t-1)}{1+\boldsymbol{\varphi}^{\mathrm{T}}(t)\boldsymbol{P}(t-1)\boldsymbol{\varphi}(t)},\quad \boldsymbol{P}(0)=p_0\boldsymbol{I}_n.\tag{3.15}$$

3.5. Initial values of the LS algorithm

According to the definitions of $\boldsymbol{\xi}(t)$ and $\boldsymbol{P}^{-1}(t)$ in (3.9)–(3.10) and their recursive relations, we have

$$\begin{aligned}
\boldsymbol{\xi}(t)&=\boldsymbol{\xi}(t-2)+\boldsymbol{\varphi}(t-1)y(t-1)+\boldsymbol{\varphi}(t)y(t)\\
&=\boldsymbol{\xi}(0)+\sum_{j=1}^{t}\boldsymbol{\varphi}(j)y(j)=\boldsymbol{\xi}(0)+\boldsymbol{H}_t^{\mathrm{T}}\boldsymbol{Y}_t,
\end{aligned}\tag{3.16}$$

$$\begin{aligned}
\boldsymbol{P}^{-1}(t)&=\boldsymbol{P}^{-1}(t-2)+\boldsymbol{\varphi}(t-1)\boldsymbol{\varphi}^{\mathrm{T}}(t-1)+\boldsymbol{\varphi}(t)\boldsymbol{\varphi}^{\mathrm{T}}(t)\\
&=\boldsymbol{P}^{-1}(0)+\sum_{j=1}^{t}\boldsymbol{\varphi}(j)\boldsymbol{\varphi}^{\mathrm{T}}(j)=\boldsymbol{P}^{-1}(0)+\boldsymbol{H}_t^{\mathrm{T}}\boldsymbol{H}_t.
\end{aligned}\tag{3.17}$$

Comparing (3.9) with (3.16), the initial value of $\boldsymbol{\xi}(t)$ should be $\boldsymbol{\xi}(0)=\mathbf{0}$; comparing (3.10) with (3.17), the initial value of $\boldsymbol{P}^{-1}(t)$ should be $\boldsymbol{P}^{-1}(0)=\mathbf{0}$, i.e., $\boldsymbol{P}(0)$ should be infinite. Thus, $\boldsymbol{P}^{-1}(0)$ is generally taken as a very small positive definite symmetric matrix, e.g., $\boldsymbol{P}^{-1}(0)=\boldsymbol{I}_n/p_0$ is very close to zero, where p_0 is a large positive number (e.g., $p_0=10^6$). This is the reason that we take $\boldsymbol{P}(0)=p_0\boldsymbol{I}_n$ with large p_0.

4. Recursive least squares (RLS) methods

In this section, we derive a recursive least squares (RLS) algorithm for estimating the parameter vector $\boldsymbol{\theta}$ of linear regressive models and establish the lemmas and theorems about the parameter estimates given by the RLS algorithm.

4.1. RLS algorithms

Lemma 4.1. *Define the gain vector $\boldsymbol{L}(t):=\boldsymbol{P}(t)\boldsymbol{\varphi}(t)\in\mathbb{R}^n$. Then the following relations hold:*

$$\boldsymbol{L}(t)=\boldsymbol{P}(t)\boldsymbol{\varphi}(t)=\frac{\boldsymbol{P}(t-1)\boldsymbol{\varphi}(t)}{1+\boldsymbol{\varphi}^T(t)\boldsymbol{P}(t-1)\boldsymbol{\varphi}(t)}, \tag{4.1}$$

$$\begin{aligned}\boldsymbol{P}(t)&=\boldsymbol{P}(t-1)-\boldsymbol{L}(t)\boldsymbol{\varphi}^T(t)\boldsymbol{P}(t-1)\\&=\boldsymbol{P}(t-1)-\frac{\boldsymbol{P}(t-1)\boldsymbol{\varphi}(t)\boldsymbol{\varphi}^T(t)\boldsymbol{P}(t-1)}{1+\boldsymbol{\varphi}^T(t)\boldsymbol{P}(t-1)\boldsymbol{\varphi}(t)}.\end{aligned} \tag{4.2}$$

Proof. Pre- and post-multiplying (3.10) by $\boldsymbol{P}(t)$ and $\boldsymbol{P}(t-1)$ give

$$\boldsymbol{P}(t-1)=\boldsymbol{P}(t)+\boldsymbol{P}(t)\boldsymbol{\varphi}(t)\boldsymbol{\varphi}^{\mathrm{T}}(t)\boldsymbol{P}(t-1). \tag{4.3}$$

Post-multiplying both sides by $\boldsymbol{\varphi}(t)$ gives

$$\begin{aligned}\boldsymbol{P}(t-1)\boldsymbol{\varphi}(t)&=\boldsymbol{P}(t)\boldsymbol{\varphi}(t)+\boldsymbol{P}(t)\boldsymbol{\varphi}(t)\boldsymbol{\varphi}^{\mathrm{T}}(t)\boldsymbol{P}(t-1)\boldsymbol{\varphi}(t)\\&=\boldsymbol{P}(t)\boldsymbol{\varphi}(t)[1+\boldsymbol{\varphi}^{\mathrm{T}}(t)\boldsymbol{P}(t-1)\boldsymbol{\varphi}(t)].\end{aligned}$$

Solving $\boldsymbol{L}(t)$ gives Eq. (4.1). From (4.3) and using (4.1), we have

$$\boldsymbol{P}(t)=\boldsymbol{P}(t-1)-\boldsymbol{P}(t)\boldsymbol{\varphi}(t)\boldsymbol{\varphi}^{\mathrm{T}}(t)\boldsymbol{P}(t-1) \tag{4.4}$$

$$\begin{aligned}&=\boldsymbol{P}(t-1)-\boldsymbol{L}(t)\boldsymbol{\varphi}^{\mathrm{T}}(t)\boldsymbol{P}(t-1)\\&=\boldsymbol{P}(t-1)-\frac{\boldsymbol{P}(t-1)\boldsymbol{\varphi}(t)\boldsymbol{\varphi}^{\mathrm{T}}(t)\boldsymbol{P}(t-1)}{1+\boldsymbol{\varphi}^{\mathrm{T}}(t)\boldsymbol{P}(t-1)\boldsymbol{\varphi}(t)},\end{aligned} \tag{4.5}$$

which can be also expressed as

$$\begin{aligned}\boldsymbol{P}(t)&=[\boldsymbol{I}_n-\boldsymbol{L}(t)\boldsymbol{\varphi}^{\mathrm{T}}(t)]\boldsymbol{P}(t-1)\\&=\boldsymbol{P}(t-1)-\boldsymbol{L}(t)[1+\boldsymbol{\varphi}^{\mathrm{T}}(t)\boldsymbol{P}(t-1)\boldsymbol{\varphi}(t)]\boldsymbol{L}^{\mathrm{T}}(t).\quad\square\end{aligned}$$

Lemma 4.2. *The least squares estimate $\hat{\boldsymbol{\theta}}(t)$ in (3.3) can be recursively computed by the following RLS algorithm:*

$$\hat{\boldsymbol{\theta}}(t)=\hat{\boldsymbol{\theta}}(t-1)+\boldsymbol{P}(t)\boldsymbol{\varphi}(t)[y(t)-\boldsymbol{\varphi}^T(t)\hat{\boldsymbol{\theta}}(t-1)],\quad \hat{\boldsymbol{\theta}}(0)=\mathbf{1}_n/p_0, \tag{4.6}$$

$$\boldsymbol{P}^{-1}(t)=\boldsymbol{P}^{-1}(t-1)+\boldsymbol{\varphi}(t)\boldsymbol{\varphi}^T(t),\quad \boldsymbol{P}(0)=p_0\boldsymbol{I}_n. \tag{4.7}$$

Proof. According to the definitions of $\boldsymbol{Y}_t$ and $\boldsymbol{H}_t$, we have

$$\boldsymbol{Y}_t:=\left[\begin{array}{c}y(1)\\y(2)\\\vdots\\y(t-1)\\\hline y(t)\end{array}\right]=\left[\begin{array}{c}\boldsymbol{Y}_{t-1}\\y(t)\end{array}\right]\in\mathbb{R}^t,\quad \boldsymbol{H}_t:=\left[\begin{array}{c}\boldsymbol{\varphi}^{\mathrm{T}}(1)\\\boldsymbol{\varphi}^{\mathrm{T}}(2)\\\vdots\\\boldsymbol{\varphi}^{\mathrm{T}}(t-1)\\\hline\boldsymbol{\varphi}^{\mathrm{T}}(t)\end{array}\right]=\left[\begin{array}{c}\boldsymbol{H}_{t-1}\\\boldsymbol{\varphi}^{\mathrm{T}}(t)\end{array}\right]\in\mathbb{R}^{t\times n}.$$

Using (3.10) and from (3.3), we have

$$\begin{aligned}\hat{\boldsymbol{\theta}}(t)&=(\boldsymbol{H}_t^{\mathrm{T}}\boldsymbol{H}_t)^{-1}\boldsymbol{H}_t^{\mathrm{T}}\boldsymbol{Y}_t=\boldsymbol{P}(t)\boldsymbol{H}_t^{\mathrm{T}}\boldsymbol{Y}_t\\&=\boldsymbol{P}(t)\left[\begin{array}{c}\boldsymbol{H}_{t-1}\\\boldsymbol{\varphi}^{\mathrm{T}}(t)\end{array}\right]^{\mathrm{T}}\left[\begin{array}{c}\boldsymbol{Y}_{t-1}\\y(t)\end{array}\right]\\&=\boldsymbol{P}(t)[\boldsymbol{H}_{t-1}^{\mathrm{T}},\boldsymbol{\varphi}(t)]\left[\begin{array}{c}\boldsymbol{Y}_{t-1}\\y(t)\end{array}\right]\\&=\boldsymbol{P}(t)[\boldsymbol{H}_{t-1}^{\mathrm{T}}\boldsymbol{Y}_{t-1}+\boldsymbol{\varphi}(t)y(t)]\end{aligned}$$

$$
\begin{aligned}
&= \boldsymbol{P}(t)[\boldsymbol{P}^{-1}(t-1)\boldsymbol{P}(t-1)\boldsymbol{H}_{t-1}^{\mathrm{T}}\boldsymbol{Y}_{t-1}+\boldsymbol{\varphi}(t)y(t)] \\
&= \boldsymbol{P}(t)[\boldsymbol{P}^{-1}(t-1)\hat{\boldsymbol{\theta}}(t-1)+\boldsymbol{\varphi}(t)y(t)] \\
&= \boldsymbol{P}(t)[\boldsymbol{P}^{-1}(t)-\boldsymbol{\varphi}(t)\boldsymbol{\varphi}^{\mathrm{T}}(t)]\hat{\boldsymbol{\theta}}(t-1)+\boldsymbol{P}(t)\boldsymbol{\varphi}(t)y(t) \\
&= \hat{\boldsymbol{\theta}}(t-1)+\boldsymbol{P}(t)\boldsymbol{\varphi}(t)[y(t)-\boldsymbol{\varphi}^{\mathrm{T}}(t)\hat{\boldsymbol{\theta}}(t-1)]. \qquad (4.8)
\end{aligned}
$$

Eqs. (4.8) and (4.2) form the recursive least squares algorithm in (4.6)–(4.7) for identifying the parameter vector $\boldsymbol{\theta}$ in (2.2). □

In order to avoid computing the inversion of the covariance matrix $\boldsymbol{P}(t)$, Eqs. (4.5) and (4.8) give the RLS algorithm without requiring matrix inversion:

$$
\begin{aligned}
\hat{\boldsymbol{\theta}}(t) &= \hat{\boldsymbol{\theta}}(t-1)+\boldsymbol{P}(t)\boldsymbol{\varphi}(t)[y(t)-\boldsymbol{\varphi}^{\mathrm{T}}(t)\hat{\boldsymbol{\theta}}(t-1)], \quad \hat{\boldsymbol{\theta}}(0)=\mathbf{1}_n/p_0, \\
\boldsymbol{P}(t) &= \boldsymbol{P}(t-1)-\frac{\boldsymbol{P}(t-1)\boldsymbol{\varphi}(t)\boldsymbol{\varphi}^{\mathrm{T}}(t)\boldsymbol{P}(t-1)}{1+\boldsymbol{\varphi}^{\mathrm{T}}(t)\boldsymbol{P}(t-1)\boldsymbol{\varphi}(t)}, \quad \boldsymbol{P}(0)=p_0\boldsymbol{I}_n.
\end{aligned}
$$

By means of the gain vector $\boldsymbol{L}(t)$, the RLS algorithm can be expressed as

$$
\begin{aligned}
\hat{\boldsymbol{\theta}}(t) &= \hat{\boldsymbol{\theta}}(t-1)+\boldsymbol{L}(t)[y(t)-\boldsymbol{\varphi}^{\mathrm{T}}(t)\hat{\boldsymbol{\theta}}(t-1)], \quad \hat{\boldsymbol{\theta}}(0)=\mathbf{1}_n/p_0, && (4.9)\\
\boldsymbol{L}(t) &= \boldsymbol{P}(t)\boldsymbol{\varphi}(t)=\boldsymbol{P}(t-1)\boldsymbol{\varphi}(t)[1+\boldsymbol{\varphi}^{\mathrm{T}}(t)\boldsymbol{P}(t-1)\boldsymbol{\varphi}(t)]^{-1}, && (4.10)\\
\boldsymbol{P}(t) &= \boldsymbol{P}(t-1)-\boldsymbol{L}(t)[1+\boldsymbol{\varphi}^{\mathrm{T}}(t)\boldsymbol{P}(t-1)\boldsymbol{\varphi}(t)]\boldsymbol{L}^{\mathrm{T}}(t) \\
&= [\boldsymbol{I}_n-\boldsymbol{L}(t)\boldsymbol{\varphi}^{\mathrm{T}}(t)]\boldsymbol{P}(t-1) && (4.11)\\
&= \boldsymbol{P}(t-1)[\boldsymbol{I}_n-\boldsymbol{\varphi}(t)\boldsymbol{L}^{\mathrm{T}}(t)] && (4.12)\\
&= \boldsymbol{P}(t-1)-\boldsymbol{L}(t)[\boldsymbol{P}(t-1)\boldsymbol{\varphi}(t)]^{\mathrm{T}}, \quad \boldsymbol{P}(0)=p_0\boldsymbol{I}_n. && (4.13)
\end{aligned}
$$

The quantity $e(t)=y(t)-\boldsymbol{\varphi}^{\mathrm{T}}(t)\hat{\boldsymbol{\theta}}(t-1)\in\mathbb{R}$ in (4.9) is called the innovation. Define the residual

$$
\hat{v}(t):=y(t)-\boldsymbol{\varphi}^{\mathrm{T}}(t)\hat{\boldsymbol{\theta}}(t)\in\mathbb{R}, \qquad (4.14)
$$

which is regarded as the estimate of the noise $v(t)$. The RLS algorithm in (4.9)–(4.13) can be extended to a multi-innovation least squares (MILS) algorithm in the next section.

To summarize, the steps involved in the RLS algorithm in (4.9)–(4.13) to recursively compute the parameter estimation vector $\hat{\boldsymbol{\theta}}(t)$ with the data length t increasing are as follows.

1. To initialize, let $t=1$, set the initial values $\boldsymbol{P}(0)=p_0\boldsymbol{I}_n$, $\hat{\boldsymbol{\theta}}(0)=\mathbf{1}_n/p_0$, $p_0=10^6$, and give the data length L_e.
2. Collect the observation data $y(t)$ and $\boldsymbol{\varphi}(t)$.
3. Compute the gain matrix $\boldsymbol{L}(t)$ using (4.10) and the covariance matrix $\boldsymbol{P}(t)$ using (4.13).
4. Update the parameter estimation vector $\hat{\boldsymbol{\theta}}(t)$ using (4.9).
5. If $t<L_e$, then increase t by 1 and go to Step 2; otherwise obtain the estimate $\hat{\boldsymbol{\theta}}(L_e)$ of the parameter vector $\boldsymbol{\theta}$ and terminate this procedure.

Lemma 4.3 (*Relation between the Innovation and the Residual*)**.** *For the RLS algorithm in (4.9)–(4.13), the residual $\hat{v}(t)$ and the innovation $e(t)$ have the following relations:*

$$
\begin{aligned}
\hat{v}(t) &= [1-\boldsymbol{\varphi}^{T}(t)\boldsymbol{P}(t)\boldsymbol{\varphi}(t)]e(t), && (4.15)\\
e(t) &= [1+\boldsymbol{\varphi}^{T}(t)\boldsymbol{P}(t-1)\boldsymbol{\varphi}(t)]\hat{v}(t). && (4.16)
\end{aligned}
$$

Proof. Substituting (4.9) into (4.14) gives

$$
\begin{aligned}
\hat{v}(t) &= y(t)-\boldsymbol{\varphi}^{\mathrm{T}}(t)[\hat{\boldsymbol{\theta}}(t-1)+\boldsymbol{L}(t)e(t)] \\
&= y(t)-\boldsymbol{\varphi}^{\mathrm{T}}(t)\hat{\boldsymbol{\theta}}(t-1)-\boldsymbol{\varphi}^{\mathrm{T}}(t)\boldsymbol{L}(t)e(t) \\
&= e(t)-\boldsymbol{\varphi}^{\mathrm{T}}(t)\boldsymbol{L}(t)e(t)=[1-\boldsymbol{\varphi}^{\mathrm{T}}(t)\boldsymbol{L}(t)]e(t) \\
&= [1-\boldsymbol{\varphi}^{\mathrm{T}}(t)\boldsymbol{P}(t)\boldsymbol{\varphi}(t)]e(t) \\
&= \left[1-\boldsymbol{\varphi}^{\mathrm{T}}(t)\frac{\boldsymbol{P}(t-1)\boldsymbol{\varphi}(t)}{1+\boldsymbol{\varphi}^{\mathrm{T}}(t)\boldsymbol{P}(t-1)\boldsymbol{\varphi}(t)}\right]e(t) \\
&= \frac{e(t)}{1+\boldsymbol{\varphi}^{\mathrm{T}}(t)\boldsymbol{P}(t-1)\boldsymbol{\varphi}(t)}. \quad \square
\end{aligned}
$$

Lemma 4.4 (*Recursive Computation of the Criterion Function [5,86,87]*)**.** *The value of the criterion function $J_1(\boldsymbol{\theta})$ at $\boldsymbol{\theta}=\hat{\boldsymbol{\theta}}(t)$ is given by*

$$J_1(t):=J_1[\hat{\boldsymbol{\theta}}(t)]=\|\boldsymbol{Y}_t-\boldsymbol{H}_t\hat{\boldsymbol{\theta}}(t)\|^2=\sum_{j=1}^{t}[y(j)-\boldsymbol{\varphi}^T(j)\hat{\boldsymbol{\theta}}(t)]^2,$$

which can be computed recursively through

$$J_1(t)=J_1(t-1)+\frac{e^2(t)}{1+\boldsymbol{\varphi}^T(t)\boldsymbol{P}(t-1)\boldsymbol{\varphi}(t)}-\frac{1}{p_0}\boldsymbol{\varphi}^T(t)\boldsymbol{P}^2(t)\boldsymbol{\varphi}(t)e^2(t).$$

Note that $J_1(t)\neq J_1(t-1)+\hat{v}^2(t)$.

Proof. Using (4.9)–(4.10) and (4.15), we have

$$\begin{aligned}
J_1(t)&=\sum_{j=1}^{t-1}[y(j)-\boldsymbol{\varphi}^{\mathrm{T}}(j)\hat{\boldsymbol{\theta}}(t)]^2+\hat{v}^2(t)\\
&=\sum_{j=1}^{t-1}\{y(j)-\boldsymbol{\varphi}^{\mathrm{T}}(j)[\hat{\boldsymbol{\theta}}(t-1)+\boldsymbol{L}(t)e(t)]\}^2+[1-\boldsymbol{\varphi}^{\mathrm{T}}(t)\boldsymbol{L}(t)]^2e^2(t)\\
&=\sum_{j=1}^{t-1}[y(j)-\boldsymbol{\varphi}^{\mathrm{T}}(j)\hat{\boldsymbol{\theta}}(t-1)-\boldsymbol{\varphi}^{\mathrm{T}}(j)\boldsymbol{L}(t)e(t)]^2+[1-\boldsymbol{\varphi}^{\mathrm{T}}(t)\boldsymbol{L}(t)]^2e^2(t)\\
&=\sum_{j=1}^{t-1}[y(j)-\boldsymbol{\varphi}^{\mathrm{T}}(j)\hat{\boldsymbol{\theta}}(t-1)]^2-2\sum_{j=1}^{t-1}[y(j)-\boldsymbol{\varphi}^{\mathrm{T}}(j)\hat{\boldsymbol{\theta}}(t-1)]\boldsymbol{\varphi}^{\mathrm{T}}(j)\boldsymbol{L}(t)e(t)\\
&\quad+\sum_{j=1}^{t-1}[\boldsymbol{\varphi}^{\mathrm{T}}(j)\boldsymbol{L}(t)e(t)]^2+e^2(t)-2\boldsymbol{\varphi}^{\mathrm{T}}(t)\boldsymbol{L}(t)e^2(t)+[\boldsymbol{\varphi}^{\mathrm{T}}(t)\boldsymbol{L}(t)e(t)]^2\\
&=J_1(t-1)-0+\boldsymbol{L}^{\mathrm{T}}(t)\sum_{j=1}^{t}\boldsymbol{\varphi}(j)\boldsymbol{\varphi}^{\mathrm{T}}(j)\boldsymbol{L}(t)e^2(t)+e^2(t)-2\boldsymbol{\varphi}^{\mathrm{T}}(t)\boldsymbol{L}(t)e^2(t)\\
&=J_1(t-1)+\boldsymbol{\varphi}^{\mathrm{T}}(t)\boldsymbol{P}(t)[\boldsymbol{P}^{-1}(t)-\boldsymbol{P}^{-1}(0)]\boldsymbol{P}(t)\boldsymbol{\varphi}(t)e^2(t)+e^2(t)-2\boldsymbol{L}^{\mathrm{T}}(t)\boldsymbol{\varphi}(t)e^2(t)\\
&=J_1(t-1)+\boldsymbol{\varphi}^{\mathrm{T}}(t)\boldsymbol{P}(t)\boldsymbol{\varphi}(t)e^2(t)-\boldsymbol{\varphi}^{\mathrm{T}}(t)\boldsymbol{P}(t)\boldsymbol{P}^{-1}(0)\boldsymbol{P}(t)\boldsymbol{\varphi}(t)e^2(t)+e^2(t)-2\boldsymbol{L}^{\mathrm{T}}(t)\boldsymbol{\varphi}(t)e^2(t)\\
&=J_1(t-1)+\boldsymbol{L}^{\mathrm{T}}(t)\boldsymbol{\varphi}(t)e^2(t)-\boldsymbol{\varphi}^{\mathrm{T}}(t)\boldsymbol{P}^2(t)\boldsymbol{\varphi}(t)e^2(t)/p_0+e^2(t)-2\boldsymbol{L}^{\mathrm{T}}(t)\boldsymbol{\varphi}(t)e^2(t)\\
&=J_1(t-1)+[1-\boldsymbol{L}^{\mathrm{T}}(t)\boldsymbol{\varphi}(t)]e^2(t)-\boldsymbol{\varphi}^{\mathrm{T}}(t)\boldsymbol{P}^2(t)\boldsymbol{\varphi}(t)e^2(t)/p_0\\
&=J_1(t-1)+\left[1-\frac{\boldsymbol{\varphi}^{\mathrm{T}}(t)\boldsymbol{P}(t-1)\boldsymbol{\varphi}(t)}{1+\boldsymbol{\varphi}^{\mathrm{T}}(t)\boldsymbol{P}(t-1)\boldsymbol{\varphi}(t)}\right]e^2(t)-\boldsymbol{\varphi}^{\mathrm{T}}(t)\boldsymbol{P}^2(t)\boldsymbol{\varphi}(t)e^2(t)/p_0\\
&=J_1(t-1)+\frac{e^2(t)}{1+\boldsymbol{\varphi}^{\mathrm{T}}(t)\boldsymbol{P}(t-1)\boldsymbol{\varphi}(t)}-\frac{1}{p_0}\boldsymbol{\varphi}^{\mathrm{T}}(t)\boldsymbol{P}^2(t)\boldsymbol{\varphi}(t)e^2(t).
\end{aligned}$$

In general, p_0 is taken to be a large positive number, $\boldsymbol{P}(t)$ will decrease with the increase of t. Thus the last term on the right-hand side can be neglected, and we have the approximate relation,

$$J_1(t)=J_1(t-1)+\frac{e^2(t)}{1+\boldsymbol{\varphi}^{\mathrm{T}}(t)\boldsymbol{P}(t-1)\boldsymbol{\varphi}(t)}.\quad\square$$

For the linear regressive systems in (2.2), when collect the observation data $y(t)$ and $\boldsymbol{\varphi}(t)$, we can use (3.3) or (3.4) to compute the LS estimate $\hat{\boldsymbol{\theta}}(t)$ of the parameter vector $\boldsymbol{\theta}$, it seems that the projection algorithm, the stochastic gradient algorithm, the LS algorithm in (3.13)–(3.15) and the RLS algorithm in (4.9)–(4.13) are not necessary. That is when the observation data are available, it is not necessary to use the recursive scheme and the iterative scheme for parameter estimation. In general, the recursive algorithm and the iterative algorithm are used in nonlinear systems or stochastic systems with colored noises for the information vector/matrix with some unknown entries.

The least squares approaches can combine some mathematical optimization approaches and statistical strategies [88–94] to study the parameter estimation problems of linear and nonlinear systems with different disturbances [95–102] and can be applied to other fields [103–113] such as information processing and engineering systems.

4.2. Basic lemmas of RLS algorithms

Lemma 4.5 (*Inequalities about symmetric matrix*). *For any real vector $\boldsymbol{x} \in \mathbb{R}^n$ and symmetric matrix $\boldsymbol{Q} \in \mathbb{R}^{n\times n}$, and real matrix $\boldsymbol{S} \in \mathbb{R}^{m\times n}$, the following inequalities hold,*

(i) $\lambda_{\min}[\boldsymbol{Q}]\|\boldsymbol{x}\|^2 \leqslant \boldsymbol{x}^T\boldsymbol{Q}\boldsymbol{x} \leqslant \lambda_{\max}[\boldsymbol{Q}]\|\boldsymbol{x}\|^2$,

(ii) $\lambda_{\min}[\boldsymbol{X}]\boldsymbol{S}^T\boldsymbol{S} \leqslant \boldsymbol{S}^T\boldsymbol{X}\boldsymbol{S} \leqslant \lambda_{\max}[\boldsymbol{X}]\boldsymbol{S}^T\boldsymbol{S}$.

Lemma 4.6 (*Eigenvalue Shift Lemma [2]*). *Suppose that the matrix $\boldsymbol{A} \in \mathbb{R}^{n\times n}$ have n eigenvalues $\lambda_i[\boldsymbol{A}]$, $i = 1, 2, \ldots, n$. Then the eigenvalues of $\boldsymbol{A} + s\boldsymbol{I}_n$ are $\lambda_i[\boldsymbol{A} + s\boldsymbol{I}] = \lambda_i[\boldsymbol{A}] + s$, $i = 1, 2, \ldots, n$, where s is a constant.*

Lemma 4.7 (*Singular Value Shift Lemma [2]*). *Suppose that $\boldsymbol{A} \in \mathbb{R}^{n\times n}$ has n singular values $\lambda_i^2[\boldsymbol{A}]$, $i = 1, 2, \ldots, n$, and let $\min_i\{\lambda_i^2[\boldsymbol{A}]\} = \alpha^2$ ($\alpha \geqslant 0$). Then for any real number s, the following inequalities hold,*

(i) $\boldsymbol{A}^T\boldsymbol{A} \geqslant \alpha^2\boldsymbol{I}, \quad \boldsymbol{A}\boldsymbol{A}^T \geqslant \alpha^2\boldsymbol{I}_n$.

(ii) $(\boldsymbol{A} + s\boldsymbol{I})^T(\boldsymbol{A} + s\boldsymbol{I}) \geqslant (\alpha + s)^2\boldsymbol{I}_n$.

(iii) $(\boldsymbol{A} + s\boldsymbol{I})(\boldsymbol{A} + s\boldsymbol{I})^T \geqslant (\alpha + s)^2\boldsymbol{I}_n$.

Lemma 4.8 (*Singular Value Shift Lemma [2]*). *Suppose that $\boldsymbol{A} \in \mathbb{R}^{m\times n}$ has $\min[m, n]$ nonzero singular values $\lambda_i^2[\boldsymbol{A}]$, $i = 1, 2, \ldots, \min[m, n]$, let $\min_i\{\lambda_i^2[\boldsymbol{A}]\} = \alpha^2$ ($\alpha > 0$). Then for any real number s, the following inequalities hold,*

(i) $(\boldsymbol{A} + s\boldsymbol{I})^T(\boldsymbol{A} + s\boldsymbol{I}) \geqslant (\alpha + s)^2\boldsymbol{I}_n, \ m > n$.

(ii) $(\boldsymbol{A} + s\boldsymbol{I})(\boldsymbol{A} + s\boldsymbol{I})^T \geqslant (\alpha + s)^2\boldsymbol{I}_m, \ m < n$.

Lemma 4.9 (*Block Diagonal Positive-Definite Matrix Lemma [2,114]*). *If the block symmetric matrix*

$$\boldsymbol{A} = \begin{bmatrix} \boldsymbol{A}_{11} & \boldsymbol{A}_{12} & \ldots & \boldsymbol{A}_{1N} \\ \boldsymbol{A}_{21} & \boldsymbol{A}_{22} & \ldots & \boldsymbol{A}_{2N} \\ \vdots & \vdots & & \vdots \\ \boldsymbol{A}_{N1} & \boldsymbol{A}_{N2} & \ldots & \boldsymbol{A}_{NN} \end{bmatrix} \in \mathbb{R}^{n\times n}, \ \boldsymbol{A}_{ij} = \boldsymbol{A}_{ji}^T \in \mathbb{R}^{n_i\times n_j}$$

satisfies $\alpha\boldsymbol{I} \leqslant \boldsymbol{A} \leqslant \beta\boldsymbol{I}_n$, where α and β are two positive numbers, then

$$\alpha\boldsymbol{I} \leqslant \begin{bmatrix} \boldsymbol{A}_{11} & \boldsymbol{0} & \ldots & \boldsymbol{0} \\ \boldsymbol{0} & \boldsymbol{A}_{22} & \ldots & \boldsymbol{0} \\ \vdots & \vdots & \ddots & \vdots \\ \boldsymbol{0} & \boldsymbol{0} & \ldots & \boldsymbol{A}_{NN} \end{bmatrix} \leqslant \beta\boldsymbol{I}.$$

Lemma 4.10 (*Block Matrix Inversion Lemma*). *For $\boldsymbol{A} \in \mathbb{R}^{m\times m}$, $\boldsymbol{B} \in \mathbb{R}^{m\times n}$, $\boldsymbol{C} \in \mathbb{R}^{n\times m}$, if $\boldsymbol{D} \in \mathbb{R}^{n\times n}$ and $\boldsymbol{Q}_1 := \boldsymbol{D} - \boldsymbol{C}\boldsymbol{A}^{-1}\boldsymbol{B} \in \mathbb{R}^{n\times n}$ are two invertible matrices, the following relation holds,*

$$\begin{bmatrix} \boldsymbol{A} & \boldsymbol{B} \\ \boldsymbol{C} & \boldsymbol{D} \end{bmatrix}^{-1} = \begin{bmatrix} \boldsymbol{A}^{-1} + \boldsymbol{A}^{-1}\boldsymbol{B}\boldsymbol{Q}_1^{-1}\boldsymbol{C}\boldsymbol{A}^{-1} & -\boldsymbol{A}^{-1}\boldsymbol{B}\boldsymbol{Q}_1^{-1} \\ -\boldsymbol{Q}_1^{-1}\boldsymbol{C}\boldsymbol{A}^{-1} & \boldsymbol{Q}_1^{-1} \end{bmatrix},$$

or

$$\begin{bmatrix} \boldsymbol{A} & \boldsymbol{B} \\ \boldsymbol{C} & \boldsymbol{D} \end{bmatrix}^{-1} = \begin{bmatrix} \boldsymbol{I} & -\boldsymbol{A}^{-1}\boldsymbol{B} \\ \boldsymbol{0} & \boldsymbol{I} \end{bmatrix}\begin{bmatrix} \boldsymbol{A}^{-1} & \boldsymbol{0} \\ \boldsymbol{0} & (\boldsymbol{D} - \boldsymbol{C}\boldsymbol{A}^{-1}\boldsymbol{B})^{-1} \end{bmatrix}\begin{bmatrix} \boldsymbol{I} & \boldsymbol{0} \\ -\boldsymbol{C}\boldsymbol{A}^{-1} & \boldsymbol{I} \end{bmatrix}.$$

Lemma 4.11 (*Block Matrix Inversion Lemma*). *For $\boldsymbol{A} \in \mathbb{R}^{m\times m}$, $\boldsymbol{B} \in \mathbb{R}^{m\times n}$, $\boldsymbol{C} \in \mathbb{R}^{n\times m}$ and $\boldsymbol{D} \in \mathbb{R}^{n\times n}$, if $\boldsymbol{A}$ and $\boldsymbol{Q}_2 := \boldsymbol{A} - \boldsymbol{B}\boldsymbol{D}^{-1}\boldsymbol{C} \in \mathbb{R}^{m\times m}$ are two invertible matrices, the following relation holds,*

$$\begin{bmatrix} \boldsymbol{A} & \boldsymbol{B} \\ \boldsymbol{C} & \boldsymbol{D} \end{bmatrix}^{-1} = \begin{bmatrix} \boldsymbol{Q}_2^{-1} & -\boldsymbol{Q}_2^{-1}\boldsymbol{B}\boldsymbol{D}^{-1} \\ -\boldsymbol{D}^{-1}\boldsymbol{C}\boldsymbol{Q}_2^{-1} & \boldsymbol{D}^{-1} + \boldsymbol{D}^{-1}\boldsymbol{C}\boldsymbol{Q}_2^{-1}\boldsymbol{B}\boldsymbol{D}^{-1} \end{bmatrix}.$$

Lemma 4.12 (*Matrix Determinant Lemma*). *For $\boldsymbol{D} \in \mathbb{R}^{m\times n}$, $\boldsymbol{E} \in \mathbb{R}^{n\times m}$, the following matrix determinant holds,*

$$\det[\boldsymbol{I}_m + \boldsymbol{D}\boldsymbol{E}] = \det[\boldsymbol{I}_n + \boldsymbol{E}\boldsymbol{D}],$$

where $\det[\boldsymbol{X}] := |\boldsymbol{X}|$ represents the determinant of the matrix $\boldsymbol{X}$. This can be proved by taking determinants on both sides of the following matrix identity,

$$\begin{bmatrix} \boldsymbol{I}_m & -\boldsymbol{D} \\ \boldsymbol{0} & \boldsymbol{E}\boldsymbol{D} + \boldsymbol{I}_n \end{bmatrix} = \begin{bmatrix} \boldsymbol{I}_m & \boldsymbol{0} \\ -\boldsymbol{E} & \boldsymbol{I}_n \end{bmatrix}\begin{bmatrix} \boldsymbol{I}_m + \boldsymbol{D}\boldsymbol{E} & -\boldsymbol{D} \\ \boldsymbol{0} & \boldsymbol{I}_n \end{bmatrix}\begin{bmatrix} \boldsymbol{I}_m & \boldsymbol{0} \\ \boldsymbol{E} & \boldsymbol{I}_n \end{bmatrix}.$$

Especially for the vectors $\boldsymbol{a} \in \mathbb{R}^n$ *and* $\boldsymbol{b} \in \mathbb{R}^n$*, we have*

$$\det[\boldsymbol{I}_n + \boldsymbol{a}\boldsymbol{b}^{T}] = 1 + \boldsymbol{b}^{T}\boldsymbol{a}.$$

Lemma 4.13. *For* $\boldsymbol{A}, \boldsymbol{B} \in \mathbb{R}^{n\times n}$, $\boldsymbol{a}, \boldsymbol{b} \in \mathbb{R}^n$*, if* $\boldsymbol{A} = \boldsymbol{B} + \boldsymbol{a}\boldsymbol{b}^{T}$*, then the following equality holds,*

$$\boldsymbol{b}^{T}\boldsymbol{A}^{-1}\boldsymbol{a} = \frac{|\boldsymbol{A}| - |\boldsymbol{B}|}{|\boldsymbol{A}|}.$$

Proof. From $\boldsymbol{A} = \boldsymbol{B} + \boldsymbol{a}\boldsymbol{b}^{\mathrm{T}}$, we have $\boldsymbol{B} = \boldsymbol{A} - \boldsymbol{a}\boldsymbol{b}^{\mathrm{T}} = \boldsymbol{A}[\boldsymbol{I}_n - \boldsymbol{A}^{-1}\boldsymbol{a}\boldsymbol{b}^{\mathrm{T}}]$. Taking the determinants on both sides and using Lemma 4.12, we have $|\boldsymbol{B}| = |\boldsymbol{A}| \det[\boldsymbol{I}_n - \boldsymbol{A}^{-1}\boldsymbol{a}\boldsymbol{b}^{\mathrm{T}}] = |\boldsymbol{A}|[1 - \boldsymbol{b}^{\mathrm{T}}\boldsymbol{A}^{-1}\boldsymbol{a}]$. This directly gives the conclusion of Lemma 4.13. □

Lemma 4.14. *For large* t*, suppose that the matrix* $\boldsymbol{P}^{-1}(t) := \sum_{j=1}^{t} \boldsymbol{\varphi}(j)\boldsymbol{\varphi}^{T}(j)$ *is invertible,* $\boldsymbol{\varphi}(t) \in \mathbb{R}^n$*. Then the following equality holds,*

$$\sum_{j=1}^{t} \boldsymbol{\varphi}^{T}(j)\boldsymbol{P}(t)\boldsymbol{\varphi}(j) = n.$$

Proof. From the given condition, we have $\sum_{j=1}^{t} \boldsymbol{P}(t)\boldsymbol{\varphi}(j)\boldsymbol{\varphi}^{\mathrm{T}}(j) = \boldsymbol{I}_n$. Taking the trace and using $\mathrm{tr}[\boldsymbol{A}\boldsymbol{B}] = \mathrm{tr}[\boldsymbol{B}\boldsymbol{A}]$ yield the conclusion of Lemma 4.14. □

Lemma 4.15 ([2]). *For the covariance matrix* $\boldsymbol{P}(t)$ *in (4.7), the following relations hold:*

(i) $[1 - \boldsymbol{\varphi}^{T}(t)\boldsymbol{P}(t)\boldsymbol{\varphi}(t)][1 + \boldsymbol{\varphi}^{T}(t)\boldsymbol{P}(t-1)\boldsymbol{\varphi}(t)] = 1,$

(ii) $\boldsymbol{\varphi}^{T}(t)\boldsymbol{P}(t)\boldsymbol{\varphi}(t) = \dfrac{\boldsymbol{\varphi}^{T}(t)\boldsymbol{P}(t-1)\boldsymbol{\varphi}(t)}{1 + \boldsymbol{\varphi}^{T}(t)\boldsymbol{P}(t-1)\boldsymbol{\varphi}(t)} \leqslant 1,$

(iii) $\boldsymbol{\varphi}^{T}(t)\boldsymbol{P}(t-1)\boldsymbol{\varphi}(t) = \dfrac{\boldsymbol{\varphi}^{T}(t)\boldsymbol{P}(t)\boldsymbol{\varphi}(t)}{1 - \boldsymbol{\varphi}^{T}(t)\boldsymbol{P}(t)\boldsymbol{\varphi}(t)},$

(iv) $\boldsymbol{\varphi}^{T}(t)\boldsymbol{P}^2(t)\boldsymbol{\varphi}(t) \leqslant \boldsymbol{\varphi}^{T}(t)\boldsymbol{P}(t)\boldsymbol{P}(t-1)\boldsymbol{\varphi}(t),$

(v) $\boldsymbol{\varphi}^{T}(t)\boldsymbol{P}^2(t-1)\boldsymbol{\varphi}(t) = \dfrac{\boldsymbol{\varphi}^{T}(t)\boldsymbol{P}(t)\boldsymbol{P}(t-1)\boldsymbol{\varphi}(t)}{1 - \boldsymbol{\varphi}^{T}(t)\boldsymbol{P}(t)\boldsymbol{\varphi}(t)},$

(vi) $\boldsymbol{\varphi}^{T}(t)\boldsymbol{P}(t-1)\boldsymbol{P}(t)\boldsymbol{\varphi}(t) = \dfrac{\boldsymbol{\varphi}^{T}(t)\boldsymbol{P}^2(t)\boldsymbol{\varphi}(t)}{1 - \boldsymbol{\varphi}^{T}(t)\boldsymbol{P}(t)\boldsymbol{\varphi}(t)}.$

Proof. (i) From (4.1), we can obtain

$$\begin{aligned}1 - \boldsymbol{\varphi}^{\mathrm{T}}(t)\boldsymbol{P}(t)\boldsymbol{\varphi}(t) &= 1 - \frac{\boldsymbol{\varphi}^{\mathrm{T}}(t)\boldsymbol{P}(t-1)\boldsymbol{\varphi}(t)}{1 + \boldsymbol{\varphi}^{\mathrm{T}}(t)\boldsymbol{P}(t-1)\boldsymbol{\varphi}(t)}\\ &= \frac{1}{1 + \boldsymbol{\varphi}^{\mathrm{T}}(t)\boldsymbol{P}(t-1)\boldsymbol{\varphi}(t)}.\end{aligned}$$

Thus we have

$$[1 - \boldsymbol{\varphi}^{\mathrm{T}}(t)\boldsymbol{P}(t)\boldsymbol{\varphi}(t)][1 + \boldsymbol{\varphi}^{\mathrm{T}}(t)\boldsymbol{P}(t-1)\boldsymbol{\varphi}(t)] = 1.$$

(ii) Pre-multiplying (4.1) by $\boldsymbol{\varphi}^{\mathrm{T}}(t)$ gives

$$\boldsymbol{\varphi}^{\mathrm{T}}(t)\boldsymbol{L}(t) = \boldsymbol{\varphi}^{\mathrm{T}}(t)\boldsymbol{P}(t)\boldsymbol{\varphi}(t) = \frac{\boldsymbol{\varphi}^{\mathrm{T}}(t)\boldsymbol{P}(t-1)\boldsymbol{\varphi}(t)}{1 + \boldsymbol{\varphi}^{\mathrm{T}}(t)\boldsymbol{P}(t-1)\boldsymbol{\varphi}(t)} \leqslant 1.$$

(iii) Multiplying the above equation by $1 + \boldsymbol{\varphi}^{\mathrm{T}}(t)\boldsymbol{P}(t-1)\boldsymbol{\varphi}(t)$ gives

$$\boldsymbol{\varphi}^{\mathrm{T}}(t)\boldsymbol{P}(t)\boldsymbol{\varphi}(t)[1 + \boldsymbol{\varphi}^{\mathrm{T}}(t)\boldsymbol{P}(t-1)\boldsymbol{\varphi}(t)] = \boldsymbol{\varphi}^{\mathrm{T}}(t)\boldsymbol{P}(t-1)\boldsymbol{\varphi}(t).$$

Solving $\boldsymbol{\varphi}^{\mathrm{T}}(t)\boldsymbol{P}(t-1)\boldsymbol{\varphi}(t)$ gives conclusion (iii).

(iv) Since $\boldsymbol{P}(t)$ is symmetric and non-negative definite, we have $\boldsymbol{x}^{\mathrm{T}}\boldsymbol{P}(t)\boldsymbol{x} \geqslant 0$ for any $\boldsymbol{x} \in \mathbb{R}^n$. Using (4.1), we have

$$\begin{aligned}0 \leqslant \boldsymbol{\varphi}^{\mathrm{T}}(t)\boldsymbol{P}^2(t)\boldsymbol{\varphi}(t) &= \boldsymbol{\varphi}^{\mathrm{T}}(t)\boldsymbol{P}(t)\boldsymbol{P}(t)\boldsymbol{\varphi}(t)\\ &= \frac{\boldsymbol{\varphi}^{\mathrm{T}}(t)\boldsymbol{P}(t)\boldsymbol{P}(t-1)\boldsymbol{\varphi}(t)}{1 + \boldsymbol{\varphi}^{\mathrm{T}}(t)\boldsymbol{P}(t-1)\boldsymbol{\varphi}(t)} \leqslant \boldsymbol{\varphi}^{\mathrm{T}}(t)\boldsymbol{P}(t)\boldsymbol{P}(t-1)\boldsymbol{\varphi}(t).\end{aligned}$$

(v) Using (i) and (4.1), we have

$$\boldsymbol{\varphi}^{\mathrm{T}}(t)\boldsymbol{P}^2(t-1)\boldsymbol{\varphi}(t) = \frac{\boldsymbol{\varphi}^{\mathrm{T}}(t)\boldsymbol{P}(t-1)\boldsymbol{P}(t-1)\boldsymbol{\varphi}(t)}{[1 + \boldsymbol{\varphi}^{\mathrm{T}}(t)\boldsymbol{P}(t-1)\boldsymbol{\varphi}(t)][1 - \boldsymbol{\varphi}^{\mathrm{T}}(t)\boldsymbol{P}(t)\boldsymbol{\varphi}(t)]}$$

$$= \frac{\boldsymbol{\varphi}^{\mathrm{T}}(t)\boldsymbol{P}(t-1)\boldsymbol{P}(t)\boldsymbol{\varphi}(t)}{1-\boldsymbol{\varphi}^{\mathrm{T}}(t)\boldsymbol{P}(t)\boldsymbol{\varphi}(t)} = \frac{\boldsymbol{\varphi}^{\mathrm{T}}(t)\boldsymbol{P}(t)\boldsymbol{P}(t-1)\boldsymbol{\varphi}(t)}{1-\boldsymbol{\varphi}^{\mathrm{T}}(t)\boldsymbol{P}(t)\boldsymbol{\varphi}(t)}.$$

(vi) Using (i) and (4.1), we can obtain

$$\begin{aligned}\boldsymbol{\varphi}^{\mathrm{T}}(t)\boldsymbol{P}(t-1)\boldsymbol{P}(t)\boldsymbol{\varphi}(t) &= \frac{\boldsymbol{\varphi}^{\mathrm{T}}(t)\boldsymbol{P}(t-1)\boldsymbol{P}(t)\boldsymbol{\varphi}(t)}{[1+\boldsymbol{\varphi}^{\mathrm{T}}(t)\boldsymbol{P}(t-1)\boldsymbol{\varphi}(t)][1-\boldsymbol{\varphi}^{\mathrm{T}}(t)\boldsymbol{P}(t)\boldsymbol{\varphi}(t)]}\\ &= \frac{\boldsymbol{\varphi}^{\mathrm{T}}(t)\boldsymbol{P}^2(t)\boldsymbol{\varphi}(t)}{1-\boldsymbol{\varphi}^{\mathrm{T}}(t)\boldsymbol{P}(t)\boldsymbol{\varphi}(t)}.\end{aligned}$$

This completes the proof of Lemma 4.15. □

Define the trace of the inverse of the covariance matrix $\boldsymbol{P}(t)$ as

$$r(t) := \mathrm{tr}[\boldsymbol{P}^{-1}(t)].$$

Using (4.7), we have

$$\begin{aligned}r(t) &= \mathrm{tr}[\boldsymbol{P}^{-1}(t-1)+\boldsymbol{\varphi}(t)\boldsymbol{\varphi}^{\mathrm{T}}(t)]\\ &= \mathrm{tr}[\boldsymbol{P}^{-1}(t-1)]+\mathrm{tr}[\boldsymbol{\varphi}(t)\boldsymbol{\varphi}^{\mathrm{T}}(t)]\\ &= r(t-1)+\|\boldsymbol{\varphi}(t)\|^2, \quad r(0)=\mathrm{tr}[\boldsymbol{P}^{-1}(0)]=n/p_0.\end{aligned}$$

Furthermore, we have

$$\begin{aligned}r(t) &= r(t-2)+\|\boldsymbol{\varphi}(t-1)\|^2+\|\boldsymbol{\varphi}(t)\|^2\\ &= r(0)+\sum_{j=1}^{t}\|\boldsymbol{\varphi}(j)\|^2\\ &= \frac{n}{p_0}+\sum_{j=1}^{t}\|\boldsymbol{\varphi}(j)\|^2.\end{aligned}$$

Lemma 4.16 ([2]). *The trace, eigenvalue and determinant have the following relations:*

(i) $r(t) = O(\lambda_{\max}[\boldsymbol{P}^{-1}(t)])$,

(ii) $\ln|\boldsymbol{P}^{-1}(t)| = O(\ln r(t))$.

Proof. Let $\lambda_i[\boldsymbol{X}]$ represent the ith eigenvalue of the matrix $\boldsymbol{X}$. Since $\boldsymbol{P}^{-1}(t)$ is non-negative definite and non-descending, we have

$$\begin{aligned}\boldsymbol{P}^{-1}(t) &\geqslant \boldsymbol{P}^{-1}(t-1)\geqslant\ldots\geqslant\boldsymbol{P}^{-1}(1)\geqslant\boldsymbol{P}^{-1}(0)=\boldsymbol{I}_n/p_0,\\ \boldsymbol{P}(t) &\leqslant \boldsymbol{P}(t-1)\leqslant\cdots\leqslant\boldsymbol{P}(1)\leqslant\boldsymbol{P}(0)=p_0\boldsymbol{I}_n,\\ r(t) &= \lambda_1[\boldsymbol{P}^{-1}(t)]+\lambda_2[\boldsymbol{P}^{-1}(t)]+\cdots+\lambda_n[\boldsymbol{P}^{-1}(t)],\\ |\boldsymbol{P}^{-1}(t)| &= \lambda_1[\boldsymbol{P}^{-1}(t)]\lambda_2[\boldsymbol{P}^{-1}(t)]\ldots\lambda_n[\boldsymbol{P}^{-1}(t)],\\ r(t) &\leqslant n\lambda_{\max}[\boldsymbol{P}^{-1}(t)]=O(\lambda_{\max}[\boldsymbol{P}^{-1}(t)]),\\ r^n(t) &\geqslant |\boldsymbol{P}^{-1}(t)|\geqslant\lambda_{\max}[\boldsymbol{P}^{-1}(t)]\left(\frac{1}{p_0}\right)^{n-1}\geqslant r(t)\frac{1}{n}\left(\frac{1}{p_0}\right)^{n-1}.\end{aligned}$$

Taking the logarithm gives

$$\ln r(t)-\ln n-(n-1)\ln p_0\leqslant\ln|\boldsymbol{P}^{-1}(t)|\leqslant n\ln r(t).$$

The last inequality means $\ln|\boldsymbol{P}^{-1}(t)| = O(\ln r(t))$. □

Lemma 4.16 indicates that the trace of a positive definite matrix is of the same order as its greatest eigenvalue. The logarithm of the determinant of a positive definite matrix is of the same order as the logarithm of its trace.

Lemma 4.17 ([2]). *For the covariance matrix $\boldsymbol{P}(t)$ in (4.7), the following inequalities hold:*

(i) $\displaystyle\sum_{t=1}^{\infty}\boldsymbol{\varphi}^{T}(t)\boldsymbol{P}(t-1)\boldsymbol{P}(t)\boldsymbol{\varphi}(t)=\sum_{t=1}^{\infty}\boldsymbol{\varphi}^{T}(t)\boldsymbol{P}(t)\boldsymbol{P}(t-1)\boldsymbol{\varphi}(t)<\infty$,

(ii) $\displaystyle\sum_{t=1}^{\infty}\boldsymbol{\varphi}^{T}(t)\boldsymbol{P}^2(t)\boldsymbol{\varphi}(t)<\infty$,

$$\text{(iii)}\quad \sum_{t=1}^{\infty}\boldsymbol{\varphi}^{T}(t)\boldsymbol{P}(t-1)\boldsymbol{P}(t)\boldsymbol{P}(t)\boldsymbol{\varphi}(t)<\infty,$$

$$\text{(iv)}\quad \sum_{t=1}^{\infty}\frac{\|\boldsymbol{P}(t-1)\boldsymbol{\varphi}(t)\|^2}{1+\boldsymbol{\varphi}^{T}(t)\boldsymbol{P}(t-1)\boldsymbol{\varphi}(t)}<\infty,$$

$$\text{(v)}\quad \sum_{t=1}^{\infty}\boldsymbol{\varphi}^{T}(t)\boldsymbol{P}(t-1)\boldsymbol{P}(t)\boldsymbol{P}(t-1)\boldsymbol{\varphi}(t)<\infty,$$

$$\text{(vi)}\quad \sum_{t=1}^{\infty}\boldsymbol{\varphi}^{T}(t)\boldsymbol{P}^{c}(t)\boldsymbol{\varphi}(t)<\infty,\ c>1,$$

$$\text{(vii)}\quad \sum_{t=1}^{\infty}\boldsymbol{\varphi}^{T}(t)\boldsymbol{P}(t)\boldsymbol{P}^{c}(t-1)\boldsymbol{\varphi}(t)<\infty,\ c>0.$$

Proof. (i) From (4.4), we have

$$\boldsymbol{P}(t)\boldsymbol{\varphi}(t)\boldsymbol{\varphi}^{\mathrm{T}}(t)\boldsymbol{P}(t-1)=\boldsymbol{P}(t-1)-\boldsymbol{P}(t).$$

Taking the trace gives

$$\begin{aligned}\boldsymbol{\varphi}^{\mathrm{T}}(t)\boldsymbol{P}(t-1)\boldsymbol{P}(t)\boldsymbol{\varphi}(t)&=\mathrm{tr}[\boldsymbol{P}(t)\boldsymbol{\varphi}(t)\boldsymbol{\varphi}^{\mathrm{T}}(t)\boldsymbol{P}(t-1)]\\&=\mathrm{tr}[\boldsymbol{P}(t-1)]-\mathrm{tr}[\boldsymbol{P}(t)].\end{aligned}$$

Summing for t from $t=1$ to $t=\infty$ gives

$$\begin{aligned}\sum_{t=1}^{\infty}\boldsymbol{\varphi}^{\mathrm{T}}(t)\boldsymbol{P}(t-1)\boldsymbol{P}(t)\boldsymbol{\varphi}(t)&=\sum_{t=1}^{\infty}\mathrm{tr}[\boldsymbol{P}(t-1)]-\mathrm{tr}[\boldsymbol{P}(t)]\\&=\mathrm{tr}[\boldsymbol{P}(0)]-\mathrm{tr}[\boldsymbol{P}(\infty)]\leqslant\mathrm{tr}[\boldsymbol{P}(0)]=np_0<\infty.\end{aligned}$$

(ii) Using (4.1) and (i), we have

$$\begin{aligned}\sum_{t=1}^{\infty}\boldsymbol{\varphi}^{\mathrm{T}}(t)\boldsymbol{P}^2(t)\boldsymbol{\varphi}(t)&=\sum_{t=1}^{\infty}\frac{\boldsymbol{\varphi}^{\mathrm{T}}(t)\boldsymbol{P}(t)\boldsymbol{P}(t-1)\boldsymbol{\varphi}(t)}{1+\boldsymbol{\varphi}^{\mathrm{T}}(t)\boldsymbol{P}(t-1)\boldsymbol{\varphi}(t)}\\&\leqslant\sum_{t=1}^{\infty}\boldsymbol{\varphi}^{\mathrm{T}}(t)\boldsymbol{P}(t)\boldsymbol{P}(t-1)\boldsymbol{\varphi}(t)<\infty.\end{aligned}$$

(iii) According to Lemma 4.5 and using (i), we have

$$\begin{aligned}\sum_{t=1}^{\infty}\boldsymbol{\varphi}^{\mathrm{T}}(t)\boldsymbol{P}(t-1)\boldsymbol{P}(t)\boldsymbol{P}(t)\boldsymbol{\varphi}(t)&=\sum_{t=1}^{\infty}\frac{\boldsymbol{\varphi}^{\mathrm{T}}(t)\boldsymbol{P}(t-1)\boldsymbol{P}(t)\boldsymbol{P}(t-1)\boldsymbol{\varphi}(t)}{1+\boldsymbol{\varphi}^{\mathrm{T}}(t)\boldsymbol{P}(t-1)\boldsymbol{\varphi}(t)}\\&\leqslant p_0\sum_{t=1}^{\infty}\frac{\boldsymbol{\varphi}^{\mathrm{T}}(t)\boldsymbol{P}(t-1)\boldsymbol{P}(t-1)\boldsymbol{\varphi}(t)}{1+\boldsymbol{\varphi}^{\mathrm{T}}(t)\boldsymbol{P}(t-1)\boldsymbol{\varphi}(t)}\\&=p_0\sum_{t=1}^{\infty}\boldsymbol{\varphi}^{\mathrm{T}}(t)\boldsymbol{P}(t-1)\boldsymbol{P}(t)\boldsymbol{\varphi}(t)<\infty.\end{aligned}$$

(iv) Similarly, we have

$$\begin{aligned}\sum_{t=1}^{\infty}\frac{\|\boldsymbol{P}(t-1)\boldsymbol{\varphi}(t)\|^2}{1+\boldsymbol{\varphi}^{\mathrm{T}}(t)\boldsymbol{P}(t-1)\boldsymbol{\varphi}(t)}&=\sum_{t=1}^{\infty}\frac{\boldsymbol{\varphi}^{\mathrm{T}}(t)\boldsymbol{P}^2(t-1)\boldsymbol{\varphi}(t)}{1+\boldsymbol{\varphi}^{\mathrm{T}}(t)\boldsymbol{P}(t-1)\boldsymbol{\varphi}(t)}\\&=\sum_{t=1}^{\infty}\boldsymbol{\varphi}^{\mathrm{T}}(t)\boldsymbol{P}(t-1)\boldsymbol{P}(t)\boldsymbol{\varphi}(t)<\infty.\quad\square\end{aligned}$$

Lemma 4.18. *For the covariance matrix $\boldsymbol{P}(t)$ in (4.7), the following inequalities hold [5,34,115,116]:*

$$\text{(i)}\quad \sum_{t=1}^{\infty}\boldsymbol{\varphi}^{T}(t-i)\boldsymbol{P}^{3}(t)\boldsymbol{\varphi}(t-i)<\infty,\quad \textit{for any } i,\ i=0,1,\ldots,p-1,$$

$$\text{(ii)}\quad \sum_{t=1}^{\infty}\boldsymbol{\varphi}^T(t-i)\boldsymbol{P}(t-1)\boldsymbol{P}(t)\boldsymbol{P}(t-1)\boldsymbol{\varphi}(t-i)<\infty,$$

$$\text{(iii)}\quad \sum_{t=1}^{\infty}\boldsymbol{\varphi}^T(t-i)\boldsymbol{P}(t)\boldsymbol{P}(t-1)\boldsymbol{P}(t)\boldsymbol{\varphi}(t-i)<\infty,$$

$$\text{(iv)}\quad \sum_{t=1}^{\infty}\frac{\boldsymbol{\varphi}^T(t-i)\boldsymbol{P}(t)\boldsymbol{\varphi}(t-i)}{r^c(t)}<\infty,\ \textit{for any } c>0,$$

$$\text{(v)}\quad \sum_{t=1}^{\infty}\frac{\boldsymbol{\varphi}^T(t-i)\boldsymbol{P}(t)\boldsymbol{\varphi}(t-i)}{[\ln r(t)|]^c}<\infty,\ \textit{for any } c>1,$$

$$\text{(vi)}\quad \sum_{t=1}^{\infty}\boldsymbol{\varphi}^T(t-i)\boldsymbol{P}^c(t)\boldsymbol{\varphi}(t-i)<\infty,\ \textit{for any } c>1.$$

Lemma 4.19 (*Inequalities about the Covariance Matrix [2,34,116,117]*). *For the RLS algorithm (4.6)–(4.7), the following inequalities hold:*

$$\text{(i)}\quad \sum_{j=1}^{t}\boldsymbol{\varphi}^T(j)\boldsymbol{P}(j)\boldsymbol{\varphi}(j)\leqslant \ln|\boldsymbol{P}^{-1}(t)|+n\ln p_0,$$

$$\text{(ii)}\quad \sum_{t=1}^{\infty}\frac{\boldsymbol{\varphi}^T(t)\boldsymbol{P}(t)\boldsymbol{\varphi}(t)}{[\ln|\boldsymbol{P}^{-1}(t)|]^c}<\infty,\ c>1,$$

$$\text{(iii)}\quad \sum_{t=1}^{\infty}\frac{\boldsymbol{\varphi}^T(t)\boldsymbol{P}(t)\boldsymbol{\varphi}(t)}{\ln|\boldsymbol{P}^{-1}(t)|[\ln\ln|\boldsymbol{P}^{-1}(t)|]^c}<\infty,\ c>1,$$

$$\text{(iv)}\quad \sum_{t=1}^{\infty}\frac{\boldsymbol{\varphi}^T(t)\boldsymbol{P}(t)\boldsymbol{\varphi}(t)}{\ln|\boldsymbol{P}^{-1}(t)|\ln\ln|\boldsymbol{P}^{-1}(t)|[\ln\ln\ln|\boldsymbol{P}^{-1}(t)|]^c}<\infty,\ c>1.$$

Proof. From (4.7), we have

$$\begin{aligned}\boldsymbol{P}^{-1}(t-1)&=\boldsymbol{P}^{-1}(t)-\boldsymbol{\varphi}(t)\boldsymbol{\varphi}^{\mathrm{T}}(t)\\&=\boldsymbol{P}^{-1}(t)[\boldsymbol{I}_n-\boldsymbol{P}(t)\boldsymbol{\varphi}(t)\boldsymbol{\varphi}^{\mathrm{T}}(t)].\end{aligned}$$

Taking the determinant and using Lemma 4.12, we have

$$\begin{aligned}|\boldsymbol{P}^{-1}(t-1)|&=|\boldsymbol{P}^{-1}(t)\|\boldsymbol{I}_n-\boldsymbol{P}(t)\boldsymbol{\varphi}(t)\boldsymbol{\varphi}^{\mathrm{T}}(t)|\\&=|\boldsymbol{P}^{-1}(t)|[1-\boldsymbol{\varphi}^{\mathrm{T}}(t)\boldsymbol{P}(t)\boldsymbol{\varphi}(t)].\end{aligned}$$

Thus we have

$$\boldsymbol{\varphi}^{\mathrm{T}}(t)\boldsymbol{P}(t)\boldsymbol{\varphi}(t)=\frac{|\boldsymbol{P}^{-1}(t)|-|\boldsymbol{P}^{-1}(t-1)|}{|\boldsymbol{P}^{-1}(t)|}.\tag{4.17}$$

(i) Replacing t in the above equation with j, and summing for j from $j=1$ to $j=t$ give [118]

$$\begin{aligned}\sum_{j=1}^{t}\boldsymbol{\varphi}^{\mathrm{T}}(j)\boldsymbol{P}(j)\boldsymbol{\varphi}(j)&=\sum_{j=1}^{t}\frac{|\boldsymbol{P}^{-1}(j)|-|\boldsymbol{P}^{-1}(j-1)|}{|\boldsymbol{P}^{-1}(j)|}\\&=\sum_{j=1}^{t}\int_{|\boldsymbol{P}^{-1}(j-1)|}^{|\boldsymbol{P}^{-1}(j)|}\frac{\mathrm{d}x}{|\boldsymbol{P}^{-1}(j)|}\\&\leqslant\int_{|\boldsymbol{P}^{-1}(0)|}^{|\boldsymbol{P}^{-1}(t)|}\frac{\mathrm{d}x}{x}=\ln|\boldsymbol{P}^{-1}(t)|-\ln|\boldsymbol{P}^{-1}(0)|\\&=\ln|\boldsymbol{P}^{-1}(t)|-\ln\left|\frac{1}{p_0}\boldsymbol{I}_n\right|\\&=\ln|\boldsymbol{P}^{-1}(t)|+n\ln p_0.\end{aligned}$$

(ii) Dividing both sides of (4.17) by $[\ln|\boldsymbol{P}^{-1}(t)]^c$, and summing for t from $t=1$ to $t=\infty$ give

$$\begin{aligned}\sum_{t=1}^{\infty}\frac{\boldsymbol{\varphi}^{\mathrm{T}}(t)\boldsymbol{P}(t)\boldsymbol{\varphi}(t)}{[\ln|\boldsymbol{P}^{-1}(t)|]^c} &= \sum_{t=1}^{\infty}\frac{|\boldsymbol{P}^{-1}(t)|-|\boldsymbol{P}^{-1}(t-1)|}{|\boldsymbol{P}^{-1}(t)|[\ln|\boldsymbol{P}^{-1}(t)|]^c}\\ &\leqslant \int_{|\boldsymbol{P}^{-1}(0)|}^{|\boldsymbol{P}^{-1}(\infty)|}\frac{\mathrm{d}x}{x[\ln x]^c}\\ &= \frac{-1}{c-1}\left.\frac{1}{[\ln x]^{c-1}}\right|_{|\boldsymbol{P}^{-1}(0)|}^{|\boldsymbol{P}^{-1}(\infty)|}\\ &= \frac{1}{c-1}\left[\frac{1}{[\ln|\boldsymbol{P}^{-1}(0)|]^{c-1}}-\frac{1}{[\ln|\boldsymbol{P}^{-1}(\infty)|]^{c-1}}\right]<\infty.\end{aligned}$$

Here supposes that $\ln|\boldsymbol{P}^{-1}(t)|>0$, otherwise, the summation starts from $t=t_0$. Similarly, we have

$$\begin{aligned}&\text{(iii)}\quad \sum_{t=1}^{\infty}\frac{\boldsymbol{\varphi}^{\mathrm{T}}(t)\boldsymbol{P}(t)\boldsymbol{\varphi}(t)}{\ln|\boldsymbol{P}^{-1}(t)|[\ln\ln|\boldsymbol{P}^{-1}(t)|]^c}\\ &\qquad\leqslant \frac{1}{c-1}\left[\frac{1}{[\ln\ln|\boldsymbol{P}^{-1}(0)|]^{c-1}}-\frac{1}{[\ln\ln|\boldsymbol{P}^{-1}(\infty)|]^{c-1}}\right]<\infty,\\ &\text{(iv)}\quad \sum_{t=1}^{\infty}\frac{\boldsymbol{\varphi}^{\mathrm{T}}(t)\boldsymbol{P}(t)\boldsymbol{\varphi}(t)}{\ln|\boldsymbol{P}^{-1}(t)\ln\ln|\boldsymbol{P}^{-1}(t)|[\ln\ln\ln|\boldsymbol{P}^{-1}(t)|]^c}\\ &\qquad\leqslant \frac{1}{c-1}\left[\frac{1}{[\ln\ln\ln|\boldsymbol{P}^{-1}(0)|]^{c-1}}-\frac{1}{[\ln\ln\ln|\boldsymbol{P}^{-1}(\infty)|]^{c-1}}\right]<\infty.\end{aligned}$$

Summation starts from $t=t_0$ such that $\ln\ln\ln|\boldsymbol{P}^{-1}(t_0)|>0$ and $\ln\ln\ln|\boldsymbol{P}^{-1}(\infty)|=\infty$. □

The first conclusion of Lemma 4.19 was given by Chinese American statistician Lai and Wei first in 1982 [118] and the other conclusions under weaker conditions are put forward by the author of this paper [2,34,116,117].

Lemma 4.20 (*Inequalities about the Covariance Matrix [2,34,35,116,117,119,120]*).
For the RLS algorithm (4.6)–(4.7), the following inequalities hold:

$$\begin{aligned}&\text{(i)}\quad \sum_{t=1}^{\infty}\frac{\boldsymbol{\varphi}^{T}(t)\boldsymbol{P}(t)\boldsymbol{\varphi}(t)}{|\boldsymbol{P}^{-1}(t)|^c}<\infty,\ c>0,\\ &\text{(ii)}\quad \sum_{t=1}^{\infty}\frac{\boldsymbol{\varphi}^{T}(t)\boldsymbol{P}(t)\boldsymbol{\varphi}(t)}{r(t)}<\infty,\\ &\text{(iii)}\quad \sum_{t=1}^{\infty}\frac{\|\boldsymbol{\varphi}(t)\|^2}{r^c(t)}<\infty,\ c>1,\\ &\text{(iv)}\quad \sum_{t=1}^{\infty}\frac{\|\boldsymbol{\varphi}(t)\|^2}{r^c(t-1)r(t)}<\infty,\ c>0.\end{aligned}$$

4.3. Basic theorems of RLS algorithms

Lemma 4.21 (*Average Value Limit Lemma [2,35]*). *Suppose that the nonnegative sequence $\{f(t)\}$ has the limit f_0, i.e., $\lim_{t\to\infty}f(t)=f_0$. Then the following limit holds,*

$$\lim_{t\to\infty}\frac{1}{t}[f(1)+f(2)+\cdots+f(t)]=f_0.$$

Lemma 4.22 (*Series Limit Lemma or Limit Series Lemma [2,116,121]*). *For the functions $f(t)\geqslant 0$ and $g(t)\geqslant 0$, the limit $\lim_{t\to\infty}f(t)=f_0<\infty$ exists, but the series $\sum_{t=1}^{\infty}g(t)$ diverges and $\sum_{t=1}^{\infty}f(t)g(t)$ converges, i.e.,*

$$\sum_{t=1}^{\infty}g(t)=\infty,\quad \sum_{t=1}^{\infty}f(t)g(t)<\infty.$$

Then we have $f_0=0$.

Proof. We prove this lemma by using the counter proof method. Assume that $f_0 \neq 0$. The limit $\lim_{t\to\infty} f(t) = f_0$ implies that for given $\varepsilon = f_0/2$, there exist a positive number t_0 such that for $t \geqslant t_0$, $|f(t) - f_0| < \varepsilon$ holds, i.e., $-\varepsilon < f(t) - f_0 < \varepsilon$. Thus we have

$$f(t) > f_0 - \varepsilon = \frac{f_0}{2}.$$

From $\sum_{t=1}^{\infty} g(t) = \infty$, we have

$$\begin{aligned}\sum_{t=1}^{\infty} f(t)g(t) &= \sum_{t=1}^{t_0} f(t)g(t) + \sum_{t=t_0+1}^{\infty} f(t)g(t)\\ &\geqslant \sum_{t=1}^{t_0} f(t)g(t) + \sum_{t=t_0+1}^{\infty} \frac{f_0}{2} g(t)\\ &= \sum_{t=1}^{t_0} f(t)g(t) + \frac{f_0}{2} \sum_{t=t_0+1}^{\infty} g(t) = \infty.\end{aligned}$$

This contradicts the assumption that $\sum_{t=1}^{\infty} f(t)g(t)$ converges. Thus $f_0 = 0$. The proof is completed. □

Lemma 4.23 (*Toeplitz Lemma [85,122]*). *For a two-index sequence $\{a_{nk}, n = 1, 2, 3, \ldots, k = 1, 2, \ldots, n\}$, $\lim_{n\to\infty} a_{nk} = 0$ for any fixed k, $\sum_{k=1}^{\infty} |a_{nk}| = O(1) < \infty$ for all n, $\{x_n\}$ is a real sequence, then (i) we have*

$$x_n \to 0 \implies y_n := \sum_{k=1}^{\infty} a_{nk} x_k \to 0.$$

(ii) If $\sum_{k=1}^{\infty} a_{nk} \to 1$, then we have

$$x_n \to x \implies y_n = \sum_{k=1}^{\infty} a_{nk} x_k \to x.$$

Proof. (i) From the assumptions $\lim_{n\to\infty} a_{nk} = 0$, $\sum_{k=1}^{\infty} |a_{nk}| \leqslant C < \infty$ and $x_n \to 0$, for given small positive number $\varepsilon > 0$, there exist a $k(\varepsilon)$ such that for $k \geqslant k(\varepsilon)$, $|x_k| < \varepsilon/C$, and

$$\begin{aligned} y_n &= \sum_{k=1}^{k(\varepsilon)} a_{nk} x_k + \sum_{k=k(\varepsilon)+1}^{\infty} a_{nk} x_k,\\ |y_n| &\leqslant \sum_{k=1}^{k(\varepsilon)} |a_{nk} x_k| + \sum_{k=k(\varepsilon)+1}^{\infty} |a_{nk}\,\|\, x_k|\\ &\leqslant \sum_{k=1}^{k(\varepsilon)} |a_{nk}\,\|\, x_k| + C\frac{\varepsilon}{C}.\end{aligned}$$

Letting $n \to \infty$ and using $\lim_{n\to\infty} a_{nk} = 0$, we have $\lim_{n\to\infty} |y_n| \leqslant \varepsilon$. Letting $\varepsilon \to 0$ establishes (i). That is $x_n \to 0$ implies $\sum_{k=1}^{\infty} a_{nk} x_k \to 0$.

(ii) From the assumption $\lim_{n\to\infty} \sum_{k=1}^{\infty} a_{nk} = 1$, we have

$$y_n = \sum_{k=1}^{\infty} a_{nk} x + \sum_{k=1}^{\infty} a_{nk}(x_k - x).$$

Taking limit and using $x_n \to x$ and part (i), we can obtain $\lim_{n\to\infty} y_n = x + 0 = x$. □

Lemma 4.24 (*Kronecker Lemma [85,122]*). *For the real sequences $\{a_k\}$ and $\{r_k\}$, if $r_k = r_{k-1} + a_k \to \infty$, $r_0 = 0$, and $\sum_{k=1}^{\infty} \frac{b_k}{r_k}$ is convergent, then we have*

$$\lim_{n\to\infty} \frac{1}{r_k} \sum_{k=1}^{n} b_k = 0.$$

The following establishes the convergence theorems of the RLS algorithms.

Define the parameter estimation error vector

$$\tilde{\boldsymbol{\theta}}(t) := \hat{\boldsymbol{\theta}}(t) - \boldsymbol{\theta}.$$

Using (2.2) and (4.6), we have

$$\begin{aligned}\tilde{\boldsymbol{\theta}}(t) &= \tilde{\boldsymbol{\theta}}(t-1)+\boldsymbol{P}(t)\boldsymbol{\varphi}(t)[\boldsymbol{\varphi}^{\mathrm{T}}(t)\boldsymbol{\theta}+v(t)-\boldsymbol{\varphi}^{\mathrm{T}}(t)\hat{\boldsymbol{\theta}}(t-1)]\\ &=: \tilde{\boldsymbol{\theta}}(t-1)+\boldsymbol{P}(t)\boldsymbol{\varphi}(t)[-\tilde{y}(t)+v(t)],\end{aligned} \tag{4.18}$$

where

$$\begin{aligned}\tilde{y}(t) &:= \boldsymbol{\varphi}^{\mathrm{T}}(t)\hat{\boldsymbol{\theta}}(t-1)-\boldsymbol{\varphi}^{\mathrm{T}}(t)\boldsymbol{\theta}\\ &= \boldsymbol{\varphi}^{\mathrm{T}}(t)\tilde{\boldsymbol{\theta}}(t-1)=\tilde{\boldsymbol{\theta}}^{\mathrm{T}}(t-1)\boldsymbol{\varphi}(t)\in\mathbb{R}.\end{aligned}$$

Define a non-negative definite function

$$T(t):=\tilde{\boldsymbol{\theta}}^{\mathrm{T}}(t)\boldsymbol{P}^{-1}(t)\tilde{\boldsymbol{\theta}}(t).$$

Lemma 4.25. *For the RLS algorithm in (4.6)–(4.7), the non-negative definite function $T(t)$ satisfies*

$$\begin{aligned}T(t) &= T(t-1)-[1-\boldsymbol{\varphi}^{T}(t)\boldsymbol{P}(t)\boldsymbol{\varphi}(t)]\tilde{y}^2(t)+\boldsymbol{\varphi}^{T}(t)\boldsymbol{P}(t)\boldsymbol{\varphi}(t)v^2(t)\\ &\quad +2[1-\boldsymbol{\varphi}^{T}(t)\boldsymbol{P}(t)\boldsymbol{\varphi}(t)]\tilde{y}(t)v(t)\\ &\leqslant T(t-1)+\boldsymbol{\varphi}^{T}(t)\boldsymbol{P}(t)\boldsymbol{\varphi}(t)v^2(t)+2[1-\boldsymbol{\varphi}^{T}(t)\boldsymbol{P}(t)\boldsymbol{\varphi}(t)]\tilde{y}(t)v(t).\end{aligned} \tag{4.19}$$

Proof. Using (4.18) and (4.7), we have

$$\begin{aligned}T(t) &= \{\tilde{\boldsymbol{\theta}}(t-1)+\boldsymbol{P}(t)\boldsymbol{\varphi}(t)[-\tilde{y}(t)+v(t)]\}^{\mathrm{T}}\boldsymbol{P}^{-1}(t)\\ &\quad\times\{\tilde{\boldsymbol{\theta}}(t-1)+\boldsymbol{P}(t)\boldsymbol{\varphi}(t)[-\tilde{y}(t)+v(t)]\}\\ &= \tilde{\boldsymbol{\theta}}^{\mathrm{T}}(t-1)\boldsymbol{P}^{-1}(t)\tilde{\boldsymbol{\theta}}(t-1)+2\tilde{\boldsymbol{\theta}}^{\mathrm{T}}(t-1)\boldsymbol{\varphi}(t)[-\tilde{y}(t)+v(t)]\\ &\quad+\boldsymbol{\varphi}^{\mathrm{T}}(t)\boldsymbol{P}(t)\boldsymbol{\varphi}(t)[-\tilde{y}(t)+v(t)]^2\\ &= \tilde{\boldsymbol{\theta}}^{\mathrm{T}}(t-1)[\boldsymbol{P}^{-1}(t-1)+\boldsymbol{\varphi}^{\mathrm{T}}(t)\boldsymbol{\varphi}(t)]\tilde{\boldsymbol{\theta}}(t-1)+2\tilde{y}(t)[-\tilde{y}(t)+v(t)]\\ &\quad+\boldsymbol{\varphi}^{\mathrm{T}}(t)\boldsymbol{P}(t)\boldsymbol{\varphi}(t)[\tilde{y}^2(t)+v^2(t)-2\tilde{y}(t)v(t)]\\ &= T(t-1)-[1-\boldsymbol{\varphi}^{\mathrm{T}}(t)\boldsymbol{P}(t)\boldsymbol{\varphi}(t)]\tilde{y}^2(t)+\boldsymbol{\varphi}^{\mathrm{T}}(t)\boldsymbol{P}(t)\boldsymbol{\varphi}(t)v^2(t)\\ &\quad+2[1-\boldsymbol{\varphi}^{\mathrm{T}}(t)\boldsymbol{P}(t)\boldsymbol{\varphi}(t)]\tilde{y}(t)v(t).\end{aligned}$$

Using (ii) in Lemma 4.15, it follows the results of Lemma 4.25. □

Theorem 4.1 (*Martingale Convergence Theorem (Lemma D.5.3 in [122])*)**.** *Suppose that $\{W(t)\}$, $\{f(t)\}$ and $\{g(t)\}$ are the nonnegative random variable sequences and adapted to the σ-algebra sequence $\{\mathcal{F}_t, t\in\mathbb{N}\}$, and they satisfy the following relation,*

$$\mathrm{E}[W(t)|\mathcal{F}_{t-1}]\leqslant W(t-1)-f(t)+g(t).$$

If $\sum_{t=1}^{\infty}g(t)<\infty$, a.s., then $W(t)$ almost surely (a.s.) converges to a finite random variable W_0, i.e., $W(t)\to W_0$, a.s., and we also have $\sum_{t=1}^{\infty}f(t)<\infty$, a.s.

Theorem 4.2 (*Martingale HyperConvergence Theorem (MHCT) [2,123–126]*)**.**

Consider the candidate stochastic Lyapunov function of the state $x(t)$, $T(t):=T[x(t)]$, which is adapted to the σ-algebra sequence $\{\mathcal{F}_t, t\in\mathbb{N}\}$. Define the set

$$R_t:=[x(t):g[x(t)]\leqslant\eta_t<\infty,\ \text{a.s.}].$$

For $x(t)\in\mathbb{R}_t^c$, suppose that the following inequality holds,

$$\Delta T(t+1):=\mathrm{E}[T(t+1)|\mathcal{F}_t]-T(t)\leqslant -b(t+1),\ \text{a.s.}, \tag{4.20}$$

where R_t^c is the complementary set of R_t, $g(x)$ is a nonnegative function, $\eta_t\geqslant 0$ is a non-decreasing bounded random variable (which implies $R_t\subset R_{t+1}$), $b(t)\geqslant 0$ is a random variable, $\{x(t),\mathcal{F}_t\}$ is an adapted sequence. If $x(t)\in R_t^c$, and $\sum_{t=t_0}^{\infty}b(t)\to\infty$, a.s. *$t_0<\infty$, then for sufficient large t, $x(t)\in R_t$ almost surely (a.s.) holds, i.e., $\lim_{t\to\infty}x(t)\in R_{\infty}$,* a.s.

Proof. Let I_t be the indicator function of $x(t)\in R_t$, and $\bar{I}_t$ be the indicator function of $x(t)\in R_t^c$. For any $t>0$, we have $x(t)\in(R_t\bigcup R_t^c)$. When $x(t)\in R_t^c$, we have

$$\begin{aligned}\mathrm{E}[T(t+1)\bar{I}_{t+1}] &= \mathrm{E}[T(t)\bar{I}_{t+1}]+\mathrm{E}[\Delta T(t+1)\bar{I}_{t+1}]\\ &\leqslant \mathrm{E}[T(t)\bar{I}_t]-\mathrm{E}[b(t)].\end{aligned}$$

That is

$$\mathrm{E}[T(t)\bar{I}_t] \leqslant \mathrm{E}[T(t_0)\bar{I}_{t_0}] - \mathrm{E}\left[\sum_{j=t_0+1}^{t} b(j)\right],$$

where $T(t_0)$ is any finite random variable. When $t \to \infty$, $\sum_{t=t_0}^{t} b(t) \to \infty$, a.s., the following inequality holds,

$$T(t_0) - \sum_{j=t_0+1}^{t} b(t) \leqslant \eta_t, \text{ a.s., for some large t.}$$

This means that $\lim_{t\to\infty} \bar{I}_t = 0$, a.s., or $\lim_{t\to\infty} I_t = 1$, a.s., i.e., $\lim_{t\to\infty} x(t) \in R_\infty$, a.s. This completes the proof of Theorem 4.2. □

Theorem 4.3 (*Convergence Theorem of the RLS algorithm [2,116,117]*). *For the linear regressive system in (2.2) and the RLS algorithm in (4.6)–(4.7), suppose that $\{v(t), \mathcal{F}_t\}$ is a difference sequence defined on the probability space, where $\{\mathcal{F}_t\}$ is the $\sigma-$algebra sequence generated by the observations up to and including time t, i.e., $\mathcal{F}_t = \sigma(y(t), \boldsymbol{\varphi}(t), y(t-1), \boldsymbol{\varphi}(t-1), \ldots, y(1), \boldsymbol{\varphi}(1))$, or $\mathcal{F}_t = \sigma(v(t), v(t-1), \ldots, v(1))$. The sequence $\{v(t)\}$ satisfies the assumptions:*

(C5) $\mathrm{E}[v(t)|\mathcal{F}_{t-1}] = 0$, a.s.,

(C6) $\mathrm{E}[v^2(t)|\mathcal{F}_{t-1}] = \sigma_v^2(t) \leqslant \bar{\sigma}_v^2 < \infty$, a.s.

This means that $\{v(t)\}$ is an uncorrelated random noise with mean zero and variance $\sigma_v^2(t)$. Then for any $c > 1$, the parameter estimation error given by the RLS algorithm satisfies

$$\text{(i)} \quad \|\hat{\boldsymbol{\theta}}(t) - \boldsymbol{\theta}\|^2 = O\left(\frac{[\ln r(t)]^c}{\lambda_{\min}[\boldsymbol{P}^{-1}(t)]}\right), \text{ a.s.}$$

$$\text{(ii)} \quad \|\hat{\boldsymbol{\theta}}(t) - \boldsymbol{\theta}\|^2 = O\left(\frac{\ln r(t)[\ln\ln r(t)]^c}{\lambda_{\min}[\boldsymbol{P}^{-1}(t)]}\right), \text{ a.s.}$$

$$\text{(iii)} \quad \|\hat{\boldsymbol{\theta}}(t) - \boldsymbol{\theta}\|^2 = O\left(\frac{\ln r(t)\ln\ln r(t)[\ln\ln\ln r(t)]^c}{\lambda_{\min}[\boldsymbol{P}^{-1}(t)]}\right), \text{ a.s.}$$

$$\text{(iv)} \quad \|\hat{\boldsymbol{\theta}}(t) - \boldsymbol{\theta}\|^2 = O\left(\frac{\ln r(t)\ln\ln r(t)\ln\ln\ln r(t)[\ln\ln\ln\ln r(t)]^c}{\lambda_{\min}[\boldsymbol{P}^{-1}(t)]}\right), \text{ a.s.}$$

Proof. Since $\tilde{y}(t)$ and $\boldsymbol{\varphi}^{\mathrm{T}}(t)\boldsymbol{P}(t)\boldsymbol{\varphi}(t)$ are uncorrelated with $v(t)$ and are $\mathcal{F}_{t-1}-$measurable, taking the conditional expectation to both sides of (4.19) with respect to $\mathcal{F}_{t-1}$ and using Assumptions (C5) and (C6), we have

$$\mathrm{E}[T(t)|\mathcal{F}_{t-1}] \leqslant T(t-1) + 2\boldsymbol{\varphi}^{\mathrm{T}}(t)\boldsymbol{P}(t)\boldsymbol{\varphi}(t)\bar{\sigma}_v^2, \text{ a.s.}$$

From (i) in Lemma 4.19, it can be seen that the summation of the second term on the right-hand side in the above equation from $t = 1$ to $t = \infty$ is infinite, so the martingale convergence theorem cannot be applied. Let

$$V(t) := \frac{T(t)}{[\ln|\boldsymbol{P}^{-1}(t)|]^c}, \quad c > 1.$$

For $t \geqslant t_0$, suppose that $\ln|\boldsymbol{P}^{-1}(t)| > 0$. Because $\ln|\boldsymbol{P}^{-1}(t)|$ is non-descending, we have

$$\begin{aligned}\mathrm{E}[V(t)|\mathcal{F}_{t-1}] &\leqslant \frac{T(t-1)}{[\ln|\boldsymbol{P}^{-1}(t)|]^c} + \frac{2\boldsymbol{\varphi}^{\mathrm{T}}(t)\boldsymbol{P}(t)\boldsymbol{\varphi}(t)}{[\ln|\boldsymbol{P}^{-1}(t)|]^c}\bar{\sigma}_v^2 \\ &\leqslant V(t-1) + \frac{2\boldsymbol{\varphi}^{\mathrm{T}}(t)\boldsymbol{P}(t)\boldsymbol{\varphi}(t)}{[\ln|\boldsymbol{P}^{-1}(t)|]^c}\bar{\sigma}_v^2, \text{ a.s.}\end{aligned}$$

According to (ii) in Lemma 4.19, it is known that the summation of the second term on the right-hand side in the above equation from $t = 1$ to $t = \infty$ is finite. Applying the martingale convergence Theorem 4.1 to the above equation, we can conclude that $V(t)$ converges to a finite random variable V_0, a.s., i.e.,

$$V(t) = \frac{T(t)}{[\ln|\boldsymbol{P}^{-1}(t)|]^c} \to V_0 < \infty, \text{ a.s.,}$$

or

$$T(t) = O([\ln|\boldsymbol{P}^{-1}(t)|]^c), \text{ a.s.} \tag{4.21}$$

According to the definition of $T(t)$ and using Lemma 4.5, we have

$$\|\tilde{\boldsymbol{\theta}}(t)\|^2 \leqslant \frac{\tilde{\boldsymbol{\theta}}^{\mathrm{T}}(t)\boldsymbol{P}^{-1}(t)\tilde{\boldsymbol{\theta}}(t)}{\lambda_{\min}[\boldsymbol{P}^{-1}(t)]}$$

$$= \frac{T(t)}{\lambda_{\min}[\boldsymbol{P}^{-1}(t)]}.$$

Using (4.21) and Lemma 4.16, it follows that

$$\begin{aligned}\|\tilde{\boldsymbol{\theta}}(t)-\boldsymbol{\theta}\|^2 &= O\left(\frac{[\ln|\boldsymbol{P}^{-1}(t)|]^c}{\lambda_{\min}[\boldsymbol{P}^{-1}(t)]}\right)\\ &= O\left(\frac{[\ln r(t)]^c}{\lambda_{\min}[\boldsymbol{P}^{-1}(t)]}\right)\\ &= O\left(\frac{\{\lambda_{\max}[\boldsymbol{P}^{-1}(t)]\}^c}{\lambda_{\min}[\boldsymbol{P}^{-1}(t)]}\right),\ \text{a.s.},\ c>1.\end{aligned}$$

This proves Conclusion (i). Let

$$V(t) := \frac{T(t)}{\ln|\boldsymbol{P}^{-1}(t)|[\ln\ln|\boldsymbol{P}^{-1}(t)|]^c};$$

$$V(t) := \frac{T(t)}{\ln|\boldsymbol{P}^{-1}(t)|\ln\ln|\boldsymbol{P}^{-1}(t)|[\ln\ln\ln|\boldsymbol{P}^{-1}(t)|]^c};$$

$$V(t) := \frac{T(t)}{\ln|\boldsymbol{P}^{-1}(t)|\ln\ln|\boldsymbol{P}^{-1}(t)|\ln\ln\ln|\boldsymbol{P}^{-1}(t)|[\ln\ln\ln\ln|\boldsymbol{P}^{-1}(t)|]^c}.$$

Similarly, we can prove other conclusions (ii), (iii) and (iv). □

Theorem 4.3 indicates that for the noise $\{v(t)\}$ with zero mean and bounded variance, the convergence speed of the RLS parameter estimation error $\|\tilde{\boldsymbol{\theta}}(t)-\boldsymbol{\theta}\|^2$ equals the ratio of the logarithm of the greatest eigenvalue or trace to the smallest eigenvalue of the matrix $\boldsymbol{P}^{-1}(t)$. From Theorem 4.3, we have the following theorem.

Theorem 4.4 (*Convergence Theorem of the RLS algorithm [2,34,116,117]*). *For the linear regressive system in (2.2) and the RLS algorithm in (4.6)–(4.7), the conditions in Theorem 4.3 hold, if there exist constants $\alpha_1>0$, $\beta_1>0$ and $\gamma>0$ such that the following generalized weak persistent excitation (GWPE) condition holds,*

$$\text{(A3)}\ \ \alpha_1\boldsymbol{I}_n \leqslant \frac{1}{t}\sum_{j=1}^{t}\boldsymbol{\varphi}(j)\boldsymbol{\varphi}^T(j) = \frac{1}{t}\boldsymbol{H}_t^T\boldsymbol{H}_t \leqslant \beta_1 t^{\gamma}\boldsymbol{I}_n,\ \text{a.s.}$$

or the generalized attenuating excitation (GAE) condition holds,

$$\text{(A4)}\ \ \alpha_1\boldsymbol{I}_n \leqslant \frac{1}{t^{\varepsilon}}\sum_{j=1}^{t}\boldsymbol{\varphi}(j)\boldsymbol{\varphi}^T(j) = \frac{1}{t^{\varepsilon}}\boldsymbol{H}_t^T\boldsymbol{H}_t \leqslant \beta_1 t^{\gamma}\boldsymbol{I}_n,\ \text{a.s.},\ \varepsilon>0.$$

Then the parameter estimation error satisfies

$$\text{(i)}\quad \|\hat{\boldsymbol{\theta}}(t)-\boldsymbol{\theta}\|^2 = O\left(\frac{[\ln t]^c}{t^{\varepsilon}}\right) \to 0,\ \text{a.s.},\ c>1.$$

$$\text{(ii)}\quad \|\hat{\boldsymbol{\theta}}(t)-\boldsymbol{\theta}\|^2 = O\left(\frac{\ln t[\ln\ln t]^c}{t^{\varepsilon}}\right) \to 0,\ \text{a.s.},\ c>1.$$

$$\text{(iii)}\quad \|\hat{\boldsymbol{\theta}}(t)-\boldsymbol{\theta}\|^2 = O\left(\frac{\ln t\ln\ln t[\ln\ln\ln t]^c}{t^{\varepsilon}}\right) \to 0,\ \text{a.s.},\ c>1.$$

$$\text{(iv)}\quad \|\hat{\boldsymbol{\theta}}(t)-\boldsymbol{\theta}\|^2 = O\left(\frac{\ln t\ln\ln t\ln\ln\ln t[\ln\ln\ln\ln t]^c}{t^{\varepsilon}}\right) \to 0,\ \text{a.s.},\ c>1.$$

The condition number of the matrix $\boldsymbol{X}$ is defined as the ratio of the maximal to minimal singular values of $\boldsymbol{X}$. For a positive-definite matrix, the condition number equals the ratio of its maximum eigenvalue to minimum eigenvalue. The condition number is equal to or more than unity. A matrix having the large condition number implies that the matrix approaches ill-conditional.

It can be seen from the definition of the condition number that when $t\to\infty$, the condition numbers of $\boldsymbol{P}^{-1}(t)$ in the strong persistent excitation condition and the weak persistent excitation condition are bounded. However, the condition numbers of $\boldsymbol{P}^{-1}(t)$ in the generalized strong persistent excitation condition and the generalized weak persistent excitation condition introduced by us are unbounded. This means that the input and output data (i.e., the information vector $\boldsymbol{\varphi}(t)$) of the system can be unbounded and the algorithm is still convergent.

Theorem 4.5 (*Convergence Theorem of the RLS algorithm [2,116]*). *For the linear regressive system in (2.2) and the RLS algorithm in (4.6)–(4.7), suppose that the martingale difference sequence $\{v(t),\mathcal{F}_t\}$ satisfies the assumptions:*

$$\text{(C7)}\quad \mathrm{E}[v(t)|\mathcal{F}_{t-1}] = 0,\ \ \text{a.s.},$$

(C8) $\mathrm{E}[v^2(t)|\mathcal{F}_{t-1}] \leqslant \sigma_v^2 r^{\epsilon}(t),\ \text{a.s.},\ 0 \leqslant \sigma_v^2 < \infty,\ 0 \leqslant \epsilon < 1.$

Then the parameter estimation error given by the RLS algorithm satisfies

$$\|\hat{\boldsymbol{\theta}}(t)-\boldsymbol{\theta}\|^2 = o\left(\frac{r(t)}{\lambda_{\min}[\boldsymbol{P}^{-1}(t)]}\right) = o\left(\frac{\lambda_{\max}[\boldsymbol{P}^{-1}(t)]}{\lambda_{\min}[\boldsymbol{P}^{-1}(t)]}\right),\ \text{a.s.}$$

where $f(t)=o(g(t))$ represents $f(t)/g(t)\to 0$ as $t\to\infty$.

Condition (C8) is only an inequality relationship in numerical terms and does not mean that $v(t)$ is correlated with $r(t)$.

Proof. Under the conditions of Theorem 4.5, following the derivation of Theorem 4.3, we have

$$\mathrm{E}[T(t)|\mathcal{F}_{t-1}] \leqslant T(t-1) + 2\sigma_v^2 \boldsymbol{\varphi}^{\mathrm{T}}(t)\boldsymbol{P}(t)\boldsymbol{\varphi}(t) r^{\epsilon}(t),\ \text{a.s.}$$

Let

$$W(t) := \frac{T(t)}{r(t)}.$$

Hence, we have

$$\mathrm{E}[W(t)|\mathcal{F}_{t-1}] \leqslant W(t-1) - W(t-1)\frac{r(t)-r(t-1)}{r(t)} + 2\sigma_v^2 \frac{\boldsymbol{\varphi}^{\mathrm{T}}(t)\boldsymbol{P}(t)\boldsymbol{\varphi}(t)}{[r(t)]^{1-\epsilon}},\ \text{a.s.} \tag{4.22}$$

According to Lemma 4.16, we have

$$\frac{1}{r(t)} \leqslant \frac{1}{|\boldsymbol{P}^{-1}(t)|^{1/n}}.$$

Let $\mu := (1-\epsilon)/n > 0$. Using (4.17), the summation of the last term on the right-hand side of (4.22) from $t=1$ to $t=\infty$ is

$$\begin{aligned}
\sum_{t=1}^{\infty} \frac{\boldsymbol{\varphi}^{\mathrm{T}}(t)\boldsymbol{P}(t)\boldsymbol{\varphi}(t)}{[r(t)]^{1-\epsilon}} &\leqslant \sum_{t=1}^{\infty} \frac{|\boldsymbol{P}^{-1}(t)|-|\boldsymbol{P}^{-1}(t-1)|}{|\boldsymbol{P}^{-1}(t)|^{1+\mu}} \\
&\leqslant \int_{|\boldsymbol{P}^{-1}(0)|}^{|\boldsymbol{P}^{-1}(\infty)|} \frac{\mathrm{d}x}{x^{1+\mu}} \\
&= \frac{1}{\mu}\left[\frac{1}{|\boldsymbol{P}^{-1}(0)|^{\mu}} - \frac{1}{|\boldsymbol{P}^{-1}(\infty)|^{\mu}}\right] < \infty.
\end{aligned}$$

Applying the martingale convergence Theorem 4.1 to (4.22), we can conclude that $W(t)$ converges to a finite random variable W_0, a.s., i.e.,

$$W(t) = \frac{T(t)}{r(t)} \to W_0 < \infty,\ \text{a.s.},$$

and

$$\sum_{t=1}^{\infty} W(t-1)\frac{r(t)-r(t-1)}{r(t)} < \infty,\ \text{a.s.} \tag{4.23}$$

According to the definition of $r(t)$, for large t_0, if there exist a small positive number $\epsilon > 0$ such that for $t > t_0$, $\|\boldsymbol{\varphi}(t)\|^2/r(t) \leqslant 1-\epsilon$ holds. Then we have

$$\begin{aligned}
\sum_{t=1}^{\infty} \frac{r(t)-r(t-1)}{r(t-1)} &= \sum_{t=1}^{\infty} \int_{r(t-1)}^{r(t)} \frac{\mathrm{d}x}{r(t-1)} \geqslant \sum_{t=1}^{\infty} \int_{r(t-1)}^{r(t)} \frac{\mathrm{d}x}{x} \\
&= \int_{r(0)}^{r(\infty)} \frac{\mathrm{d}x}{x} = \ln r(\infty) - \ln r(0) \to \infty,\ \text{as } r(t)\to\infty, \\
\sum_{t=1}^{\infty} \frac{r(t)-r(t-1)}{r(t)} &= \sum_{t=1}^{\infty} \left[1-\frac{\|\boldsymbol{\varphi}(t)\|^2}{r(t)}\right]\frac{r(t)-r(t-1)}{r(t-1)} \\
&= \sum_{t=1}^{t_0} \left[1-\frac{\|\boldsymbol{\varphi}(t)\|^2}{r(t)}\right]\frac{r(t)-r(t-1)}{r(t-1)}
\end{aligned}$$

$$
\begin{aligned}
&+\sum_{t=t_0+1}^{\infty}\left[1-\frac{\|\boldsymbol{\varphi}(t)\|^2}{r(t)}\right]\frac{r(t)-r(t-1)}{r(t-1)}\\
&\geqslant \sum_{t=1}^{t_0}\left[1-\frac{\|\boldsymbol{\varphi}(t)\|^2}{r(t)}\right]\frac{r(t)-r(t-1)}{r(t-1)}\\
&+\sum_{t=1}^{\infty}\epsilon\frac{r(t)-r(t-1)}{r(t-1)}=\infty.
\end{aligned} \tag{4.24}
$$

Applying Lemma 4.22, from (4.23) and (4.24), we have $W_0=0$, a.s., i.e.,

$$
\frac{T(t)}{r(t)} \to 0,\ \text{a.s.} \tag{4.25}
$$

Using Lemma 4.16: $r(t)\leqslant n\lambda_{\max}[\boldsymbol{P}^{-1}(t)]$, from (4.25), we have

$$
\frac{\lambda_{\min}[\boldsymbol{P}^{-1}(t)]\|\tilde{\boldsymbol{\theta}}(t)\|^2}{n\lambda_{\max}[\boldsymbol{P}^{-1}(t)]}\leqslant\frac{\lambda_{\min}[\boldsymbol{P}^{-1}(t)]\|\tilde{\boldsymbol{\theta}}(t)\|^2}{r(t)}\leqslant\frac{T(t)}{r(t)}\to 0,\ \text{a.s.}
$$

This proves Theorem 4.5. □

From Theorem 4.5, we can conclude that under the generalized weak persistent excitation (GWPE) condition (A3), the parameter estimation error almost surely converges to zero, i.e., $\lim_{t\to\infty}\|\hat{\boldsymbol{\theta}}(t)-\boldsymbol{\theta}\|^2=0$, a.s.

Theorem 4.5 shows that the parameter estimation error almost surely converges to zero even for unbounded noise variance. The convergence of the RLS algorithm does not require that the disturbance noise has the normal distribution and data stationarity and ergodicity.

The parameter estimation convergence theorems established in the paper hold for the linear-parameter systems like (2.2) when the disturbance noise in the systems has mean zero and is uncorrelated with the information vector $\boldsymbol{\varphi}(t)$. Such systems include notably the AR models, CAR models, and FIR models.

Some lemmas and theorems about the RLS algorithm can be applied to the systems: (i) multiple-input equation-error systems, (ii) multi-input multi-output equation-error systems, (iii) linear-parameter equation-error systems. Some examples are as follows.

(i) **The autoregressive (AR) model**

$$
\begin{aligned}
&A(z)y(t)=v(t),\\
&A(z):=1+a_1z^{-1}+a_2z^{-2}+\cdots+a_{n_a}z^{-n_a}.
\end{aligned}
$$

This AR model can be expressed as

$$
\begin{aligned}
y(t)&=[1-A(z)]y(t)+v(t)\\
&=-a_1y(t-1)-a_2y(t-2)-\cdots-a_{n_a}y(t-n_a)+v(t)\\
&=\boldsymbol{\varphi}^{\mathrm{T}}(t)\boldsymbol{\theta}+v(t),\\
\boldsymbol{\varphi}(t)&:=[-y(t-1),-y(t-2),\ldots,-y(t-n_a)]^{\mathrm{T}}\in\mathbb{R}^{n_a},\\
\boldsymbol{\theta}&:=[a_1,a_2,\ldots,a_{n_a}]^{\mathrm{T}}\in\mathbb{R}^{n_a}.
\end{aligned}
$$

(ii) **The finite impulse response (FIR) model**

$$
\begin{aligned}
y(t)&=B(z)u(t)+v(t),\\
B(z)&:=b_1z^{-1}+b_2z^{-2}+\cdots+b_{n_b}z^{-n_b}.
\end{aligned}
$$

This FIR model can be expressed as

$$
\begin{aligned}
y(t)&=b_1u(t-1)+b_2u(t-2)+\cdots+b_{n_b}u(t-n_b)+v(t)\\
&=\boldsymbol{\varphi}^{\mathrm{T}}(t)\boldsymbol{\theta}+v(t),\\
\boldsymbol{\varphi}(t)&:=[u(t-1),u(t-2),\ldots,u(t-n_b)]^{\mathrm{T}}\in\mathbb{R}^{n_b},\\
\boldsymbol{\theta}&:=[b_1,b_2,\ldots,b_{n_b}]^{\mathrm{T}}\in\mathbb{R}^{n_b}.
\end{aligned}
$$

(iii) **The controlled autoregressive (CAR) model**, i.e., **equation-error (EE) model**

$$
A(z)y(t)=B(z)u(t)+v(t).
$$

This CAR model can be expressed as

$$
\begin{aligned}
y(t)&=[1-A(z)]y(t)+B(z)u(t)+v(t)\\
&=\boldsymbol{\varphi}^{\mathrm{T}}(t)\boldsymbol{\theta}+v(t),
\end{aligned}
$$

$$\begin{aligned}
\boldsymbol{\varphi}(t) &:= [-y(t-1), -y(t-2), \ldots, -y(t-n_a),\\
&\quad u(t-1), u(t-2), \ldots, u(t-n_b)]^{\mathrm{T}} \in \mathbb{R}^{n_a+n_b},\\
\boldsymbol{\theta} &:= [a_1, a_2, \ldots, a_{n_a}, b_1, b_2, \ldots, b_{n_b}]^{\mathrm{T}} \in \mathbb{R}^{n_a+n_b}.
\end{aligned}$$

(iv) **The multi-input equation-error model**, i.e., **multi-input CAR model**

$$\begin{aligned}
A(z)y(t) &= \sum_{j=1}^{r} B_j(z)u_j(t) + v(t),\\
B_j(z) &:= b_j(1)z^{-1} + b_j(2)z^{-2} + \cdots + b_j(n_j)z^{-n_j},
\end{aligned}$$

where $\boldsymbol{u}(t) := [u_1(t), u_2(t), \ldots, u_r(t)]^{\mathrm{T}} \in \mathbb{R}^r$ is the input vector of the system. The parameter vector to be identified is

$$\begin{aligned}
\boldsymbol{\theta} &:= [a_1, a_2, \ldots, a_{n_a}, b_1(1), b_1(2), \ldots, b_1(n_1), b_2(1), b_2(2), \ldots, b_2(n_2), \ldots,\\
&\quad b_r(1), b_r(2), \ldots, b_r(n_r)]^{\mathrm{T}} \in \mathbb{R}^n, \ n := n_a + n_1 + n_2 + \cdots + n_r.
\end{aligned}$$

The RLS algorithm of identifying $\boldsymbol{\theta}$ is given by

$$\begin{aligned}
\hat{\boldsymbol{\theta}}(t) &= \hat{\boldsymbol{\theta}}(t-1) + \boldsymbol{P}(t)\boldsymbol{\varphi}(t)[y(t) - \boldsymbol{\varphi}^{\mathrm{T}}(t)\hat{\boldsymbol{\theta}}(t-1)], \ \hat{\boldsymbol{\theta}}(0) = \mathbf{1}_n/p_0,\\
\boldsymbol{P}(t) &= \boldsymbol{P}(t-1) + \frac{\boldsymbol{P}(t-1)\boldsymbol{\varphi}(t)\boldsymbol{\varphi}^{\mathrm{T}}(t)\boldsymbol{P}(t-1)}{1 + \boldsymbol{\varphi}^{\mathrm{T}}(t)\boldsymbol{P}(t-1)\boldsymbol{\varphi}(t)}, \ \boldsymbol{P}(0) = p_0\boldsymbol{I}_n,\\
\boldsymbol{\varphi}(t) &:= [-y(t-1), \ldots, -y(t-n_a), u_1(t-1), \ldots, u_1(t-n_1),\\
&\quad u_2(t-1), \ldots, u_2(t-n_2), \ldots, u_r(t-1), \ldots, u_r(t-n_r)]^{\mathrm{T}}.
\end{aligned}$$

(v) **The multi-input multi-output (MIMO) equation-error model**, i.e., **MIMO CAR model**

$$\begin{aligned}
y_i(t) &= \sum_{j=1}^{m} A_{ij}(z)y_j(t) + \sum_{j=1}^{r} B_{ij}(z)u_j(t) + v_i(t), \ i = 1, 2, \ldots, m,\\
A_{ij}(z) &:= a_{ij}(1)z^{-1} + a_{ij}(2)z^{-2} + \cdots + a_{ij}(n_{ij})z^{-n_{ij}},\\
B_{ij}(z) &:= b_{ij}(1)z^{-1} + b_{ij}(2)z^{-2} + \cdots + b_{ij}(n_{ij})z^{-n_{ij}},
\end{aligned}$$

where $\boldsymbol{y}(t) := [y_1(t), y_2(t), \ldots, y_m(t)]^{\mathrm{T}} \in \mathbb{R}^m$ is the output vector of the systems, and $\boldsymbol{v}(t) := [v_1(t), v_2(t), \ldots, v_m(t)]^{\mathrm{T}} \in \mathbb{R}^m$ is the white noise vector with zero mean. The parameter vector to be identified is

$$\begin{aligned}
\boldsymbol{\theta}_i &:= [a_{i1}(1), a_{i1}(2), \ldots, a_{i1}(n_{i1}), a_{i2}(1), a_{i2}(2), \ldots, a_{i2}(n_{i2}), \ldots,\\
&\quad a_{im}(1), a_{im}(2), \ldots, a_{im}(n_{im}), b_{i1}(1), b_{i1}(2), \ldots, b_{i1}(n_{i1}),\\
&\quad b_{i2}(1), b_{i2}(2), \ldots, b_{i2}(n_{i2}), \ldots, b_{ir}(1), b_{ir}(2), \ldots, b_{im}(n_{ir})]^{\mathrm{T}} \in \mathbb{R}^{n_i}.
\end{aligned}$$

The corresponding RLS algorithm of identifying $\boldsymbol{\theta}$ is given by

$$\begin{aligned}
\hat{\boldsymbol{\theta}}_i(t) &= \hat{\boldsymbol{\theta}}_i(t-1) + \boldsymbol{P}_i(t)\boldsymbol{\varphi}_i(t)[y_i(t) - \boldsymbol{\varphi}_i^{\mathrm{T}}(t)\hat{\boldsymbol{\theta}}_i(t-1)], \ \hat{\boldsymbol{\theta}}_i(0) = \mathbf{1}_{n_i}/p_0,\\
\boldsymbol{P}_i(t) &= \boldsymbol{P}_i(t-1) + \frac{\boldsymbol{P}_i(t-1)\boldsymbol{\varphi}_i(t)\boldsymbol{\varphi}_i^{\mathrm{T}}(t)\boldsymbol{P}_i(t-1)}{1 + \boldsymbol{\varphi}_i^{\mathrm{T}}(t)\boldsymbol{P}_i(t-1)\boldsymbol{\varphi}_i(t)}, \ \boldsymbol{P}_i(0) = p_0\boldsymbol{I}_{n_i},\\
\boldsymbol{\varphi}_i(t) &:= [y_1(t-1), \ldots, y_1(t-n_{i1}), y_2(t-1), \ldots, y_2(t-n_{i2}), \ldots,\\
&\quad y_m(t-1), \ldots, y_m(t-n_{im}), u_1(t-1), \ldots, u_1(t-n_{i1}),\\
&\quad u_2(t-1), \ldots, u_2(t-n_{i2}), \ldots, u_r(t-1), \ldots, u_r(t-n_{ir})]^{\mathrm{T}}.
\end{aligned}$$

(vi) **The linear-parameter finite impulse response (LP-FIR) model** [1]:

$$\begin{aligned}
y(t) &= f(\boldsymbol{\theta}, u(t), z) + v(t)\\
&= \boldsymbol{\theta}^{\mathrm{T}}\boldsymbol{\varphi}(u(t), u(t-1), \ldots, u(t-n_b)) + v(t).
\end{aligned}$$

The corresponding RLS algorithm is given by

$$\begin{aligned}
\hat{\boldsymbol{\theta}}(t) &= \hat{\boldsymbol{\theta}}(t-1) + \boldsymbol{L}(t)[y(t) - \boldsymbol{\varphi}^{\mathrm{T}}(u(t), u(t-1), \ldots, u(t-n_b))\hat{\boldsymbol{\theta}}(t-1)],\\
\boldsymbol{L}(t) &= \boldsymbol{P}(t)\boldsymbol{\varphi}(u(t), u(t-1), \ldots, u(t-n_b))\\
&= \frac{\boldsymbol{P}(t-1)\boldsymbol{\varphi}(u(t), u(t-1), \ldots, u(t-n_b))}{1 + \boldsymbol{\varphi}^{\mathrm{T}}(u(t), \ldots, u(t-n_b))\boldsymbol{P}(t-1)\boldsymbol{\varphi}(u(t), \ldots, u(t-n_b))},\\
\boldsymbol{P}(t) &= [\boldsymbol{I}_{n_b} - \boldsymbol{L}(t)\boldsymbol{\varphi}^{\mathrm{T}}(u(t), u(t-1), \ldots, u(t-n_b))]\boldsymbol{P}(t-1), \ \boldsymbol{P}(0) = p_0\boldsymbol{I}_{n_b}.
\end{aligned}$$

(vii) **The linear-parameter equation-error (LP-EE) model** [1,2]:

$$A(z)y(t) = f(\boldsymbol{\theta}, u(t), z) + v(t),$$

where $f(\boldsymbol{\theta}, u(t), z) := f(\boldsymbol{\theta}, u(t), u(t-1), \ldots, u(t-n_b))$ is the linear function of $\boldsymbol{\theta}$ and the nonlinear function of the inputs $u(t), u(t-1), \ldots, u(t-n_b)$, i.e., $f(\boldsymbol{\theta}, u(t), z) := \boldsymbol{\theta}^{\mathrm{T}}\boldsymbol{\varphi}(u(t), u(t-1), \ldots, u(t-n_b))$. The LP-EE system can be expressed as

$$\begin{aligned}
y(t) &= [1-A(z)]y(t) + f(\boldsymbol{\theta}, u(t), z) + v(t) \\
&= [1-A(z)]y(t) + \boldsymbol{\theta}^{\mathrm{T}}\boldsymbol{\varphi}(u(t), u(t-1), \ldots, u(t-n_b)) + v(t) \\
&= -\sum_{i=1}^{n_a} a_i y(t-i) + \boldsymbol{\theta}^{\mathrm{T}}\boldsymbol{\varphi}(u(t), u(t-1), \ldots, u(t-n_b)) + v(t) \\
&= \boldsymbol{\phi}^{\mathrm{T}}(t)\boldsymbol{\vartheta} + v(t), \\
\boldsymbol{\phi}(t) &:= [-y(t-1), -y(t-2), \ldots, -y(t-n_a), \\
&\qquad \boldsymbol{\varphi}^{\mathrm{T}}(u(t), u(t-1), \ldots, u(t-n_b))] \in \mathbb{R}^{n_a+n_b}, \\
\boldsymbol{\vartheta} &:= \begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_{n_a} \\ \boldsymbol{\theta} \end{bmatrix} \in \mathbb{R}^{n_a+n_b}.
\end{aligned}$$

One extension is the nonlinear function of the outputs $y(t-1), y(t-2), \ldots, y(t-n_a)$ and can be expressed as

$$\begin{aligned}
y(t) &= A'(z)g(y(t)) + B(z)u(t) + v(t) \\
&= \boldsymbol{\phi}^{\mathrm{T}}(t)\boldsymbol{\vartheta} + v(t), \\
\boldsymbol{\phi}(t) &:= [g(y(t-1)), g(y(t-2)), \ldots, g(y(t-n_a)), \\
&\qquad u(t-1), u(t-2), \ldots, u(t-n_b)] \in \mathbb{R}^{n_a+n_b}, \\
\boldsymbol{\vartheta} &:= [a_1, a_2, \ldots, a_{n_a}, b_1, b_2, \ldots, b_{n_b}]^{\mathrm{T}} \in \mathbb{R}^{n_a+n_b},
\end{aligned}$$

and the other is the nonlinear function of the inputs and outputs:

$$\begin{aligned}
y(t) &= A'(z)g(y(t)) + f(\boldsymbol{\theta}, u(t), z) + v(t) \\
&= A'(z)g(y(t)) + B(z)f(u(t)) + v(t) \\
&= \boldsymbol{\phi}^{\mathrm{T}}(t)\boldsymbol{\vartheta} + v(t), \\
\boldsymbol{\phi}(t) &:= [g(y(t-1)), g(y(t-2)), \ldots, g(y(t-n_a)), \\
&\qquad f(u(t-1)), f(u(t-2)), \ldots, f(u(t-n_b))] \in \mathbb{R}^{n_a+n_b}, \\
\boldsymbol{\vartheta} &:= [a_1, a_2, \ldots, a_{n_a}, b_1, b_2, \ldots, b_{n_b}]^{\mathrm{T}} \in \mathbb{R}^{n_a+n_b}.
\end{aligned}$$

Hence, we have the general linear-parameter input–output systems [1]:

$$\begin{aligned}
y(t) &= g(\boldsymbol{\vartheta}, y(t), z) + f(\boldsymbol{\theta}, u(t), z) + v(t) \\
&= \boldsymbol{\vartheta}^{\mathrm{T}}\boldsymbol{\psi}(y(t-1), y(t-2), \ldots, y(t-n_a)) \\
&\quad + \boldsymbol{\theta}^{\mathrm{T}}\boldsymbol{\varphi}(u(t), u(t-1), \ldots, u(t-n_b)) + v(t).
\end{aligned}$$

This is a class of simple Hammerstein–Wiener nonlinear equation-error systems. (viii) **The multi-input linear-parameter equation-error model** [1]:

$$A(z)y(t) = \sum_{j=1}^{r} B_j(z)u_j(t)y(t-j) + v(t).$$

5. Multi-innovation least squares (MILS) methods

The multi-innovation recursive least squares method is referred to as the multi-innovation least squares (MILS) method, whose basic idea is expanding the scalar innovation to an innovation vector and/or the vector innovation to an innovation matrix through introducing the innovation length p [5,37–39]. This section derives the MILS estimates and the MILS algorithm for estimating the parameter vector $\boldsymbol{\theta}$ of linear regressive models, and explore the convergence properties of the MILS algorithm.

5.1. MILS estimates

Use the newest p observation data from $j = t - p + 1$ to $j = t$ to define the stacked output vector $\boldsymbol{Y}(p, t)$, the stacked information matrix $\boldsymbol{\Phi}(p, t)$ and the stacked noise vector $\boldsymbol{V}(p, t)$ as

$$\boldsymbol{Y}(p, t) := \begin{bmatrix} y(t) \\ y(t-1) \\ \vdots \\ y(t-p+1) \end{bmatrix} \in \mathbb{R}^p, \quad \boldsymbol{V}(p, t) := \begin{bmatrix} v(t) \\ v(t-1) \\ \vdots \\ v(t-p+1) \end{bmatrix} \in \mathbb{R}^p,$$
$$\boldsymbol{\Phi}(p, t) := [\boldsymbol{\varphi}(t), \boldsymbol{\varphi}(t-1), \ldots, \boldsymbol{\varphi}(t-p+1)] \in \mathbb{R}^{n\times p}.$$

From (2.2), we have the matrix equation:

$$\boldsymbol{Y}(p, t) = \boldsymbol{\Phi}^{\mathrm{T}}(p, t)\boldsymbol{\theta} + \boldsymbol{V}(p, t),$$

which is called the multi-innovation identification model. Define the criterion function

$$\begin{aligned} J_2(\boldsymbol{\theta}) &:= \sum_{j=1}^{t} [\boldsymbol{Y}(p, j) - \boldsymbol{\Phi}^{\mathrm{T}}(p, j)\boldsymbol{\theta}]^{\mathrm{T}} [\boldsymbol{Y}(p, j) - \boldsymbol{\Phi}^{\mathrm{T}}(p, j)\boldsymbol{\theta}] \\ &= \sum_{j=1}^{t} \|\boldsymbol{Y}(p, j) - \boldsymbol{\Phi}^{\mathrm{T}}(p, j)\boldsymbol{\theta}\|^2. \end{aligned}$$

Define the stacked vector $\boldsymbol{Z}_t$ and the information matrix $\boldsymbol{\Omega}_t$ as

$$\boldsymbol{Z}_t := \begin{bmatrix} \boldsymbol{Y}(p, 1) \\ \boldsymbol{Y}(p, 2) \\ \vdots \\ \boldsymbol{Y}(p, t) \end{bmatrix} \in \mathbb{R}^{pt}, \quad \boldsymbol{\Omega}_t := \begin{bmatrix} \boldsymbol{\Phi}^{\mathrm{T}}(p, 1) \\ \boldsymbol{\Phi}^{\mathrm{T}}(p, 2) \\ \vdots \\ \boldsymbol{\Phi}^{\mathrm{T}}(p, t) \end{bmatrix} \in \mathbb{R}^{(pt)\times n}.$$

Then the cost function $J_2(\boldsymbol{\theta})$ can be written as

$$J_2(\boldsymbol{\theta}) = (\boldsymbol{Z}_t - \boldsymbol{\Omega}_t\boldsymbol{\theta})^{\mathrm{T}}(\boldsymbol{Z}_t - \boldsymbol{\Omega}_t\boldsymbol{\theta}).$$

Letting $\boldsymbol{\theta} = \hat{\boldsymbol{\theta}}(t)$ make $J_2(\boldsymbol{\theta})\Big|_{\hat{\boldsymbol{\theta}}(t)} = \min$. Letting the partial derivative of $J_2(\boldsymbol{\theta})$ with respect to $\boldsymbol{\theta}$ at $\boldsymbol{\theta} = \hat{\boldsymbol{\theta}}(t)$ be zero gives

$$\begin{aligned} \frac{\partial J_2(\boldsymbol{\theta})}{\partial \boldsymbol{\theta}}\bigg|_{\boldsymbol{\theta}=\hat{\boldsymbol{\theta}}(t)} &= \frac{\partial[(\boldsymbol{Z}_t - \boldsymbol{\Omega}_t\boldsymbol{\theta})^{\mathrm{T}}(\boldsymbol{Z}_t - \boldsymbol{\Omega}_t\boldsymbol{\theta})]}{\partial \boldsymbol{\theta}}\bigg|_{\boldsymbol{\theta}=\hat{\boldsymbol{\theta}}(t)} \\ &= -2\boldsymbol{\Omega}_t^{\mathrm{T}}(\boldsymbol{Z}_t - \boldsymbol{\Omega}_t\boldsymbol{\theta})\Big|_{\boldsymbol{\theta}=\hat{\boldsymbol{\theta}}(t)} = \mathbf{0}. \end{aligned}$$

That is

$$(\boldsymbol{\Omega}_t^{\mathrm{T}}\boldsymbol{\Omega}_t)\hat{\boldsymbol{\theta}}(t) = \boldsymbol{\Omega}_t^{\mathrm{T}}\boldsymbol{Z}_t,$$

or

$$\begin{aligned} \hat{\boldsymbol{\theta}}(t) &= (\boldsymbol{\Omega}_t^{\mathrm{T}}\boldsymbol{\Omega}_t)^{-1}\boldsymbol{\Omega}_t^{\mathrm{T}}\boldsymbol{Z}_t \\ &= \left[\sum_{j=1}^{t} \boldsymbol{\Phi}(p, j)\boldsymbol{\Phi}^{\mathrm{T}}(p, j)\right]^{-1} \left[\sum_{j=1}^{t} \boldsymbol{\Phi}(p, j)\boldsymbol{Y}(p, j)\right]. \end{aligned} \tag{5.1}$$

This is the multi-innovation least squares (MILS) estimate.

5.2. MILS algorithm with recursive covariance

Define the vector $\boldsymbol{\xi}(t)$ and the covariance matrix $\boldsymbol{P}(t)$ as

$$\begin{aligned} \boldsymbol{\xi}(t) &:= \boldsymbol{\Omega}_t^{\mathrm{T}}\boldsymbol{Z}_t = \sum_{j=1}^{t} \boldsymbol{\Phi}(p, j)\boldsymbol{Y}(p, j) \\ &= \boldsymbol{\xi}(t-1) + \boldsymbol{\Phi}(p, t)\boldsymbol{Y}(p, t), \end{aligned} \tag{5.2}$$

$$\boldsymbol{P}^{-1}(t) := \boldsymbol{\Omega}_t^{\mathrm{T}}\boldsymbol{\Omega}_t = \sum_{j=1}^{t} \boldsymbol{\Phi}(p, j)\boldsymbol{\Phi}^{\mathrm{T}}(p, j)$$

$$= \boldsymbol{P}^{-1}(t-1) + \Phi(p,t)\Phi^{\mathrm{T}}(p,t). \tag{5.3}$$

Eq. (3.3) can be written as

$$\hat{\boldsymbol{\theta}}(t) = \boldsymbol{P}(t)\boldsymbol{\xi}(t). \tag{5.4}$$

Applying Lemma 3.8 to (5.3) gives

$$\begin{aligned}\boldsymbol{P}(t) &= \boldsymbol{P}(t-1) - \boldsymbol{P}(t-1)\Phi(p,t)\\ &\quad \times [\boldsymbol{I}_p + \Phi^{\mathrm{T}}(p,t)\boldsymbol{P}(t-1)\Phi(p,t)]^{-1}\Phi^{\mathrm{T}}(p,t)\boldsymbol{P}(t-1).\end{aligned} \tag{5.5}$$

Combining (5.4), (5.2) and (5.5) gives the multi-innovation least squares (LS) algorithm with the recursive covariance matrix [5,38,39]:

$$\begin{aligned}\hat{\boldsymbol{\theta}}(t) &= \boldsymbol{P}(t)\boldsymbol{\xi}(t),\\ \boldsymbol{\xi}(t) &= \boldsymbol{\xi}(t-1) + \Phi(p,t)\boldsymbol{Y}(p,t), \quad \boldsymbol{\xi}(0) = \mathbf{0},\\ \boldsymbol{P}(t) &= \boldsymbol{P}(t-1) - \boldsymbol{P}(t-1)\Phi(p,t)\\ &\quad \times [\boldsymbol{I}_p + \Phi^{\mathrm{T}}(p,t)\boldsymbol{P}(t-1)\Phi(p,t)]^{-1}\Phi^{\mathrm{T}}(p,t)\boldsymbol{P}(t-1), \quad \boldsymbol{P}(0) = p_0\boldsymbol{I}_n.\end{aligned}$$

5.3. Recursive MILS algorithms

To simplify, we call the recursive MILS algorithm as the MILS algorithm.

Imitating the derivation of the RLS algorithm and based on the MILS estimate in (5.1), we can obtain the multi-innovation least squares (MILS) algorithm [5,38,39]:

$$\hat{\boldsymbol{\theta}}(t) = \hat{\boldsymbol{\theta}}(t-1) + \boldsymbol{P}(t)\Phi(p,t)[\boldsymbol{Y}(p,t) - \Phi^{\mathrm{T}}(p,t)\hat{\boldsymbol{\theta}}(t-1)], \quad \hat{\boldsymbol{\theta}}(0) = \mathbf{1}_n/p_0, \tag{5.6}$$

$$\boldsymbol{P}^{-1}(t) = \boldsymbol{P}^{-1}(t-1) + \Phi(p,t)\Phi^{\mathrm{T}}(p,t), \quad \boldsymbol{P}(0) = p_0\boldsymbol{I}_n, \tag{5.7}$$

$$\boldsymbol{Y}(p,t) = [y(t), y(t-1), \ldots, y(t-p+1)]^{\mathrm{T}}, \tag{5.8}$$

$$\Phi(p,t) = [\boldsymbol{\varphi}(t), \boldsymbol{\varphi}(t-1), \ldots, \boldsymbol{\varphi}(t-p+1)]. \tag{5.9}$$

Introducing the gain matrix $\boldsymbol{L}(t) := \boldsymbol{P}(t)\Phi(p,t) \in \mathbb{R}^{n\times p}$, the MILS algorithm can be written as

$$\begin{aligned}\hat{\boldsymbol{\theta}}(t) &= \hat{\boldsymbol{\theta}}(t-1) + \boldsymbol{L}(t)[\boldsymbol{Y}(p,t) - \Phi^{\mathrm{T}}(p,t)\hat{\boldsymbol{\theta}}(t-1)], \quad \hat{\boldsymbol{\theta}}(0) = \mathbf{1}_n/p_0,\\ \boldsymbol{L}(t) &= \boldsymbol{P}(t-1)\Phi(p,t)[\boldsymbol{I}_p + \Phi^{\mathrm{T}}(p,t)\boldsymbol{P}(t-1)\Phi(p,t)]^{-1},\\ \boldsymbol{P}(t) &= \boldsymbol{P}(t-1) - \boldsymbol{P}(t-1)\Phi(p,t)\\ &\quad \times [\boldsymbol{I}_p + \Phi^{\mathrm{T}}(p,t)\boldsymbol{P}(t-1)\Phi(p,t)]^{-1}\Phi^{\mathrm{T}}(p,t)\boldsymbol{P}(t-1)\\ &= \boldsymbol{P}(t-1) - \boldsymbol{L}(t)\Phi^{\mathrm{T}}(p,t)\boldsymbol{P}(t-1), \quad \boldsymbol{P}(0) = p_0\boldsymbol{I}_n,\\ \boldsymbol{Y}(p,t) &= [y(t), y(t-1), \ldots, y(t-p+1)]^{\mathrm{T}},\\ \Phi(p,t) &= [\boldsymbol{\varphi}(t), \boldsymbol{\varphi}(t-1), \ldots, \boldsymbol{\varphi}(t-p+1)],\end{aligned}$$

where $\boldsymbol{P}(t) \in \mathbb{R}^{n\times n}$ is the covariance matrix, $p \geqslant 1$ is the innovation length, and $\hat{\boldsymbol{\theta}}(t)$ represents the estimate of $\boldsymbol{\theta}$ at time t. When $p = 1$, the MILS algorithm degenerates to the recursive least squares algorithm. The least squares algorithm and the multi-innovation least squares algorithm for linear regressive models in this paper can combine some estimation algorithms for studying new methods for linear and nonlinear stochastic systems with colored noises [127–133] and can be applied to other fields [134–142] such as control and schedule systems [143–155], the information processing and transportation communication systems and so on.

Lemma 5.1 (*Relation between the Innovation and the Residual for the MILS Algorithm*)**.** *In the MILS algorithm in (5.6)–(5.9), $\boldsymbol{E}(p,t) := \boldsymbol{Y}(p,t) - \Phi^{T}(p,t)\hat{\boldsymbol{\theta}}(t-1) \in \mathbb{R}^p$ is called the innovation vector. Define the residual vector $\hat{\boldsymbol{V}}(p,t) := \boldsymbol{Y}(p,t) - \Phi^{T}(p,t)\hat{\boldsymbol{\theta}}(t) \in \mathbb{R}^p$. Then the residual vector $\hat{\boldsymbol{V}}(p,t)$ and the innovation vector $\boldsymbol{E}(p,t)$ have the following relations:*

$$\begin{aligned}\hat{\boldsymbol{V}}(p,t) &= [\boldsymbol{I}_p - \Phi^{T}(p,t)\boldsymbol{P}(t)\Phi(p,t)]\boldsymbol{E}(p,t),\\ \boldsymbol{E}(p,t) &= [\boldsymbol{I}_p + \Phi^{T}(p,t)\boldsymbol{P}(t-1)\Phi(p,t)]\hat{\boldsymbol{V}}(p,t).\end{aligned}$$

Proof. Using (5.6), we have

$$\begin{aligned}\hat{\boldsymbol{V}}(p,t) &= \boldsymbol{Y}(p,t) - \Phi^{\mathrm{T}}(p,t)[\hat{\boldsymbol{\theta}}(t-1) + \boldsymbol{L}(t)\boldsymbol{E}(p,t)]\\ &= \boldsymbol{Y}(p,t) - \Phi^{\mathrm{T}}(p,t)\hat{\boldsymbol{\theta}}(t-1) - \Phi^{\mathrm{T}}(p,t)\boldsymbol{L}(t)\boldsymbol{E}(p,t)\\ &= \boldsymbol{E}(p,t) - \Phi^{\mathrm{T}}(p,t)\boldsymbol{L}(t)\boldsymbol{E}(p,t)\\ &= [\boldsymbol{I}_p - \Phi^{\mathrm{T}}(p,t)\boldsymbol{L}(t)]\boldsymbol{E}(p,t)\end{aligned}$$

$$\begin{aligned}&=[\boldsymbol{I}_p-\Phi^{\mathrm{T}}(p,t)\boldsymbol{P}(t)\Phi(p,t)]\boldsymbol{E}(p,t)\\&=\{\boldsymbol{I}_p-\Phi^{\mathrm{T}}(p,t)\boldsymbol{P}(t-1)\Phi(p,t)\\&\quad\times[\boldsymbol{I}_p+\Phi^{\mathrm{T}}(p,t)\boldsymbol{P}(t-1)\Phi(p,t)]^{-1}\}\boldsymbol{E}(p,t)\\&=[\boldsymbol{I}_p+\Phi^{\mathrm{T}}(p,t)\boldsymbol{P}(t-1)\Phi(p,t)]^{-1}\boldsymbol{E}(p,t).\quad\square\end{aligned}$$

5.4. Analysis of the MILS algorithms

Here establishes some lemmas and theorem about the convergence of the MILS algorithm.

Lemma 5.2. *For the MILS algorithm in (5.6)–(5.9), let $r(t):=\mathrm{tr}[\boldsymbol{P}^{-1}(t)]$. for any i ($i=0,1,\ldots,p-1$), the following inequalities hold [5,34,115,116]:*

$$\text{(i)}\quad \sum_{t=1}^{\infty}\boldsymbol{\varphi}^T(t-i)\boldsymbol{P}^3(t)\boldsymbol{\varphi}(t-i)<\infty,$$

$$\text{(ii)}\quad \sum_{t=1}^{\infty}\boldsymbol{\varphi}^T(t-i)\boldsymbol{P}(t-1)\boldsymbol{P}(t)\boldsymbol{P}(t-1)\boldsymbol{\varphi}(t-i)<\infty,$$

$$\text{(iii)}\quad \sum_{t=1}^{\infty}\boldsymbol{\varphi}^T(t-i)\boldsymbol{P}(t)\boldsymbol{P}(t-1)\boldsymbol{P}(t)\boldsymbol{\varphi}(t-i)<\infty,$$

$$\text{(iv)}\quad \sum_{t=1}^{\infty}\frac{\boldsymbol{\varphi}^T(t-i)\boldsymbol{P}(t)\boldsymbol{\varphi}(t-i)}{r^c(t)}<\infty,\ \textit{for any } c>0,$$

$$\text{(v)}\quad \sum_{t=1}^{\infty}\frac{\boldsymbol{\varphi}^T(t-i)\boldsymbol{P}(t)\boldsymbol{\varphi}(t-i)}{[\ln r(t)]^c}<\infty,\ \textit{for any } c>1,$$

$$\text{(vi)}\quad \sum_{t=1}^{\infty}\boldsymbol{\varphi}^T(t-i)\boldsymbol{P}^c(t)\boldsymbol{\varphi}(t-i)<\infty,\ \textit{for any } c>1.$$

Lemma 5.3 ([5,38]). *For the MILS algorithm in (5.6)–(5.9), and each i ($i=0,1,\ldots,p-1$), the following inequality holds:*

$$\sum_{t=1}^{\infty}\frac{\boldsymbol{\varphi}^T(t-i)\boldsymbol{P}(t)\boldsymbol{\varphi}(t-i)}{r^{\varepsilon}(t)}<\infty,\ \text{a.s.},\ \textit{for any } \varepsilon>0.$$

Proof. As in [34,116], from the definition of $\boldsymbol{P}(t)$ in (5.7), we have

$$\begin{aligned}\boldsymbol{P}^{-1}(t-1)&=\boldsymbol{P}^{-1}(t)-\Phi^{\mathrm{T}}(t)\Phi(t)\\&\leqslant \boldsymbol{P}^{-1}(t)-\boldsymbol{\varphi}(t-i)\boldsymbol{\varphi}^{\mathrm{T}}(t-i)\\&=\boldsymbol{P}^{-1}(t)[\boldsymbol{I}-\boldsymbol{P}(t)\boldsymbol{\varphi}(t-i)\boldsymbol{\varphi}^{\mathrm{T}}(t-i)].\end{aligned}$$

Taking determinants on both sides and using Lemma 4.5 yield

$$\begin{aligned}|\boldsymbol{P}^{-1}(t-1)|&\leqslant|\boldsymbol{P}^{-1}(t)\,\|\,\boldsymbol{I}-\boldsymbol{P}(t)\boldsymbol{\varphi}(t-i)\boldsymbol{\varphi}^{\mathrm{T}}(t-i)|\\&=|\boldsymbol{P}^{-1}(t)|[1-\boldsymbol{\varphi}^{\mathrm{T}}(t-i)\boldsymbol{P}(t)\boldsymbol{\varphi}(t-i)].\end{aligned}$$

Hence

$$\boldsymbol{\varphi}^{\mathrm{T}}(t-i)\boldsymbol{P}(t)\boldsymbol{\varphi}(t-i)\leqslant\frac{|\boldsymbol{P}^{-1}(t)|-|\boldsymbol{P}^{-1}(t-1)|}{|\boldsymbol{P}^{-1}(t)|}.\tag{5.10}$$

Dividing (5.10) by $r^{\varepsilon}(t)$ and summing for t give

$$\begin{aligned}\sum_{t=1}^{\infty}\frac{\boldsymbol{\varphi}^{\mathrm{T}}(t-i)\boldsymbol{P}(t)\boldsymbol{\varphi}(t-i)}{r^{\varepsilon}(t)}&\leqslant\sum_{t=1}^{\infty}\frac{|\boldsymbol{P}^{-1}(t)|-|\boldsymbol{P}^{-1}(t-1)|}{|\boldsymbol{P}^{-1}(t)|^{1+\varepsilon/n_0}}\\&=\sum_{t=1}^{\infty}\int_{|\boldsymbol{P}^{-1}(t-1)|}^{|\boldsymbol{P}^{-1}(t)|}\frac{\mathrm{d}x}{|\boldsymbol{P}^{-1}(t)|^{1+\varepsilon/n_0}}\\&\leqslant\int_{|\boldsymbol{P}^{-1}(0)|}^{|\boldsymbol{P}^{-1}(\infty)|}\frac{\mathrm{d}x}{x^{1+\varepsilon/n_0}}=\frac{-n_0}{\varepsilon}\left.\frac{1}{x^{\varepsilon/n_0}}\right|_{|\boldsymbol{P}^{-1}(0)|}^{|\boldsymbol{P}^{-1}(\infty)|}\end{aligned}$$

$$= \frac{n_0}{\varepsilon}\left(\frac{1}{|\boldsymbol{P}^{-1}(0)|^{\varepsilon/n_0}} - \frac{1}{|\boldsymbol{P}^{-1}(\infty)|^{\varepsilon/n_0}}\right) < \infty, \text{ a.s.} \quad \square$$

Lemma 5.4 (*[5,38]*). *For the MILS algorithm in (5.6)–(5.9), for any i $(i = 0, 1, \dots, p-1)$ and $t_0 < \infty$, the following inequalities hold:*

$$\text{(i)} \quad \sum_{j=1}^{t} \boldsymbol{\varphi}^T(j-i)\boldsymbol{P}(j)\boldsymbol{\varphi}(j-i) \leqslant \ln|\boldsymbol{P}^{-1}(t)| + n\ln p_0,$$

$$\text{(ii)} \quad \sum_{t=t_0}^{\infty} \frac{\boldsymbol{\varphi}^T(t-i)\boldsymbol{P}(t)\boldsymbol{\varphi}(t-i)}{[\ln|\boldsymbol{P}^{-1}(t)|]^c} < \infty, \text{ a.s., } \textit{for any} \ \ c > 1.$$

The proof can be done by a similar way in [2,34,41,115,116,156].

Theorem 5.1 (*[5,38]*). *For the MILS algorithm in (5.6)–(5.9), suppose that $\{v(t), \mathcal{F}_t\}$ is a difference sequence defined on the probability space, where $\{\mathcal{F}_t\}$ is the $\sigma-$algebra sequence generated by the observations up to and including time t, i.e., $\mathcal{F}_t = \sigma(y(t), \boldsymbol{\varphi}(t), y(t-1), \boldsymbol{\varphi}(t-1), \dots, y(1)\boldsymbol{\varphi}(1))$, or $\mathcal{F}_t = \sigma(v(t), v(t-1), \dots, v(1))$. The sequence $\{v(t)\}$ satisfies the assumptions:*

$$\text{(A5)} \quad \mathrm{E}[v(t)|\mathcal{F}_{t-1}] = 0, \text{ a.s.,}$$

$$\text{(A6)} \quad \mathrm{E}[v^2(t)|\mathcal{F}_{t-1}] \leqslant \sigma^2 < \infty, \text{ a.s.}$$

There exist constants $\alpha > 0$, $\beta > 0$ and $\alpha_0 \geqslant 0$ such that the following generalized persistent excitation (GPE) condition holds:

$$\text{(A7)} \quad \alpha\boldsymbol{I}_n \leqslant \frac{1}{t}\sum_{j=1}^{t} \boldsymbol{\Phi}(p,j)\boldsymbol{\Phi}^T(p,j) \leqslant \beta t^{\alpha_0}\boldsymbol{I}_n, \text{ a.s.}$$

Then the parameter estimation error consistently converges to zero, i.e., $\lim_{t\to\infty} \|\hat{\boldsymbol{\theta}}(t) - \boldsymbol{\theta}\|^2 = 0$, a.s. $\square$

6. Conclusions

This work investigates some properties of the least squares and multi-innovation least squares algorithms for linear regressive systems and gives some lemmas and theorems about their convergence. Many parameter estimation methods have been proposed for linear stochastic systems and nonlinear stochastic systems with colored noises [157–163] based on the gradient search and least squares search and Newton search [164–170] and can be used to model industrial processes and agriculture network systems [171–177] by means of some mathematical tools and information processing approaches. The basic lemma and theorems established for linear regressive systems in this paper can extended to multivariate stochastic systems and multiple-input multiple-output systems [178–183] and can be applied to other literatures [184–192] such as information processing [193–195].

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

References

[1] F. Ding, System Identification - New Theory and Methods, Science Press, Beijing, 2013.
[2] F. Ding, System Identification - Performances Analysis for Identification Methods, Science Press, Beijing, 2014.
[3] F. Ding, System Identification - Auxiliary Model Identification Idea and Methods, Science Press, Beijing, 2017.
[4] F. Ding, System Identification - Iterative Search Principle and Identification Methods, Science Press, Beijing, 2018.
[5] F. Ding, System Identification - Multi-Innovation Identification Theory and Methods, Science Press, Beijing, 2016.
[6] F. Ding, System Identification - Hierarchical Identification Principle and Methods, Science Press, Beijing, 2024.
[7] J. Pan, X. Jiang, X.K. Wan, W. Ding, A filtering based multi-innovation extended stochastic gradient algorithm for multivariable control systems, Int. J. Control Autom. Syst. 15 (3) (2017) 1189–1197.
[8] H. Ma, J. Pan, W. Ding, Partially-coupled least squares based iterative parameter estimation for multi-variable output-error-like autoregressive moving average systems, IET Control Theory Appl. 13 (18) (2019) 3040–3051.
[9] J. Pan, H. Ma, J. Sheng, Recursive coupled projection algorithms for multivariable output-error-like systems with coloured noises, IET Signal Process 14 (7) (2020) 455–466.

[10] Y.J. Liu, F. Ding, Y. Shi, An efficient hierarchical identification method for general dual-rate sampled-data systems, Automatica 50 (3) (2014) 962–970.
[11] M.H. Li, X. Liu, The least squares based iterative algorithms for parameter estimation of a bilinear system with autoregressive noise using the data filtering technique, Signal Process. 147 (2018) 23–34.
[12] M.H. Li, X. Liu, Maximum likelihood least squares based iterative estimation for a class of bilinear systems using the data filtering technique, Int. J. Control Autom. Syst. 18 (6) (2020) 1581–1592.
[13] M.H. Li, X. Liu, Maximum likelihood hierarchical least squares-based iterative identification for dual-rate stochastic systems, Internat. J. Adapt. Control Signal Process. 35 (2) (2021) 240–261.
[14] M.H. Li, X. Liu, Iterative identification methods for a class of bilinear systems by using the particle filtering technique, Internat. J. Adapt. Control Signal Process. 35 (10) (2021) 2056–2074.
[15] M.H. Li, X. Liu, Particle filtering-based iterative identification methods for a class of nonlinear systems with interval-varying measurements, Int. J. Control Autom. Syst. 20 (7) (2022) 2239–2248.
[16] Y. Ji, C. Zhang, Z. Kang, T. Yu, Parameter estimation for block-oriented nonlinear systems using the key term separation, Internat. J. Robust Nonlinear Control 30 (9) (2020) 3727–3752.
[17] Y. Ji, X.K. Jiang, L.J. Wan, Hierarchical least squares parameter estimation algorithm for two-input Hammerstein finite impulse response systems, J. Franklin Inst. 357 (8) (2020) 5019–5032.
[18] Y. Ji, Z. Kang, C. Zhang, Two-stage gradient-based recursive estimation for nonlinear models by using the data filtering, Int. J. Control Autom. Syst. 19 (8) (2021) 2706–2715.
[19] Y.M. Fan, X. Liu, Auxiliary model-based multi-innovation recursive identification algorithms for an input nonlinear controlled autoregressive moving average system with variable-gain nonlinearity, Internat. J. Adapt. Control Signal Process. 36 (3) (2022) 521–540.
[20] J.M. Li, F. Ding, T. Hayat, A novel nonlinear optimization method for fitting a noisy Gaussian activation function, Internat. J. Adapt. Control Signal Process. 36 (3) (2022) 690–707.
[21] J.W. Wang, Y. Ji, C. Zhang, Iterative parameter and order identification for fractional-order nonlinear finite impulse response systems using the key term separation, Internat. J. Adapt. Control Signal Process. 35 (8) (2021) 1562–1577.
[22] J.W. Wang, Y. Ji, X. Zhang, L. Xu, Two-stage gradient-based iterative algorithms for the fractional-order nonlinear systems by using the hierarchical identification principle, Internat. J. Adapt. Control Signal Process. 36 (7) (2022) 1778–1796.
[23] Y.M. Fan, X. Liu, Two-stage auxiliary model gradient-based iterative algorithm for the input nonlinear controlled autoregressive system with variable-gain nonlinearity, Internat. J. Robust Nonlinear Control 30 (14) (2020) 5492–5509.
[24] X. Liu, Y.M. Fan, Maximum likelihood extended gradient-based estimation algorithms for the input nonlinear controlled autoregressive moving average system with variable-gain nonlinearity, Internat. J. Robust Nonlinear Control 31 (9) (2021) 4017–4036.
[25] L. Xu, L. Chen, W.L. Xiong, Parameter estimation and controller design for dynamic systems from the step responses based on the Newton iteration, Nonlinear Dynam. 79 (3) (2015) 2155–2163.
[26] L. Xu, The damping iterative parameter identification method for dynamical systems based on the sine signal measurement, Signal Process. 120 (2016) 660–667.
[27] L. Xu, The parameter estimation algorithms based on the dynamical response measurement data, Adv. Mech. Eng. 9 (11) (2017) 1687814017730003.
[28] L.J. Wan, F. Ding, Decomposition- and gradient-based iterative identification algorithms for multivariable systems using the multi-innovation theory, Circuits Syst. Signal Process. 38 (7) (2019) 2971–2991.
[29] L. Xu, G.L. Song, A recursive parameter estimation algorithm for modeling signals with multi-frequencies, Circuits Syst. Signal Process. 39 (8) (2020) 4198–4224.
[30] L. Xu, F. Ding, E.F. Yang, Auxiliary model multiinnovation stochastic gradient parameter estimation methods for nonlinear sandwich systems, Internat. J. Robust Nonlinear Control 31 (1) (2021) 148–165.
[31] L. Xu, F.Y. Chen, T. Hayat, Hierarchical recursive signal modeling for multi-frequency signals based on discrete measured data, Internat. J. Adapt. Control Signal Process. 35 (5) (2021) 676–693.
[32] L. Xu, F. Ding, Q. Zhu, Decomposition strategy-based hierarchical least mean square algorithm for control systems from the impulse responses, Int. J. Sci. Syst. 52 (9) (2021) 1806–1821.
[33] L. Xu, W.L. Xiong, A. Alsaedi, T. Hayat, Hierarchical parameter estimation for the frequency response based on the dynamical window data, Int. J. Control Autom. Syst. 16 (4) (2018) 1756–1764.
[34] F. Ding, T. Chen, Combined parameter and output estimation of dual-rate systems using an auxiliary model, Automatica 40 (10) (2004) 1739–1748.
[35] F. Ding, T. Chen, Parameter estimation of dual-rate stochastic systems by using an output error method, IEEE Trans. Automat. Control 50 (9) (2005) 1436–1441.
[36] Y.J. Wang, F. Ding, Novel data filtering based parameter identification for multiple-input multiple-output systems using the auxiliary model, Automatica 71 (2016) 308–313.
[37] F. Ding, T. Chen, Performance analysis of multi-innovation gradient type identification methods, Automatica 43 (1) (2007) 1–14.
[38] F. Ding, X. Liu, G. Liu, Multiinnovation least squares identification for linear and pseudo-linear regression models, IEEE Trans. Syst. Man Cybern. Part B: Cybern. 40 (3) (2010) 767–778.
[39] F. Ding, Several multi-innovation identification methods, Digit. Signal Process. 20 (4) (2010) 1027–1039.
[40] F. Ding, Hierarchical multi-innovation stochastic gradient algorithm for Hammerstein nonlinear system modeling, Appl. Math. Model. 37 (4) (2013) 1694–1704.
[41] F. Ding, T. Chen, Hierarchical gradient-based identification of multivariable discrete-time systems, Automatica 41 (2) (2005) 315–325.
[42] F. Ding, T. Chen, Hierarchical least squares identification methods for multivariable systems, IEEE Trans. Automat. Control 50 (3) (2005) 397–402.
[43] F. Ding, H. Ma, J. Pan, E.F. Yang, Hierarchical gradient- and least squares-based iterative algorithms for input nonlinear output-error systems using the key term separation, J. Franklin Inst. 358 (9) (2021) 5113–5135.
[44] X. Zhang, F. Ding, Hierarchical parameter and state estimation for bilinear systems, Internat. J. Systems Sci. 51 (2) (2020) 275–290.
[45] H.Q. Han, L. Xie, F. Ding, X.G. Liu, Hierarchical least squares based iterative identification for multivariable systems with moving average noises, Math. Comput. Modelling 51 (9–10) (2010) 1213–1220.
[46] Z.N. Zhang, F. Ding, X.G. Liu, Hierarchical gradient based iterative parameter estimation algorithm for multivariable output error moving average systems, Comput. Math. Appl. 61 (3) (2011) 672–682.
[47] Y.H. Zhou, F. Ding, T. Hayat, Aitken-based acceleration estimation algorithms for a nonlinear model with exponential terms by using the decomposition, Int. J. Control Autom. Syst. 19 (11) (2021) 3720–3730.
[48] X. Meng, Y. Ji, J. Wang, Iterative parameter estimation for photovoltaic cell models by using the hierarchical principle, Int. J. Control Autom. Syst. 20 (8) (2022) 2583–2593.

[49] Y. Gu, W. Dai, Q. Zhu, H. Nouri, Hierarchical multi-innovation stochastic gradient identification algorithm for estimating a bilinear state-space model with moving average noise, J. Comput. Appl. Math. 420 (2023) 114794.
[50] F. Ding, G. Liu, X. Liu, Partially coupled stochastic gradient identification methods for non-uniformly sampled systems, IEEE Trans. Automat. Control 55 (8) (2010) 1976–1981.
[51] F. Ding, Coupled-least-squares identification for multivariable systems, IET Control Theory Appl. 7 (1) (2013) 68–79.
[52] Y.H. Zhou, X. Zhang, F. Ding, Partially-coupled nonlinear parameter optimization algorithm for a class of multivariate hybrid models, Appl. Math. Comput. 414 (2022) 126663.
[53] T. Cui, F. Ding, T. Hayat, Moving data window-based partially-coupled estimation approach for modeling a dynamical system involving unmeasurable states, ISA Trans. 128 (2022) 437–452.
[54] F. Ding, Y.J. Wang, J. Ding, Recursive least squares parameter identification algorithms for systems with colored noise using the filtering technique and the auxiliary model, Digit. Signal Process. 37 (2015) 100–108.
[55] F. Ding, Combined state and least squares parameter estimation algorithms for dynamic systems, Appl. Math. Model. 38 (1) (2014) 403–412.
[56] F. Ding, State filtering and parameter estimation for state space systems with scarce measurements, Signal Process. 104 (2014) 369–380.
[57] F. Ding, X. Liu, H.B. Chen, G.Y. Yao, Hierarchical gradient based and hierarchical least squares based iterative parameter identification for CARARMA systems, Signal Process. 97 (2014) 31–39.
[58] F. Ding, X. Liu, G. Liu, Gradient based and least-squares based iterative identification methods for OE and OEMA systems, Digit. Signal Process. 20 (3) (2010) 664–677.
[59] F. Ding, Two-stage least squares based iterative estimation algorithm for CARARMA system modeling, Appl. Math. Model. 37 (7) (2013) 4798–4808.
[60] F. Ding, Decomposition based fast least squares algorithm for output error systems, Signal Process. 93 (5) (2013) 1235–1242.
[61] F. Ding, X.P. Liu, G. Liu, Identification methods for Hammerstein nonlinear systems, Digit. Signal Process. 21 (2) (2011) 215–238.
[62] F. Ding, X.G. Liu, J. Chu, Gradient-based and least-squares-based iterative algorithms for Hammerstein systems using the hierarchical identification principle, IET Control Theory Appl. 7 (2) (2013) 176–184.
[63] Q.L. Liu, F.Y. Chen, T. Hayat, Recursive least squares estimation methods for a class of nonlinear systems based on non-uniform sampling, Internat. J. Adapt. Control Signal Process. 35 (8) (2021) 1612–1632.
[64] Q.L. Liu, F. Ding, Gradient-based recursive parameter estimation for a periodically nonuniformly sampled-data Hammerstein-Wiener system based on the key-term separation, Internat. J. Adapt. Control Signal Process. 35 (10) (2021) 1970–1989.
[65] J. Ding, L.J. Chen, Z.X. Cao, H.H. Guo, Convergence analysis of the modified adaptive extended Kalman filter for the parameter estimation of a brushless DC motor, Internat. J. Robust Nonlinear Control 31 (16) (2021) 7606–7620.
[66] H. Ma, F. Ding, Y. Wang, Multi-innovation Newton recursive methods for solving the support vector machine regression problems, Internat. J. Robust Nonlinear Control 31 (15) (2021) 7239–7260.
[67] J. Ding, Z.X. Cao, J.Z. Chen, G.P. Jiang, Weighted parameter estimation for Hammerstein nonlinear ARX systems, Circuits Syst. Signal Process. 39 (4) (2020) 2178–2192.
[68] M. Gan, H.T. Zhu, G.Y. Chen, C.L.P. Chen, Weighted generalized cross validation based regularization for broad learning system, IEEE Trans. Cybern. 52 (5) (2022) 4064–4072.
[69] M. Gan, Y. Guan, G.Y. Chen, C.L.P. Chen, Recursive variable projection algorithm for a class of separable nonlinear models, IEEE Trans. Neural Netw. Learn. Syst. 32 (11) (2021) 4971–4982.
[70] M. Gan, G. Chen, L. Chen, C.L.P. Chen, Term selection for a class of separable nonlinear models, IEEE Trans. Neural Netw. Learn. Syst. 31 (2) (2020) 445–451.
[71] G. Chen, M. Gan, C.L.P. Chen, H.X. Li, A regularized variable projection algorithm for separable nonlinear least-squares problems, IEEE Trans. Automat. Control 64 (2) (2019) 526–537.
[72] M. Gan, C.L.P. Chen, G.Y. Chen, L. Chen, On some separated algorithms for separable nonlinear squares problems, IEEE Trans. Cybern. 48 (10) (2018) 2866–2874.
[73] L. Xu, F. Ding, L.J. Wan, J. Sheng, Separable multi-innovation stochastic gradient estimation algorithm for the nonlinear dynamic responses of systems, Internat. J. Adapt. Control Signal Process. 34 (7) (2020) 937–954.
[74] L. Xu, F. Ding, E.F. Yang, Separable recursive gradient algorithm for dynamical systems based on the impulse response signals, Int. J. Control Autom. Syst. 18 (12) (2020) 3167–3177.
[75] L. Xu, Separable Newton recursive estimation method through system responses based on dynamically discrete measurements with increasing data length, Int. J. Control Autom. Syst. 20 (2) (2022) 432–443.
[76] L. Xu, Separable multi-innovation Newton iterative modeling algorithm for multi-frequency signals based on the sliding measurement window, Circuits Syst. Signal Process. 41 (2) (2022) 805–830.
[77] L. Xu, F. Ding, Q. Zhu, Separable synchronous multi-innovation gradient-based iterative signal modeling from on-line measurements, IEEE Trans. Instrum. Meas. 71 (2022) 6501313.
[78] F. Ding, T. Chen, Iterative least squares solutions of coupled Sylvester matrix equations, Systems Control Lett. 54 (2) (2005) 95–107.
[79] F. Ding, T. Chen, On iterative solutions of general coupled matrix equations, SIAM J. Control Optim. 44 (6) (2006) 2269–2284.
[80] F. Ding, X.P. Liu, J. Ding, Iterative solutions of the generalized Sylvester matrix equations by using the hierarchical identification principle, Appl. Math. Comput. 197 (1) (2008) 41–50.
[81] F. Ding, H.M. Zhang, Gradient-based iterative algorithm for a class of the coupled matrix equations related to control systems, IET Control Theory Appl. 8 (15) (2014) 1588–1595.
[82] L. Xie, J. Ding, F. Ding, Gradient based iterative solutions for general linear matrix equations, Comput. Math. Appl. 58 (7) (2009) 1441–1448.
[83] L. Xie, Y.J. Liu, H.Z. Yang, Gradient based and least squares based iterative algorithms for matrix equations $AXB+CX^{T}D=F$, Appl. Math. Comput. 217 (5) (2010) 2191–2199.
[84] J. Ding, Y.J. Liu, F. Ding, Iterative solutions to matrix equations of form AiXBi=Fi, Comput. Math. Appl. 59 (11) (2010) 3500–3507.
[85] T. Söderström, P. Stoica, System Identification, Prentice Hall Inc., Englewood Cliffs, New Jersey, 1989.
[86] J.X. Ma, R. Ding, Recursive computational formulas of the least squares criterion functions for scalar system identification, Appl. Math. Model. 38 (1) (2014) 1–11.
[87] J.X. Ma, F. Ding, Recursive relations of the cost functions for the least squares algorithms for multivariable systems, Circuits Syst. Signal Process. 32 (1) (2013) 83–101.
[88] C.C. Yin, Y.Z. Wen, An extension of Paulsen-Gjessing's risk model with stochastic return on investments, Insurance Math. Econom. 52 (3) (2013) 469–476.
[89] C.C. Yin, J.S. Zhao, Nonexponential asymptotics for the solutions of renewal equations, with applications, J. Appl. Probab. 43 (3) (2006) 815–824.
[90] C.C. Yin, K.C. Yuen, Optimality of the threshold dividend strategy for the compound Poisson model, Statist. Probab. Lett. 81 (12) (2011) 1841–1846.
[91] C.C. Yin, K.C. Yuen, Optimal dividend problems for a jump-diffusion model with capital injections and proportional transaction costs, J. Ind. Manag. Optim. 11 (4) (2015) 1247–1262.

[92] H. Wang, H. Fan, J. Pan, A true three-scroll chaotic attractor coined, Discret. Contin. Dynam. Syst. Ser. B 27 (5) (2022) 2891–2915.
[93] F.Z. Geng, X.Y. Wu, A novel kernel functions algorithm for solving impulsive boundary value problems, Appl. Math. Lett. 134 (2022) 108318.
[94] X.Y. Li, B.Y. Wu, A kernel regression approach for identification of first order differential equations based on functional data, Appl. Math. Lett. 127 (2022) 107832.
[95] Y. Gu, Q. Zhu, H. Nouri, Identification and U-control of a state-space system with time-delay, Internat. J. Adapt. Control Signal Process. 36 (1) (2022) 138–154.
[96] J.L. Ding, W.H. Zhang, Finite-time adaptive control for nonlinear systems with uncertain parameters based on the command filters, Internat. J. Adapt. Control Signal Process. 35 (9) (2021) 1754–1767.
[97] P. Ma, L. Wang, Filtering-based recursive least squares estimation approaches for multivariate equation-error systems by using the multi innovation theory, Internat. J. Adapt. Control Signal Process. 35 (9) (2021) 1898–1915.
[98] J. Chen, M. Gan, Q. Zhu, Y. Mao, Varying infimum gradient descent algorithm for agent-server systems with uncertain communication network, IEEE Trans. Instrum. Meas. 70 (2021) 9510511.
[99] J. Chen, Q. Zhu, Y.J. Liu, Modified Kalman filtering based multi-step-length gradient iterative algorithm for ARX models with random missing outputs, Automatica 118 (2020) 109034.
[100] J. Pan, W. Li, H.P. Zhang, Control algorithms of magnetic suspension systems based on the improved double exponential reaching law of sliding mode control, Int. J. Control Autom. Syst. 16 (6) (2018) 2878–2887.
[101] J.X. Xiong, J. Pan, G.Y. Chen, X. Zhang, F. Ding, Sliding mode dual-channel disturbance rejection attitude control for a quadrotor, IEEE Trans. Ind. Electron. 69 (10) (2022) 10489–10499.
[102] J. Pan, Q. Chen, J. Xiong, G. Chen, A novel quadruple boost nine level switched capacitor inverter, J. Electr. Eng. Technol. 18 (1) (2023) 467–480.
[103] J. Pan, S.D. Liu, J. Shu, X.K. Wan, Hierarchical recursive least squares estimation algorithm for secondorder Volterra nonlinear systems, Int. J. Control Autom. Syst. 20 (12) (2022) 3940–3950.
[104] M. Li, G. Xu, Q. Lai, J. Chen, A chaotic strategy-based quadratic opposition-based learning adaptive variable-speed whale optimization algorithm, Math. Comput. Simulation 193 (2022) 71–99.
[105] M.D. Li, G.H. Xu, L. Zeng, Q. Lai, Hybrid whale optimization algorithm based on symbiosis strategy for global optimization, Appl. Intell. (2023) http://dx.doi.org/10.1007/s10489-022-04132-9.
[106] J. Hou, F.W. Chen, P.H. Li, Z.Q. Zhu, Gray-box parsimonious subspace identification of Hammerstein-type systems, IEEE Trans. Ind. Electron. 68 (10) (2021) 9941–9951.
[107] J. Hou, H. Su, C.P. Yu, F.W. Chen, P.H. Li, Bias-correction errors-in-variables Hammerstein model identification, IEEE Trans. Ind. Electron. (2022) http://dx.doi.org/10.1109/TIE.2022.3199931.
[108] J. Hou, H. Su, C.P. Yu, F.W. Chen, P.H. Li, H.F. Xie, T.F. Li, Consistent subspace identification of errors-in-variables Hammerstein systems, IEEE Trans. Syst. Man Cybern. Syst. (2022) http://dx.doi.org/10.1109/TSMC.2022.3213809.
[109] Y.F. Chen, C. Zhang, C.Y. Liu, Y.M. Wang, X.K. Wan, Atrial fibrillation detection using feedforward neural network, J. Med. Biolog. Eng. 42 (1) (2022) 63–73.
[110] Y. Wang, G.C. Yang, S.B. Li, Y. Li, L. He, D. Liu, Arrhythmia classification algorithm based on multi-head self-attention mechanism, Biomed. Signal Process. Control 79 (2023) 104206.
[111] Y. Li, G.C. Yang, Z.D. Su, Y. Wang, Human activity recognition based on multienvironment sensor data, Inf. Fusion 91 (2023) 47–63.
[112] J. Lin, Y. Li, G.C. Yang, FPGAN: Face de-identification method with generative adversarial networks for social robots, Neural Netw. 133 (2021) 132–147.
[113] G.C. Yang, Z.J. Chen, Y. Li, Z.D. Su, Rapid relocation method for mobile robot based on improved ORB-SLAM2 algorithm, Remote Sens. 11 (2) (2019) 149, http://dx.doi.org/10.3390/rs11020149.
[114] F. Ding, J.B. Yang, Hierarchical identification of large scale systems, Acta Automat. Sinica 25 (5) (1999) 647–654.
[115] F. Ding, H.B. Chen, M. Li, Multi-innovation least squares identification methods based on the auxiliary model for MISO systems, Appl. Math. Comput. 187 (2) (2007) 658–668.
[116] F. Ding, X. Liu, Y. Shi, Convergence analysis of estimation algorithms for dual-rate stochastic systems, Appl. Math. Comput. 176 (1) (2006) 245–261.
[117] F. Ding, T. Chen, Identification of Hammerstein nonlinear ARMAX systems, Automatica 41 (9) (2005) 1479–1489.
[118] T. Lai, C. Wei, Least squares estimates in stochastic regression models with applications to identification and control of dynamic systems, Ann. Stat. 10 (1) (1982) 154–166.
[119] F. Ding, H.Z. Yang, F. Liu, Performance analysis of stochastic gradient algorithms under weak conditions, Sci. China Ser. F–Inf. Sci. 51 (9) (2008) 1269–1280.
[120] F. Ding, G. Liu, X. Liu, Parameter estimation with scarce measurements, Automatica 47 (8) (2011) 1646–1655.
[121] F. Ding, X. Liu, H.Z. Yang, Parameter identification and intersample output estimation for dual-rate systems, IEEE Trans. Syst. Man Cybern. Part A: Syst. Humans 38 (4) (2008) 966–975.
[122] G.C. Goodwin, K.S. Sin, Adaptive Filtering Prediction and Control, Prentice Hall, Englewood Cliffs, New Jersey, 1984.
[123] F. Ding, Time-Varying Parameter System Identification and Its Applications, Ph.D. Dissertation, Department of Automation, Tsinghua University, Beijing, 1994.
[124] F. Ding, Martingale hyperconvergence theorem and convergence of the forgetting factor least squares algorithm, Control Theory Appl. 14 (1) (1997) 90–95.
[125] F. Ding, Remarks on the martingale hyperconvergence theorem and convergence analysis of the forgetting factor least squares algorithms, Control Theory Appl. 16 (4) (1999) 569–572.
[126] F. Ding, J.B. Yang, Y.M. Xu, Convergence analysis of forgetting gradient algorithms by using the martingale hyperconvergence theorem, Tsinghua Sci. Technol. 5 (2) (2000) 187–192.
[127] Y.J. Wang, S.H. Tang, M.Q. Deng, Modeling nonlinear systems using the tensor network B-spline and the multi-innovation identification theory, Internat. J. Robust Nonlinear Control 32 (13) (2022) 7304–7318.
[128] Y.J. Wang, S.H. Tang, X.B. Gu, Parameter estimation for nonlinear Volterra systems by using the multi-innovation identification theory and tensor decomposition, J. Franklin Inst. 359 (2) (2022) 1782–1802.
[129] Y.J. Wang, L. Yang, An efficient recursive identification algorithm for multilinear systems based on tensor decomposition, Internat. J. Robust Nonlinear Control 31 (16) (2021) 7920–7936.
[130] Z. Kang, Y. Ji, X. Liu, Hierarchical recursive least squares algorithms for Hammerstein nonlinear autoregressive output-error systems, Internat. J. Adapt. Control Signal Process. 35 (11) (2021) 2276–2295.
[131] S.J. Fan, L. Xu, T. Hayat, Correlation analysis-based stochastic gradient and least squares identification methods for errors-in-variables systems using the multi-innovation, Int. J. Control Autom. Syst. 19 (1) (2021) 289–300.
[132] S.Y. Liu, Y.L. Zhang, L. Xu, Extended gradient-based iterative algorithm for bilinear state-space systems with moving average noises by using the filtering technique, Int. J. Control Autom. Syst. 19 (4) (2021) 1597–1606.

[133] C. Zhang, H.B. Liu, Y. Ji, Gradient parameter estimation of a class of nonlinear systems based on the maximum likelihood principle, Int. J. Control Autom. Syst. 20 (5) (2022) 1393–1404.
[134] H.B. Liu, J.W. Wang, Y. Ji, Maximum likelihood recursive generalized extended least squares estimation methods for a bilinear-parameter systems with ARMA noise based on the over-parameterization model, Int. J. Control Autom. Syst. 20 (8) (2022) 2606–2615.
[135] H. Ma, Y. Wang, Z. Ji, F. Ding, A novel three-stage quality oriented data-driven nonlinear industrial process monitoring strategy, IEEE Trans. Instrum. Meas. 71 (2022) 3524711.
[136] H. Ma, F. Ding, Y. Wang, A novel multi-innovation gradient support vector machine regression method, ISA Trans. 130 (2022) 343–359.
[137] Y. Cao, Y. An, S. Su, G. Xie, Y. Sun, A statistical study of railway safety in China and Japan 1990–2020, Accid. Anal. Prev. 175 (2022) 106764.
[138] Y. Cao, Y.R. Yang, L.C. Ma, J.K. Wen, Research on virtual coupled train control method based on GPC & VAPF, Chin. J. Electron. 31 (5) (2022) 897–905.
[139] Y.K. Sun, Y. Cao, P. Li, Contactless fault diagnosis for railway point machines based on multi-scale fractional wavelet packet energy entropy and synchronous optimization strategy, IEEE Trans. Veh. Technol. 71 (6) (2022) 5906–5914.
[140] Y. Cao, Y.S. Ji, Y.K. Sun, S. Su, The fault diagnosis of a switch machine based on deep random forest fusion, IEEE Intell. Trans. Syst. Mag. (2023) http://dx.doi.org/10.1109/MITS.2022.3174238.
[141] X. Wang, S. Su, Y. Cao, X.L. Wang, Robust control for dynamic train regulation in fully automatic operation system under uncertain wireless transmissions, IEEE Trans. Intell. Transp. Syst. (2022) http://dx.doi.org/10.1109/TITS.2022.3170950.
[142] Y. Cao, L.C. Ma, S. Xiao, Standard analysis for transfer delay in CTCS-3, Chin. J. Electron. 26 (5) (2017) 1057–1063.
[143] Y. Cao, Z.X. Zhang, F.L. Cheng, S. Su, Trajectory optimization for high-speed trains via a mixed integer linear programming approach, IEEE Trans. Intell. Transp. Syst. 23 (10) (2022) 17666–17676.
[144] Y. Cao, Y.K. Sun, G. Xie, P. Li, A sound-based fault diagnosis method for railway point machines based on two-stage feature selection strategy and ensemble classifier, IEEE Trans. Intell. Transp. Syst. 23 (8) (2022) 12074–12083.
[145] Y. Cao, J.K. Wen, A. Hobiny, P. Li, T. Wen, Parameter-varying artificial potential field control of virtual coupling system with nonlinear dynamics, Fractals 30 (2) (2022) 2240099.
[146] Y. Cao, J.K. Wen, L.C. Ma, Tracking and collision avoidance of virtual coupling train control system, Alex. Eng. J. 60 (2) (2021) 2115–2125.
[147] Y.K. Sun, Y. Cao, L.C. Ma, A fault diagnosis method for train plug doors via sound signals, IEEE Intell. Trans. Syst. Mag. 13 (3) (2021) 107–117.
[148] Y.K. Sun, Y. Cao, G. Xie, T. Wen, Sound based fault diagnosis for RPMs based on multi-scale fractional permutation entropy and two-scale algorithm, IEEE Trans. Veh. Technol. 70 (11) (2021) 11184–11192.
[149] S. Su, X.K. Wang, Y. Cao, J.T. Yin, An energy-efficient train operation approach by integrating the metro timetabling and eco-driving, IEEE Trans. Intell. Transp. Syst. 21 (10) (2020) 4252–4268.
[150] Y. Cao, Z. Wang, F. Liu, P. Li, G. Xie, Bio-inspired speed curve optimization and sliding mode tracking control for subway trains, IEEE Trans. Veh. Technol. 68 (7) (2019) 6331–6342.
[151] Y. Cao, Y.K. Sun, G. Xie, T. Wen, Fault diagnosis of train plug door based on a hybrid criterion for IMFs selection and fractional wavelet package energy entropy, IEEE Trans. Veh. Technol. 68 (8) (2019) 7544–7551.
[152] S. Su, J.F. She, K.C. Li, X. Wang, Y. Zhou, A nonlinear safety equilibrium spacing based model predictive control for virtually coupled train set over gradient terrains, IEEE Trans. Transp. Electrif. 8 (2) (2022) 2810–2824.
[153] S. Su, Q.Y. Zhu, J.Q. Liu, T. Tang, Q.L. Wei, Y. Cao, Eco-driving of trains with a data-driven iterative learning approach, IEEE Trans. Ind. Inform. (2022) http://dx.doi.org/10.1109/TII.2022.3195888.
[154] S. Su, T. Tang, J. Xun, F. Cao, Y.H. Wang, Design of running grades for energy-efficient train regulation: A case study for Beijing Yizhuang line, IEEE Intell. Trans. Syst. Mag. 13 (2) (2021) 189–200.
[155] S. Su, X.K. Wang, T. Tang, G. Wang, Y. Cao, Energy-efficient operation by cooperative control among trains: A multi-agent reinforcement learning approach, Control Eng. Pract. 116 (2021) 104901.
[156] F. Ding, Y. Shi, T. Chen, Performance analysis of estimation algorithms of non-stationary ARMA processes, IEEE Trans. Signal Process. 54 (3) (2006) 1041–1053.
[157] S.Y. Liu, X. Zhang, L. Xu, F. Ding, Expectation–maximization algorithm for bilinear systems by using the Rauch-Tung-Striebel smoother, Automatica 142 (2022) 110365.
[158] J.M. Li, F. Ding, Fitting nonlinear signal models using the increasing-data criterion, IEEE Signal Process. Lett. 29 (2022) 1302–1306.
[159] H. Xu, F. Ding, B. Champagne, Joint parameter and time-delay estimation for a class of nonlinear time-series models, IEEE Signal Process. Lett. 29 (2022) 947–951.
[160] L. Xu, A proportional differential control method for a time-delay system using the Taylor expansion approximation, Appl. Math. Comput. 236 (2014) 391–399.
[161] L. Xu, Application of the Newton iteration algorithm to the parameter estimation for dynamical systems, J. Comput. Appl. Math. 288 (2015) 33–43.
[162] X. Zhang, F. Ding, Optimal adaptive filtering algorithm by using the fractional-order derivative, IEEE Signal Process. Lett. 29 (2022) 399–403.
[163] Y.H. Zhou, F. Ding, Modeling nonlinear processes using the radial basis function-based state-dependent autoregressive models, IEEE Signal Process. Lett. 27 (2020) 1600–1604.
[164] Y.H. Zhou, F. Ding, Hierarchical estimation approach for RBF-AR models with regression weights based on the increasing data length, IEEE Trans. Circuits Syst. II: Express Briefs 68 (12) (2021) 3597–3601.
[165] Y. Ji, Z. Kang, X. Zhang, L. Xu, Model recovery for multi-input signal-output nonlinear systems based on the compressed sensing recovery theory, J. Franklin Inst. 359 (5) (2022) 2317–2339.
[166] Y. Ji, Z. Kang, Three-stage forgetting factor stochastic gradient parameter estimation methods for a class of nonlinear systems, Internat. J. Robust Nonlinear Control 31 (3) (2021) 971–987.
[167] J.M. Li, F. Ding, Synchronous optimization schemes for dynamic systems through the kernel-based nonlinear observer canonical form, IEEE Trans. Instrum. Meas. 71 (2022) 3210952.
[168] Y. Ji, Z. Kang, X. Liu, The data filtering based multiple-stage Levenberg–Marquardt algorithm for Hammerstein nonlinear systems, Internat. J. Robust Nonlinear Control 31 (15) (2021) 7007–7025.
[169] F. Ding, X. Liu, X. Ma, Kalman state filtering based least squares iterative parameter estimation for observer canonical state space systems using decomposition, J. Comput. Appl. Math. 301 (2016) 135–143.
[170] F. Ding, X. Zhang, L. Xu, The innovation algorithms for multivariable state-space models, Internat. J. Adapt. Control Signal Process. 33 (11) (2019) 1601–1618.
[171] C. Xu, H. Xu, Z. Guan, Y. Ge, Observer-based dynamic event-triggered semi-global bipartite consensus of linear multi-agent systems with input saturation, IEEE Trans. Cybern. (2022) http://dx.doi.org/10.1109/TCYB.2022.3164048.
[172] C. Xu, Y. Qin, H. Su, Observer-based dynamic event-triggered bipartite consensus of discrete-time multi-agent systems, IEEE Trans. Circuits Syst. II: Express Briefs (2023) http://dx.doi.org/10.1109/TCSII.2022.3217918.
[173] C. Wei, X. Zhang, L. Xu, Overall recursive least squares and overall stochastic gradient algorithms and their convergence for feedback nonlinear controlled autoregressive systems, Internat. J. Robust Nonlinear Control 32 (9) (2022) 5534–5554.

[174] J.Y. You, C.P. Yu, J. Sun, J. Chen, Generalized maximum entropy based identification of graphical ARMA models, Automatica 141 (2022) 110319.
[175] J.M. Li, Y.J. Rong, C. Wang, F. Ding. X.L. Li, Fitting the nonlinear systems based on the kernel functions through recursive search, Int. J. Control Autom. Syst. 20 (6) (2022) 1849–1860.
[176] C.P. Yu, Y. Li, H. Fang, J. Chen, System identification approach for inverse optimal control of finite-horizon linear quadratic regulators, Automatica 129 (2021) 109636.
[177] X. Wang, Y.J. Rong, C. Wang, Gradient-based iterative parameter estimation for a finite impulse response system with saturation nonlinearity, Int. J. Control Autom. Syst. 20 (1) (2022) 73–83.
[178] L. Xu, F. Ding, Separable synthesis estimation methods and convergence analysis for multivariable systems, J. Comput. Appl. Math. (2023) 115104, http://dx.doi.org/10.1016/j.cam.2023.115104.
[179] L. Xu, F. Ding, X. Lu, L.J. Wan, J. Sheng, Hierarchical multi-innovation generalised extended stochastic gradient methods for multivariable equation-error autoregressive moving average systems, IET Control Theory Appl. 14 (10) (2020) 1276–1286.
[180] F. Ding, X.P. Liu, Auxiliary model based stochastic gradient algorithm for multivariable output error systems, Acta Automat. Sinica 36 (7) (2010) 993–998.
[181] Y.J. Wang, F. Ding, M.H. Wu, Recursive parameter estimation algorithm for multivariate output-error systems, J. Franklin Inst. 355 (12) (2018) 5163–5181.
[182] J.X. Ma, W.L. Xiong, J. Chen, F. Ding, Hierarchical identification for multivariate Hammerstein systems by using the modified Kalman filter, IET Control Theory Appl. 11 (6) (2017) 857–869.
[183] P. Ma, F. Ding, New gradient based identification methods for multivariate pseudo-linear systems using the multi-innovation and the data filtering, J. Franklin Inst. 354 (3) (2017) 1568–1583.
[184] Z.W. Shi, H.D. Yang, M. Dai, The data-filtering based bias compensation recursive least squares identification for multi-input single-output systems with colored noises, J. Franklin Inst. (360) (2023) http://dx.doi.org/10.1016/j.jfranklin.2023.01.040.
[185] J. Pan, Y.Q. Liu, J. Shu, Gradient-based parameter estimation for an exponential nonlinear autoregressive time-series model by using the multi-innovation, Int. J. Control Autom. Syst. 21 (1) (2023) 140–150.
[186] X.H. Wang, F. Ding, Modified particle filtering-based robust estimation for a networked control system corrupted by impulsive noise, Internat. J. Robust Nonlinear Control 32 (2) (2022) 830–850.
[187] F. Ding, L. Lv, J. Pan, X.K. Wan, X.B. Jin, Two-stage gradient-based iterative estimation methods for controlled autoregressive systems using the measurement data, Int. J. Control Autom. Syst. 18 (4) (2020) 886–896.
[188] S.Y. Liu, F. Ding, L. Xu, T. Hayat, Hierarchical principle-based iterative parameter estimation algorithm for dual-frequency signals, Circuits Syst. Signal Process. 38 (7) (2019) 3251–3268.
[189] X. Zhang, F. Ding, E.F. Yang, State estimation for bilinear systems through minimizing the covariance matrix of the state estimation errors, Internat. J. Adapt. Control Signal Process. 33 (7) (2019) 1157–1173.
[190] M.H. Li, X.M. Liu, F. Ding, The filtering-based maximum likelihood iterative estimation algorithms for a special class of nonlinear systems with autoregressive moving average noise using the hierarchical identification principle, Internat. J. Adapt. Control Signal Process. 33 (7) (2019) 1189–1211.
[191] J.X. Ma, F. Ding, Filtering-based multistage recursive identification algorithm for an input nonlinear output-error autoregressive system by using the key term separation technique, Circuits Syst. Signal Process. 36 (2) (2017) 577–599.
[192] F. Ding, F.F. Wang, L. Xu, M.H. Wu, Decomposition based least squares iterative identification algorithm for multivariate pseudo-linear ARMA systems using the data filtering, J. Franklin Inst. 354 (3) (2017) 1321–1339.
[193] X. Jin, Wang Z., Kong J., Bai Y., Su T., Ma H., Chakrabarti P., Deep spatio-temporal graph network with self-optimization for air quality prediction, Entropy 25 (2023) 247, http://dx.doi.org/10.3390/e25020247.
[194] X. Jin, Wang Z., Gong W., Kong J., Bai Y., Su T., Ma H., Chakrabarti P., Variational bayesian network with information interpretability filtering for air quality forecasting, Mathematics 11 (4) (2023) 837, http://dx.doi.org/10.3390/math11040837.
[195] N. Zhao, A. Wu, Y. Pei, Y. Liang, D. Niyato, Patial-temporal aggregation graph convolution network for efficient mobile cellular traffic prediction, IEEE Commun. Lett. 26 (3) (2022) 587–591.